

(12) **United States Patent**
Park et al.

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(54) **BANDWIDTH PART FAILURE INDICATION
IN RADIO LINK FAILURE REPORT**

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U.S.C. 154(b) by 0 days.

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claimer.

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Related U.S. Application Data

(63) Continuation of application No. 18/233,460, filed on
Aug. 14, 2023, now Pat. No. 12,035,402, which is a
(Continued)

(51) **Int. Cl.**
H04W 76/18 (2018.01)
H04L 5/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04W 76/18** (2018.02); **H04L 5/0055**
(2013.01); **H04W 72/0453** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC . H04W 76/18; H04W 72/0453; H04W 72/23;
H04W 76/19; H04W 76/27;
(Continued)

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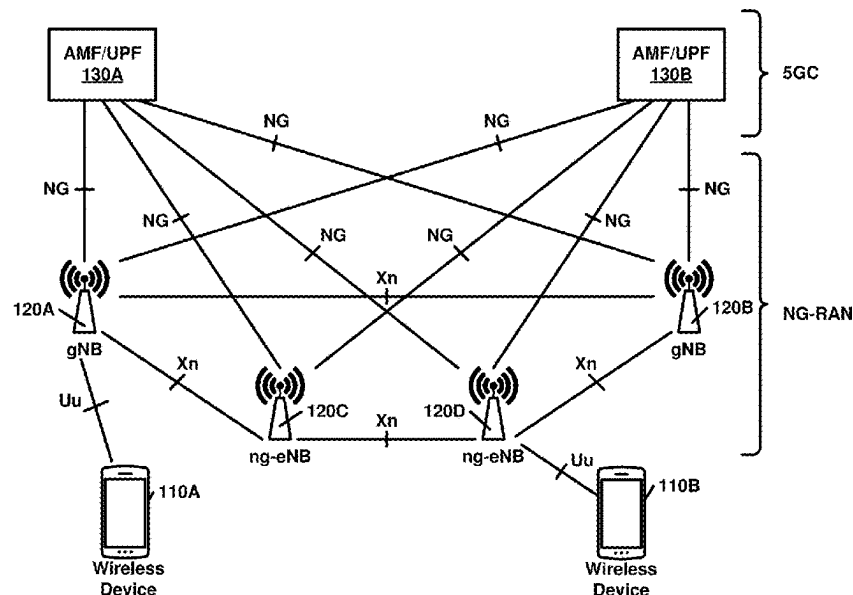
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(57) **ABSTRACT**

A base station transmits, to a wireless device, radio resource
control (RRC) configuration parameters of a cell comprising
a bandwidth part (BWP). The base station receive a radio
link failure report indicating a connection failure during a
random access procedure of the wireless device. The radio
link failure report comprises a cell identifier of the cell and
a parameter indicating a bandwidth of the BWP associated
with the connection failure.

20 Claims, 28 Drawing Sheets



Related U.S. Application Data

continuation of application No. 18/047,018, filed on Oct. 17, 2022, now Pat. No. 11,729,850, which is a continuation of application No. 16/992,656, filed on Aug. 13, 2020, now Pat. No. 11,516,869, which is a continuation of application No. 16/267,322, filed on Feb. 4, 2019, now Pat. No. 10,750,562.

- (60) Provisional application No. 62/625,637, filed on Feb. 2, 2018.

- (51) **Int. Cl.**
H04W 72/0453 (2023.01)
H04W 72/23 (2023.01)
H04W 76/11 (2018.01)
H04W 76/19 (2018.01)
H04W 76/27 (2018.01)

- (52) **U.S. Cl.**
 CPC **H04W 72/23** (2023.01); **H04W 76/19** (2018.02); **H04W 76/27** (2018.02); **H04L 5/001** (2013.01); **H04W 76/11** (2018.02)

- (58) **Field of Classification Search**
 CPC H04W 76/11; H04L 5/0055; H04L 5/001; H04L 5/0094
 See application file for complete search history.

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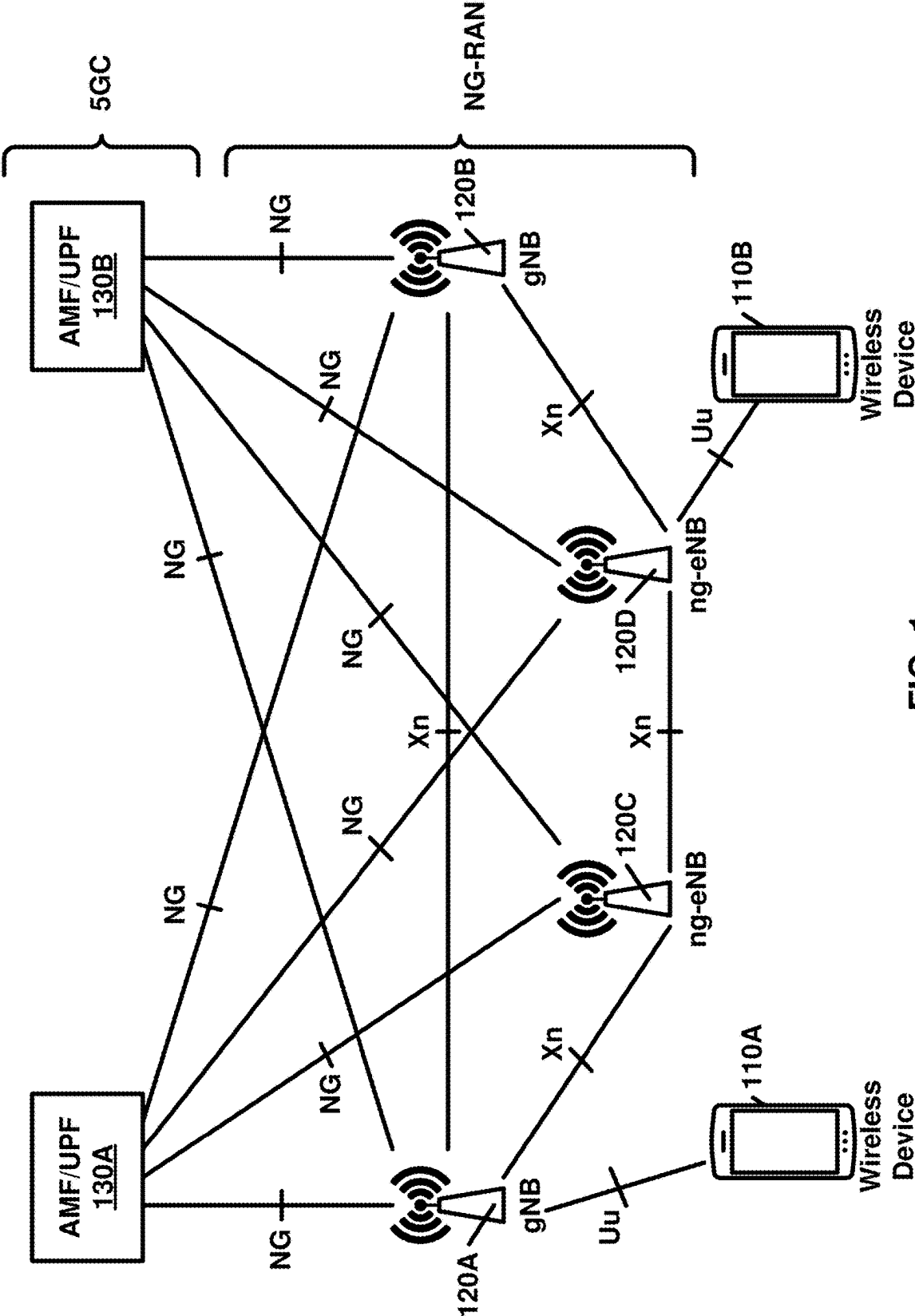


FIG. 1

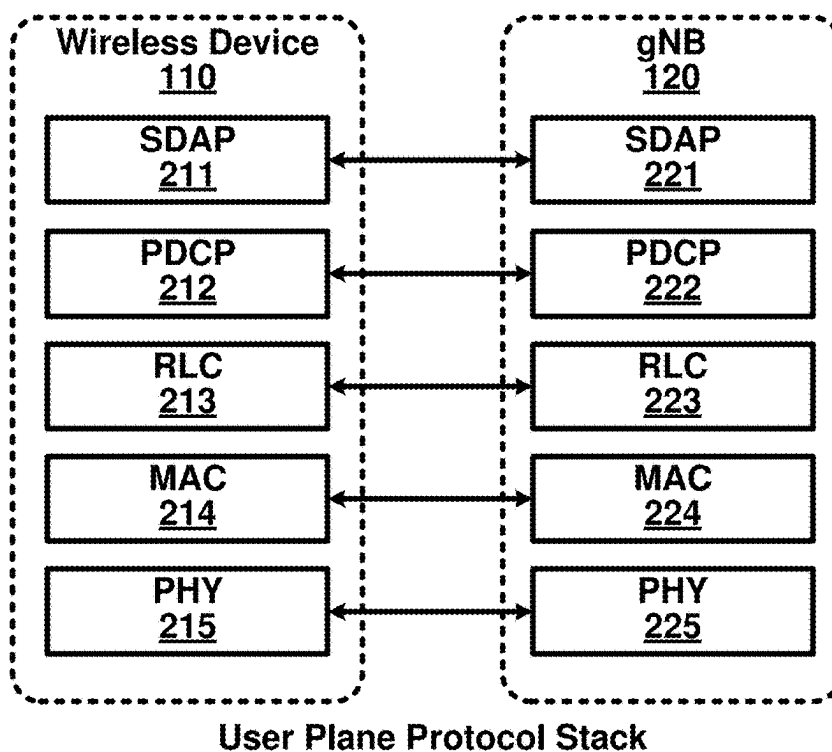


FIG. 2A

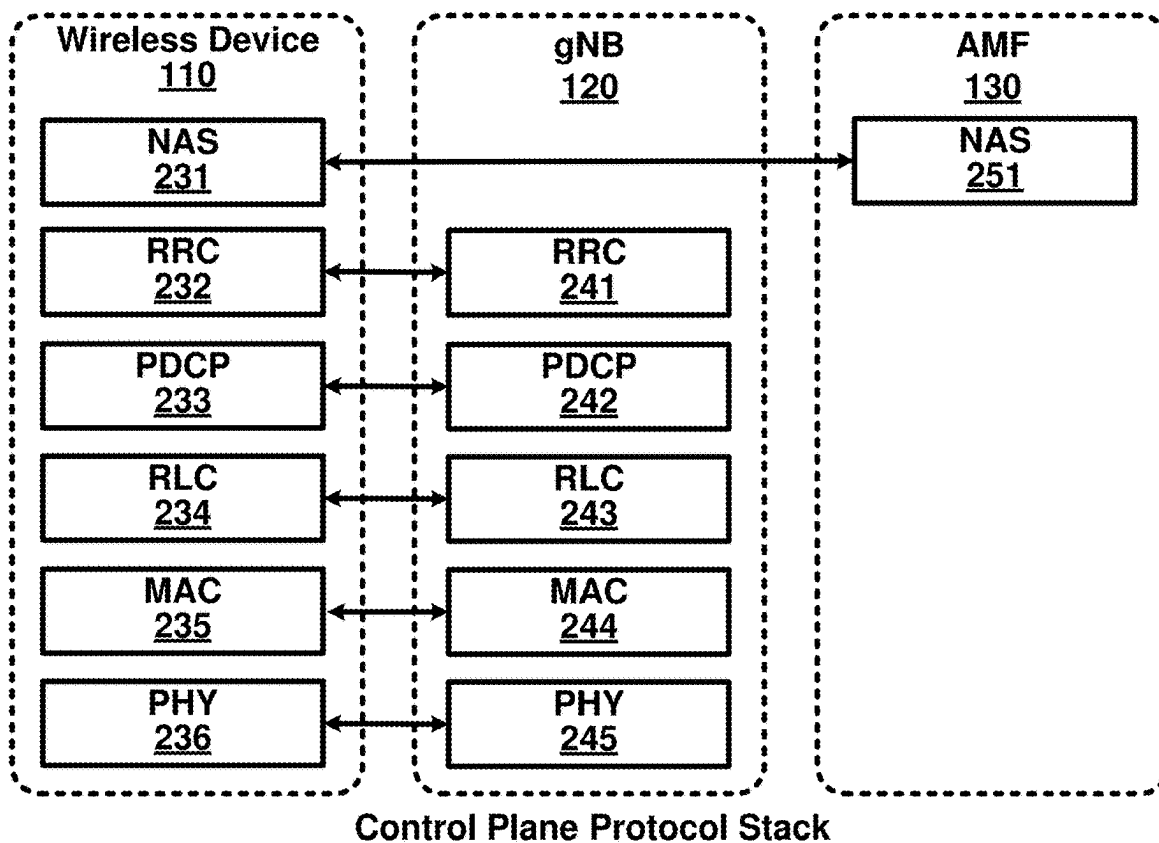


FIG. 2B

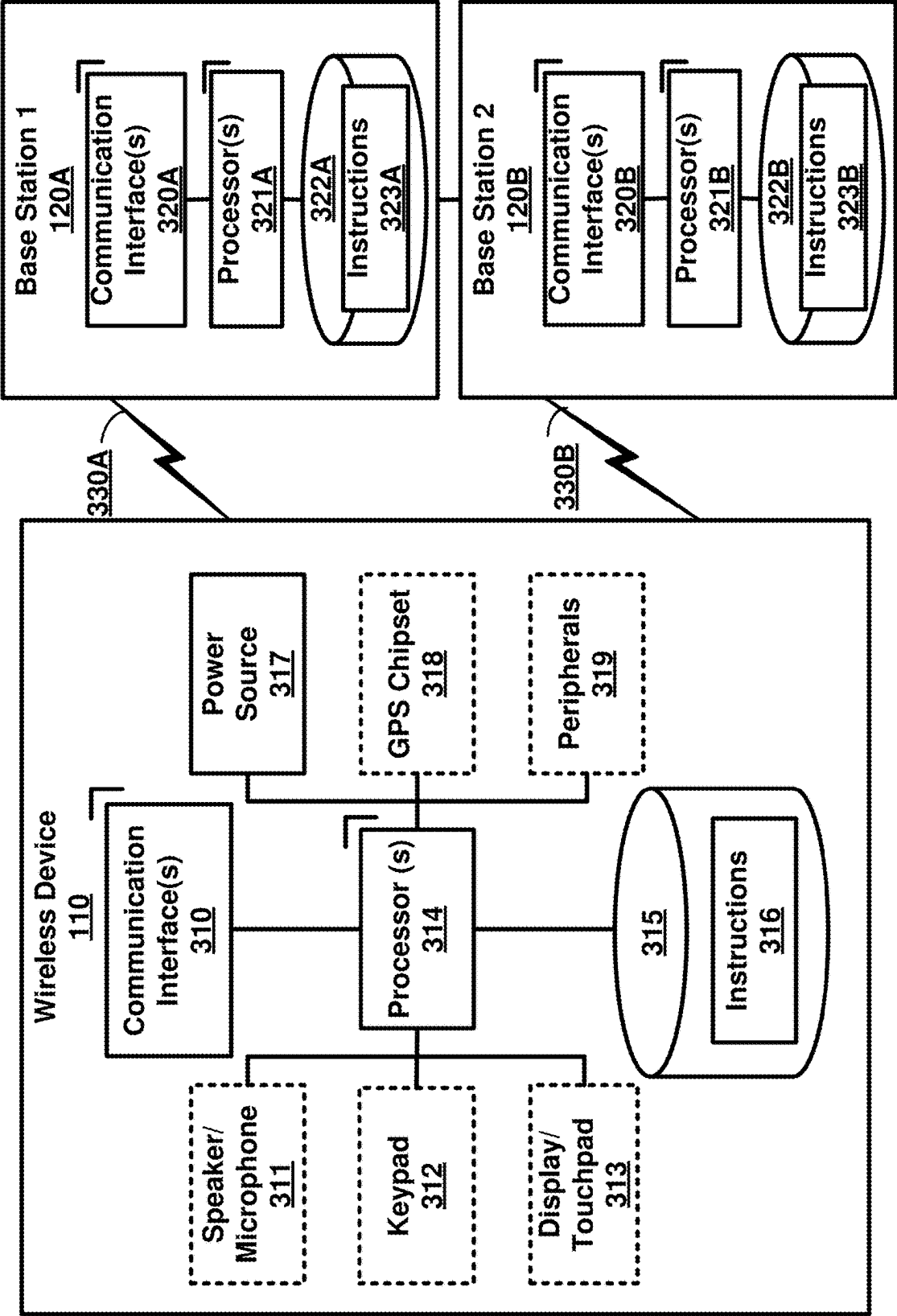


FIG. 3

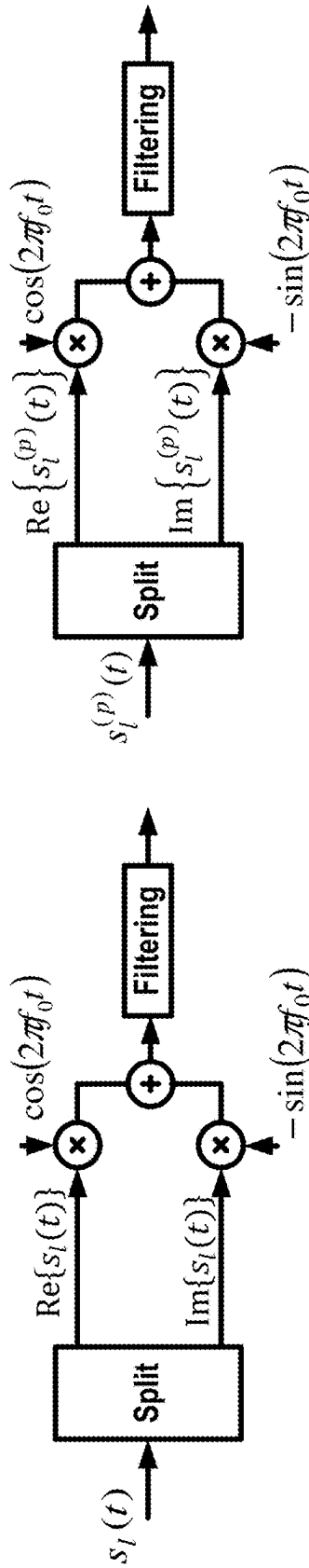
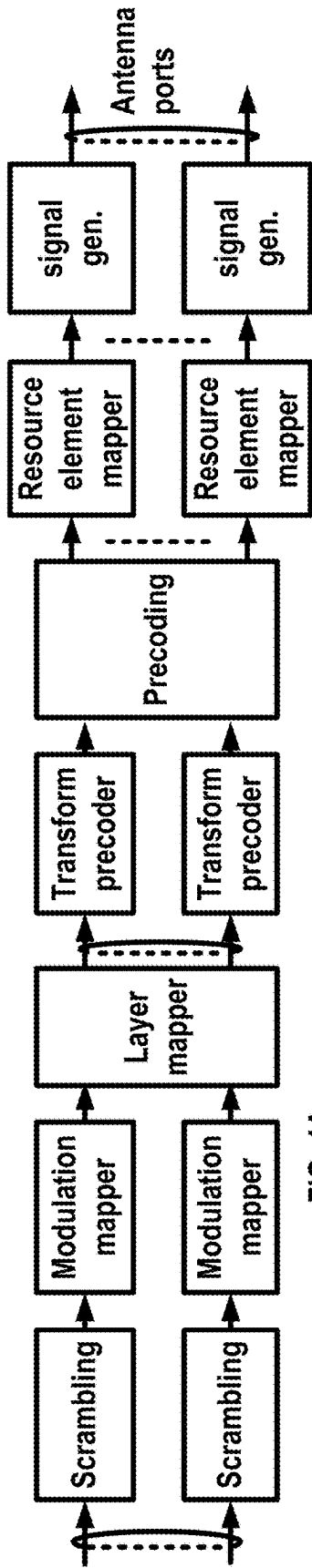
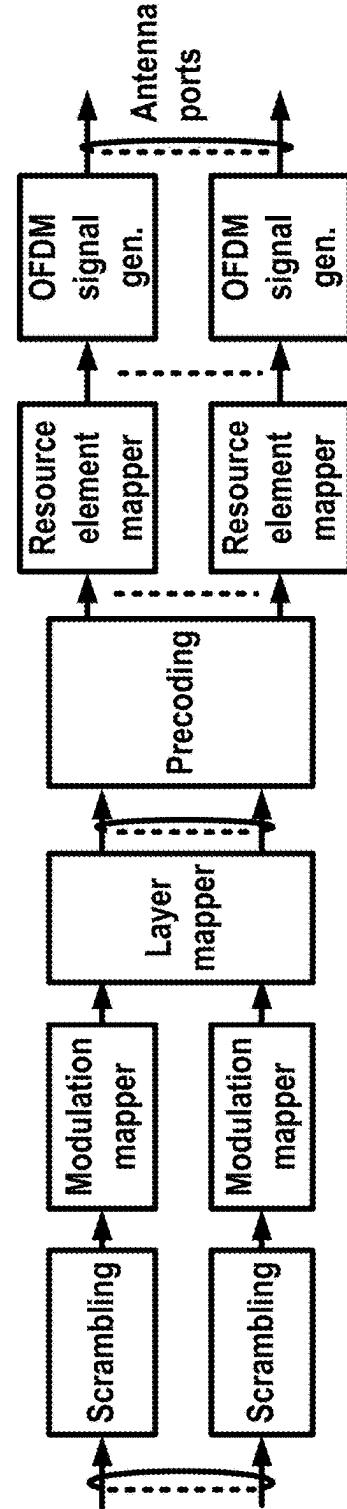


FIG. 4D



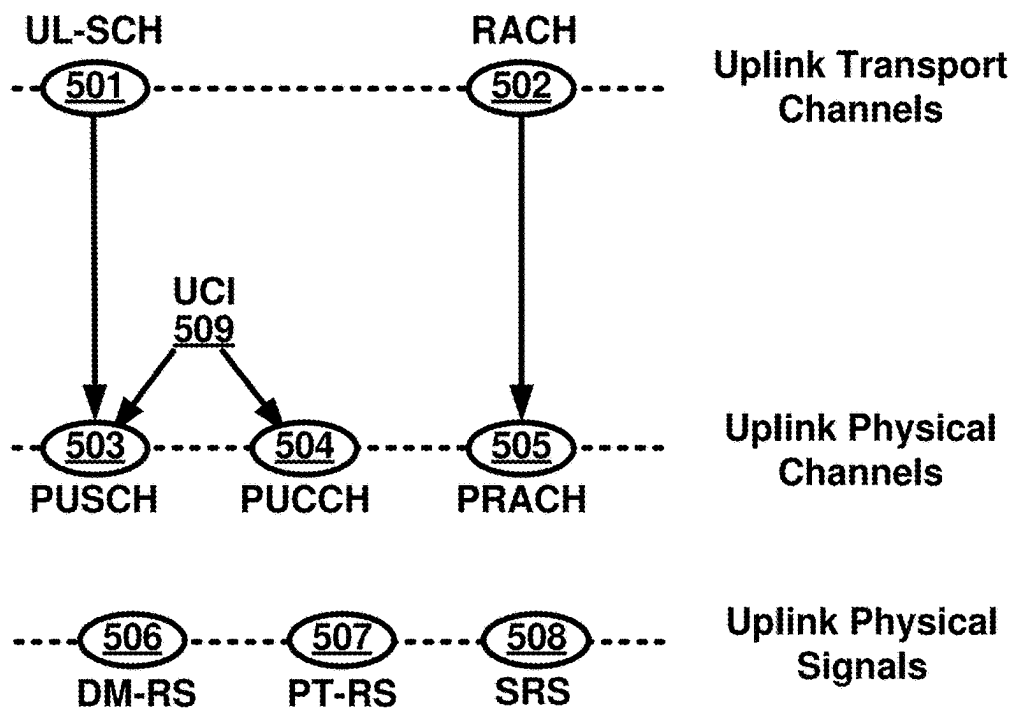


FIG. 5A

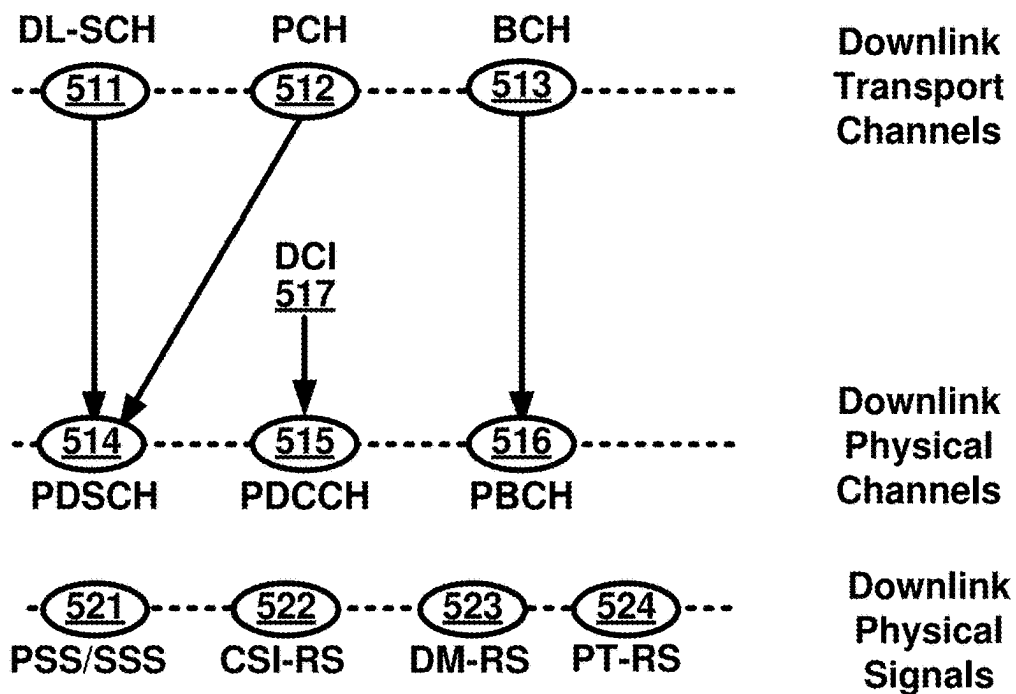


FIG. 5B

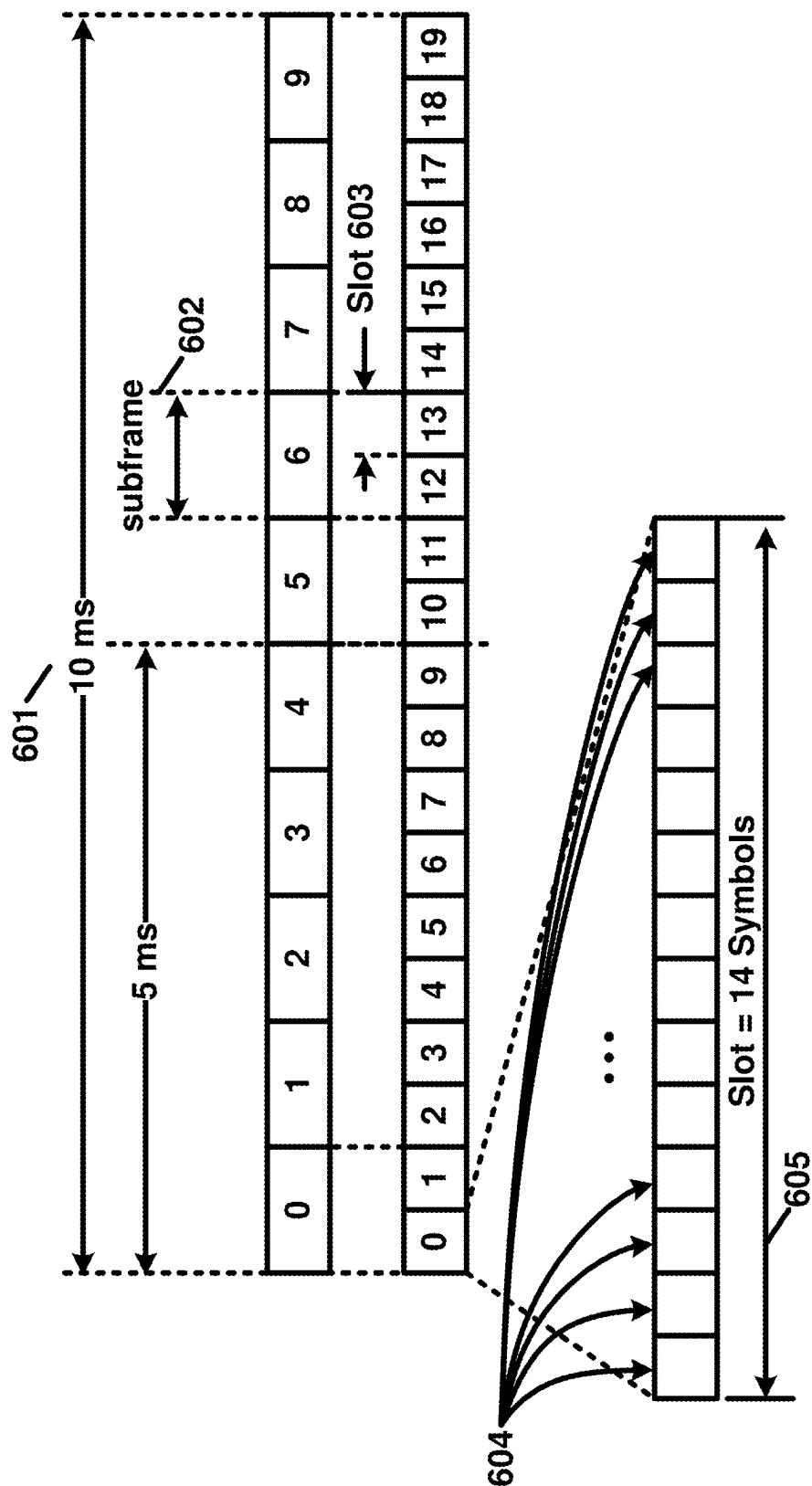


FIG. 6

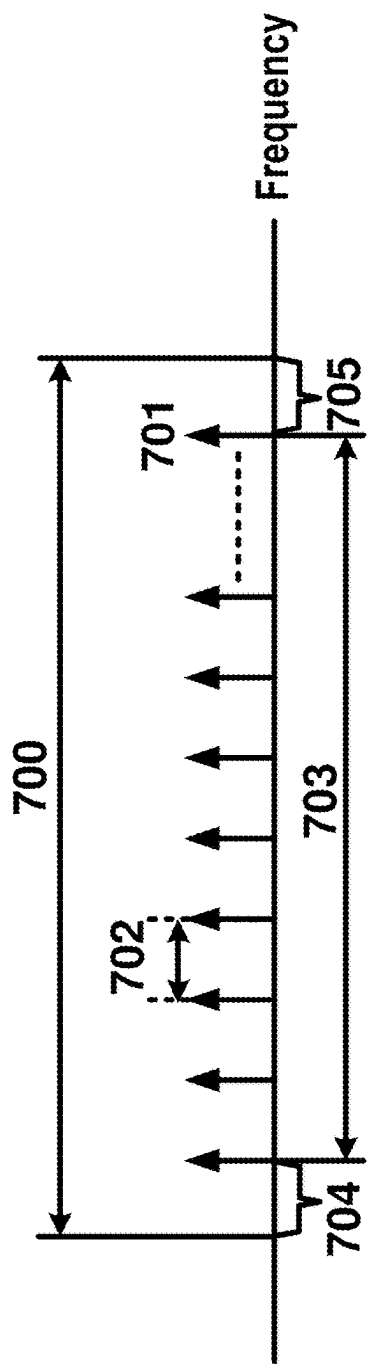


FIG. 7A

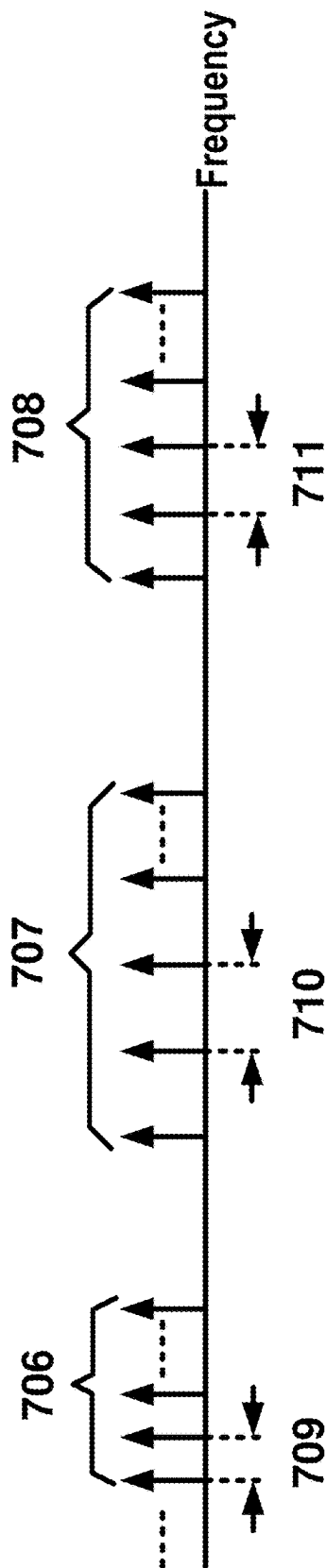


FIG. 7B

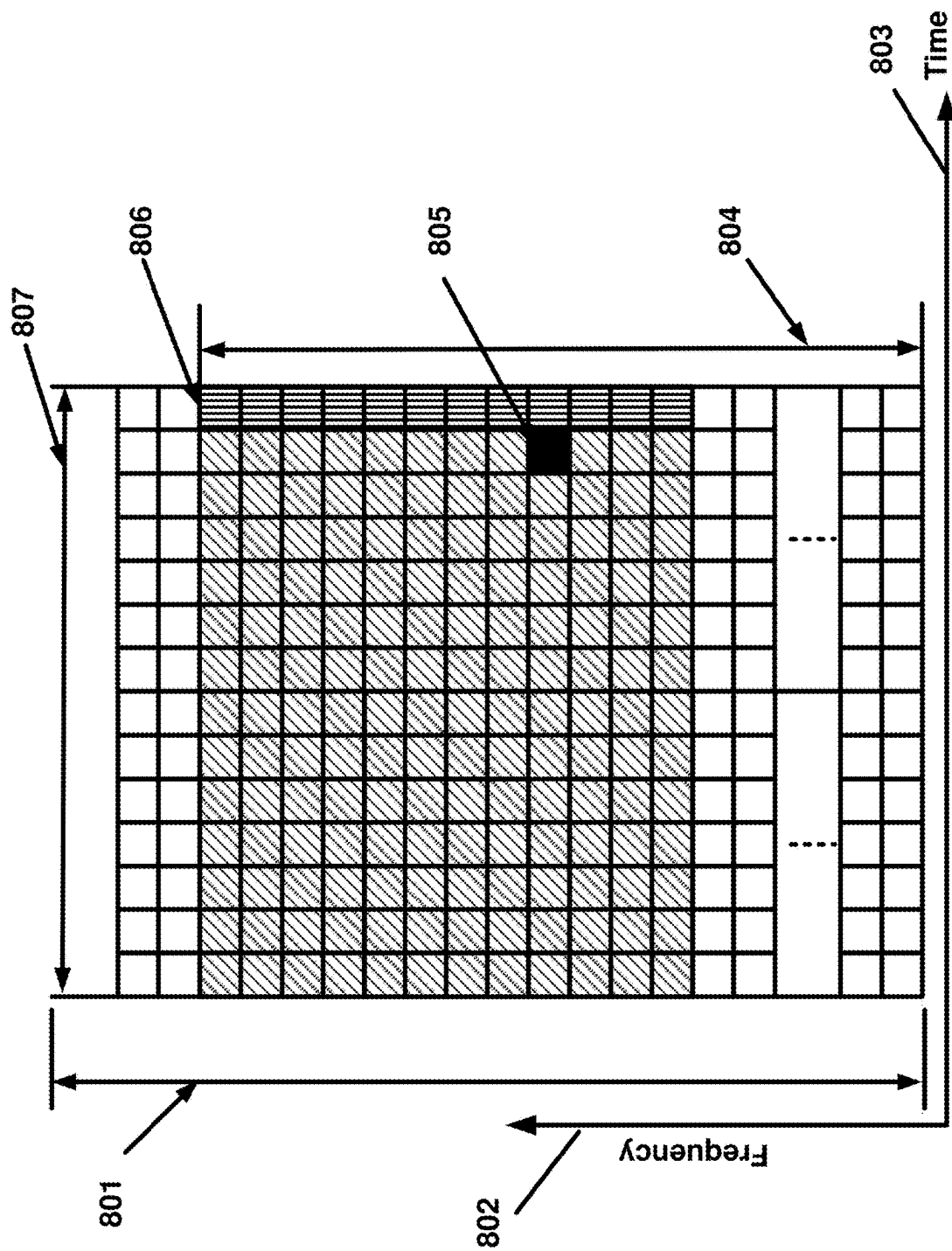
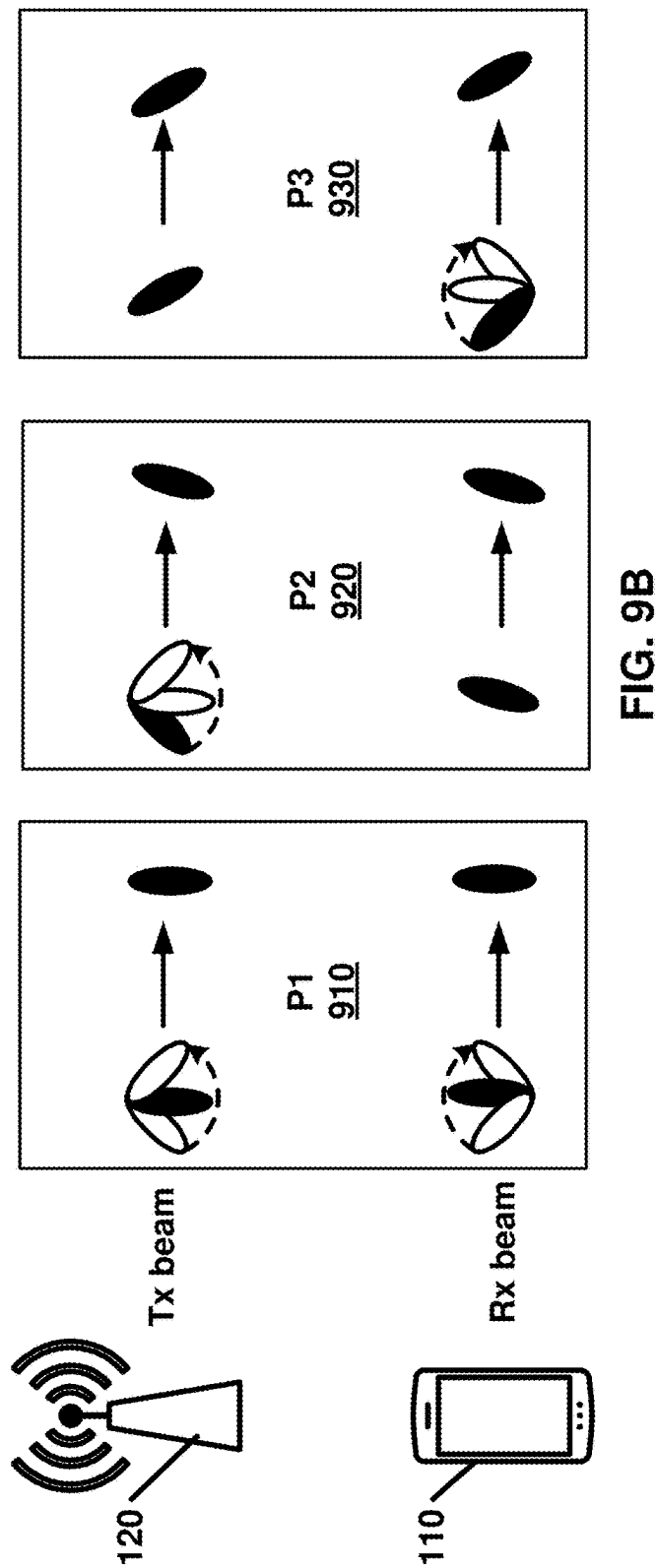
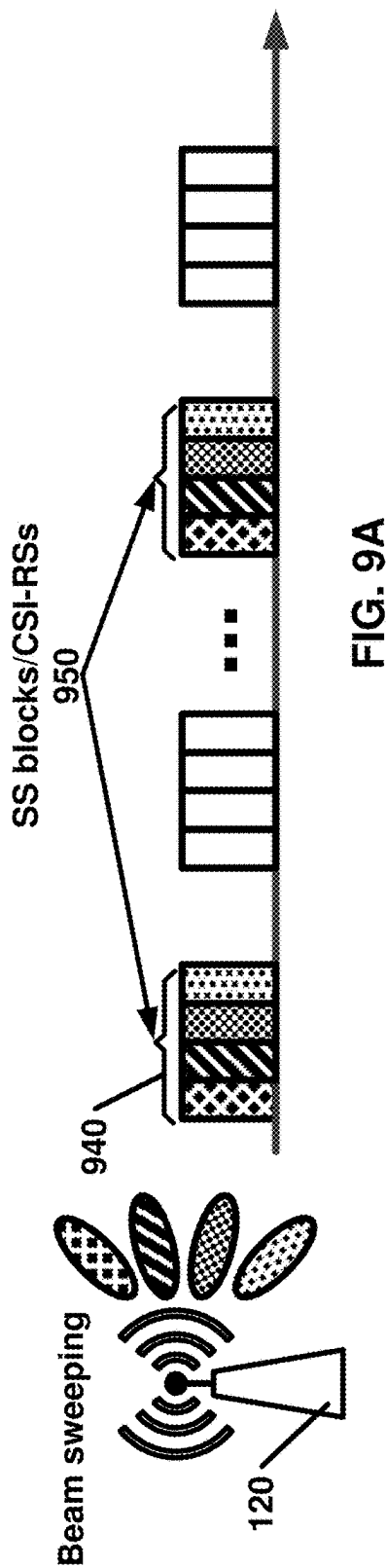


FIG. 8



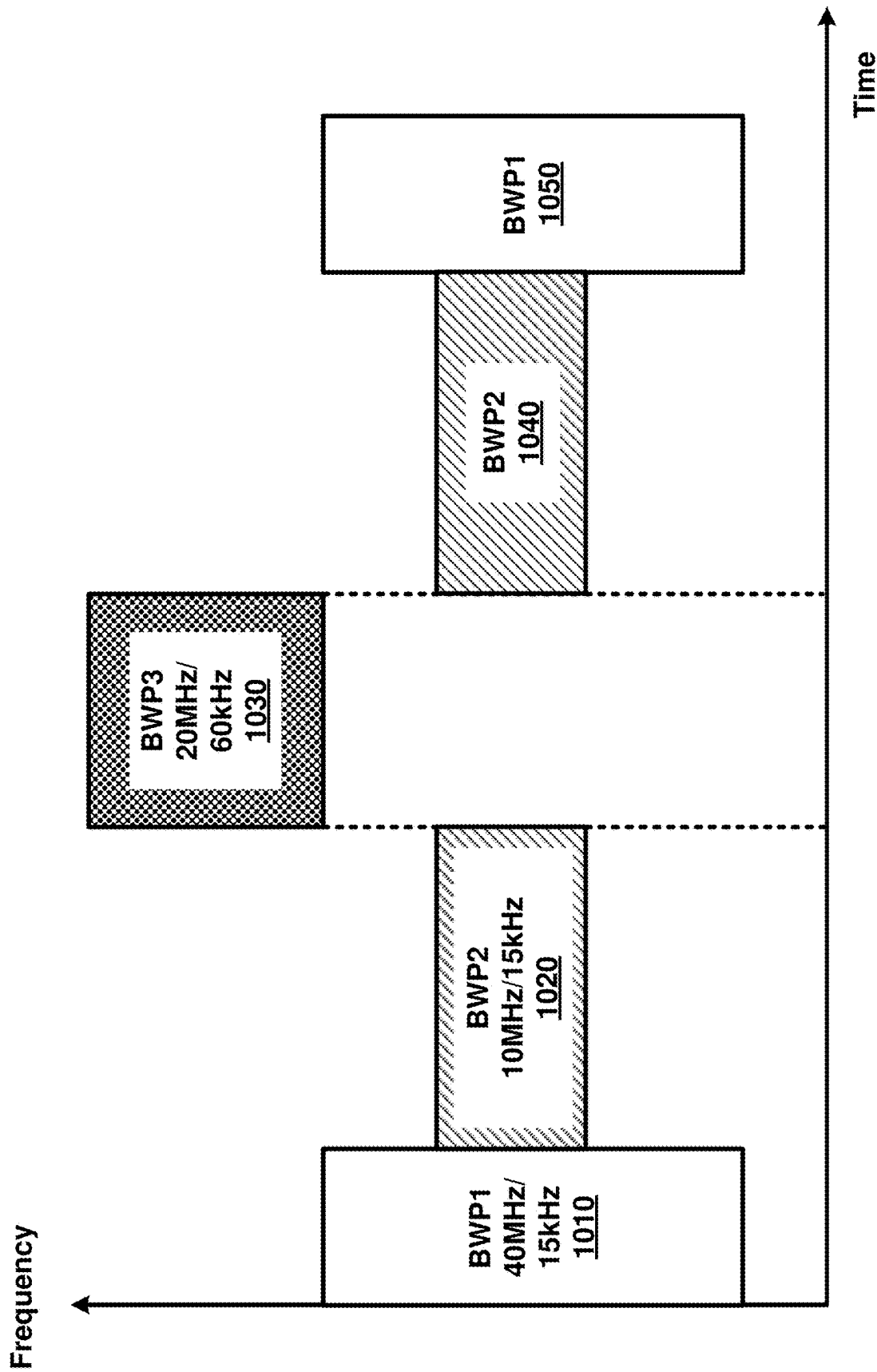


FIG. 10

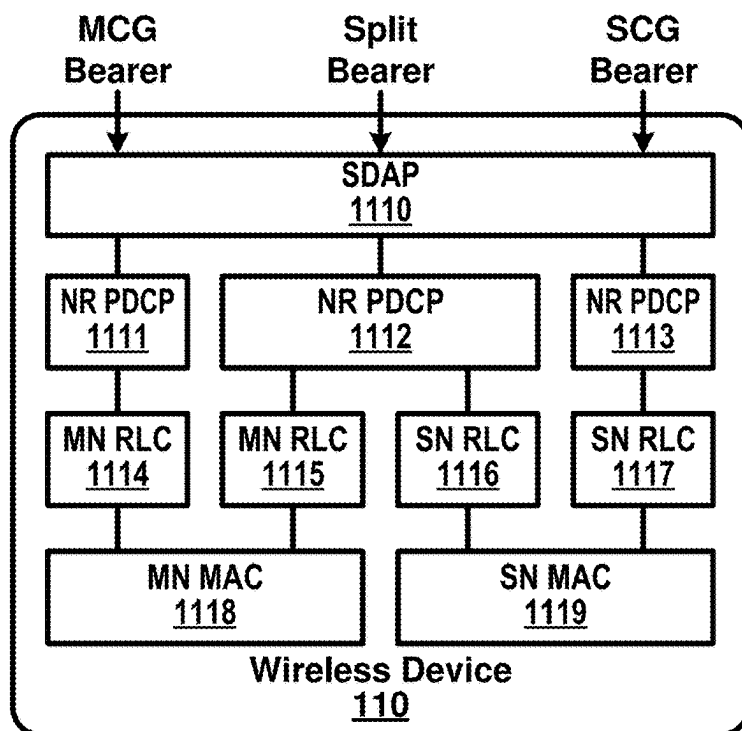


FIG. 11A

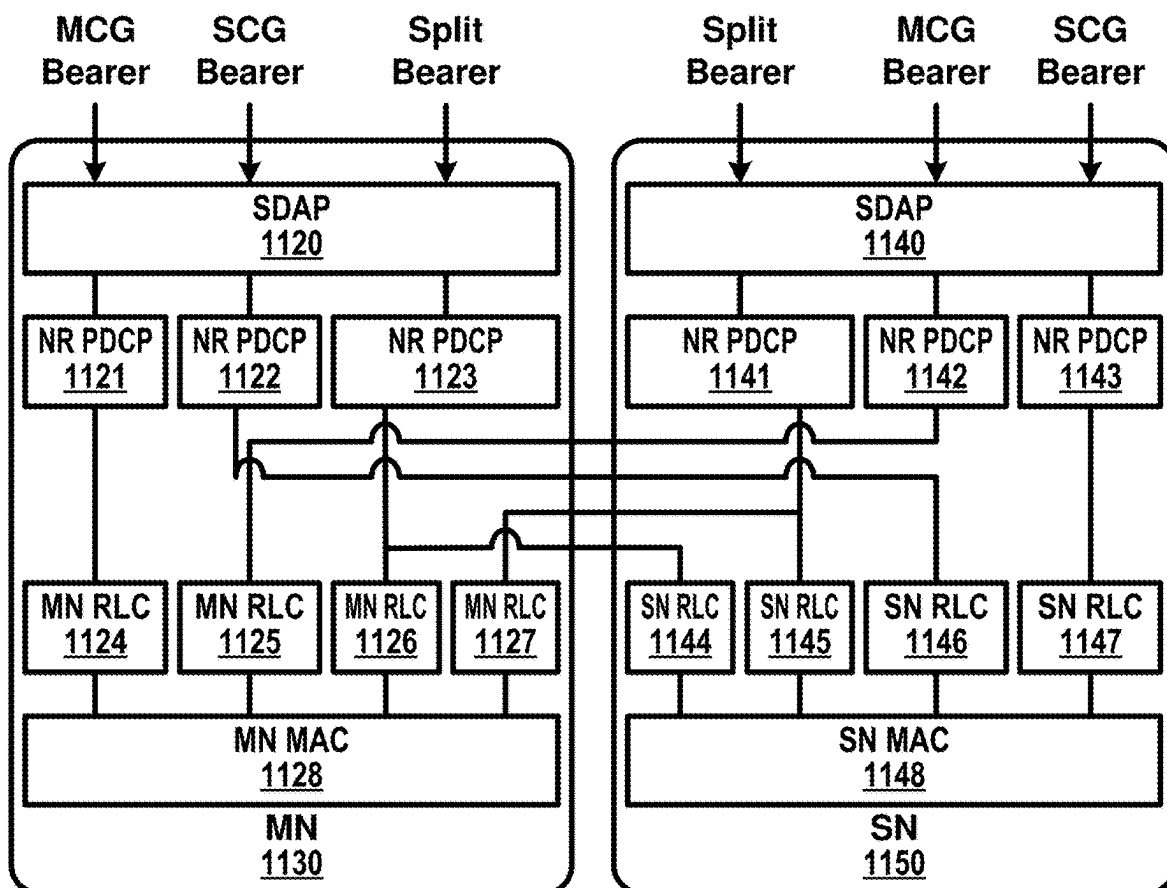


FIG. 11B

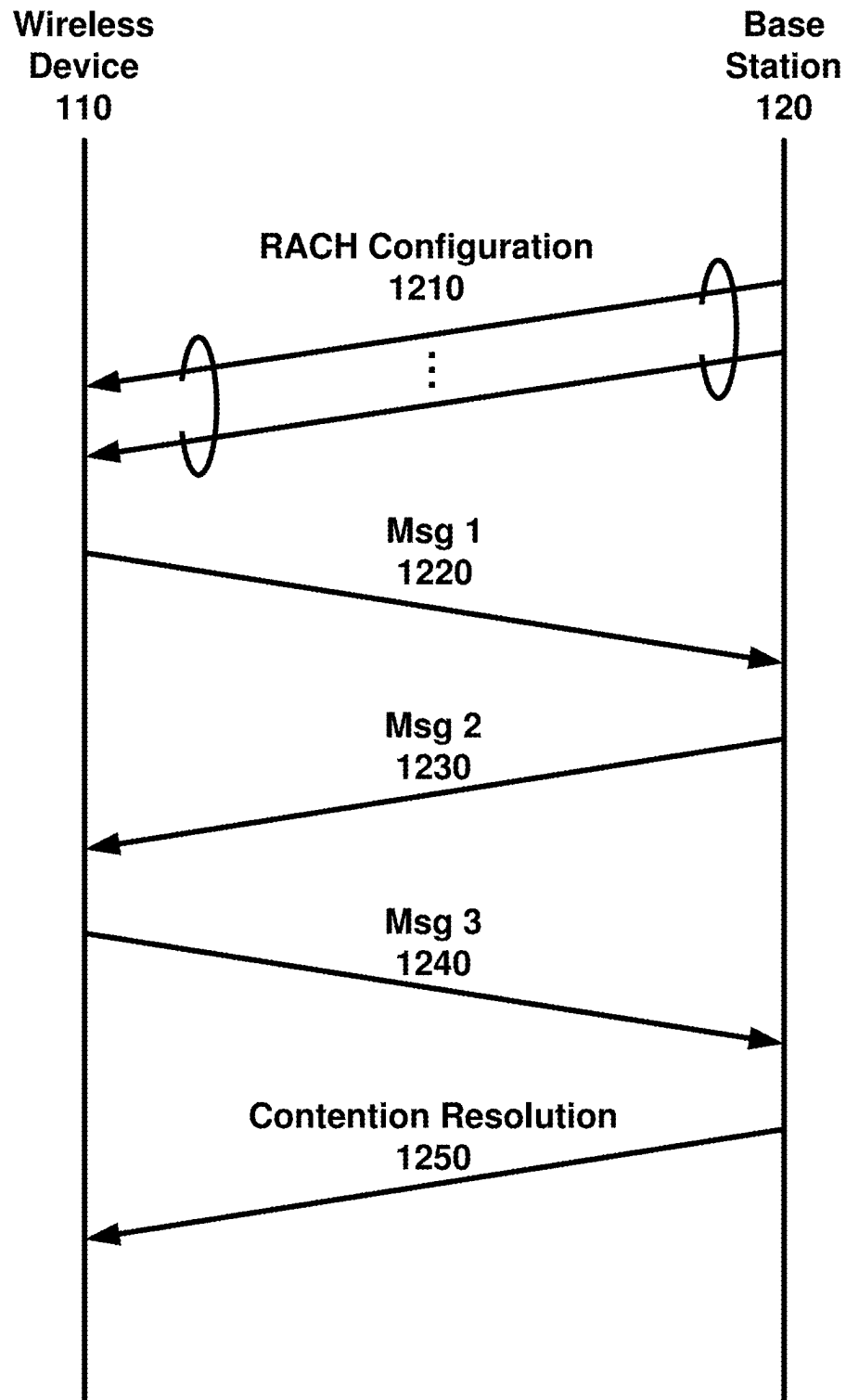


FIG. 12

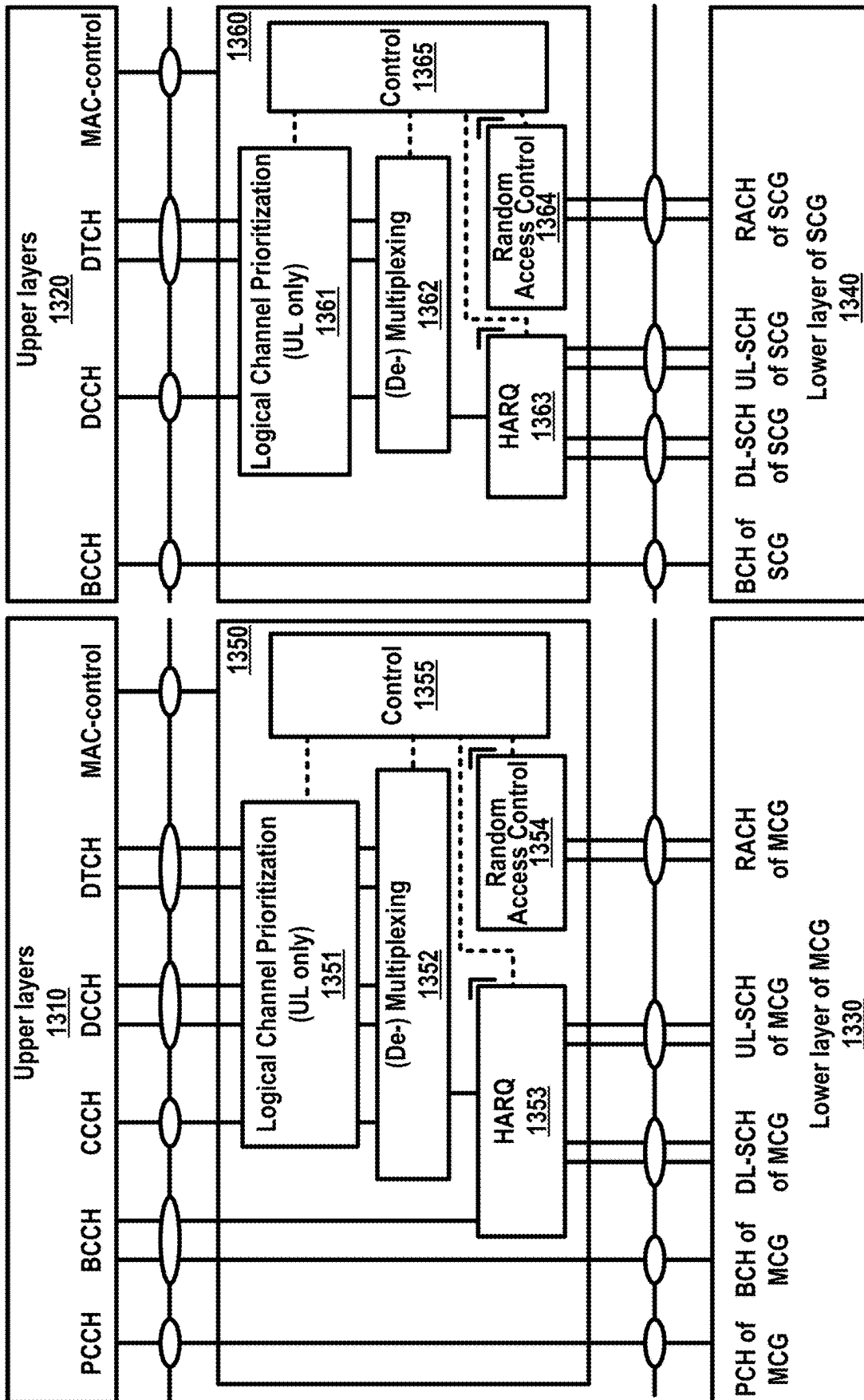


FIG. 13

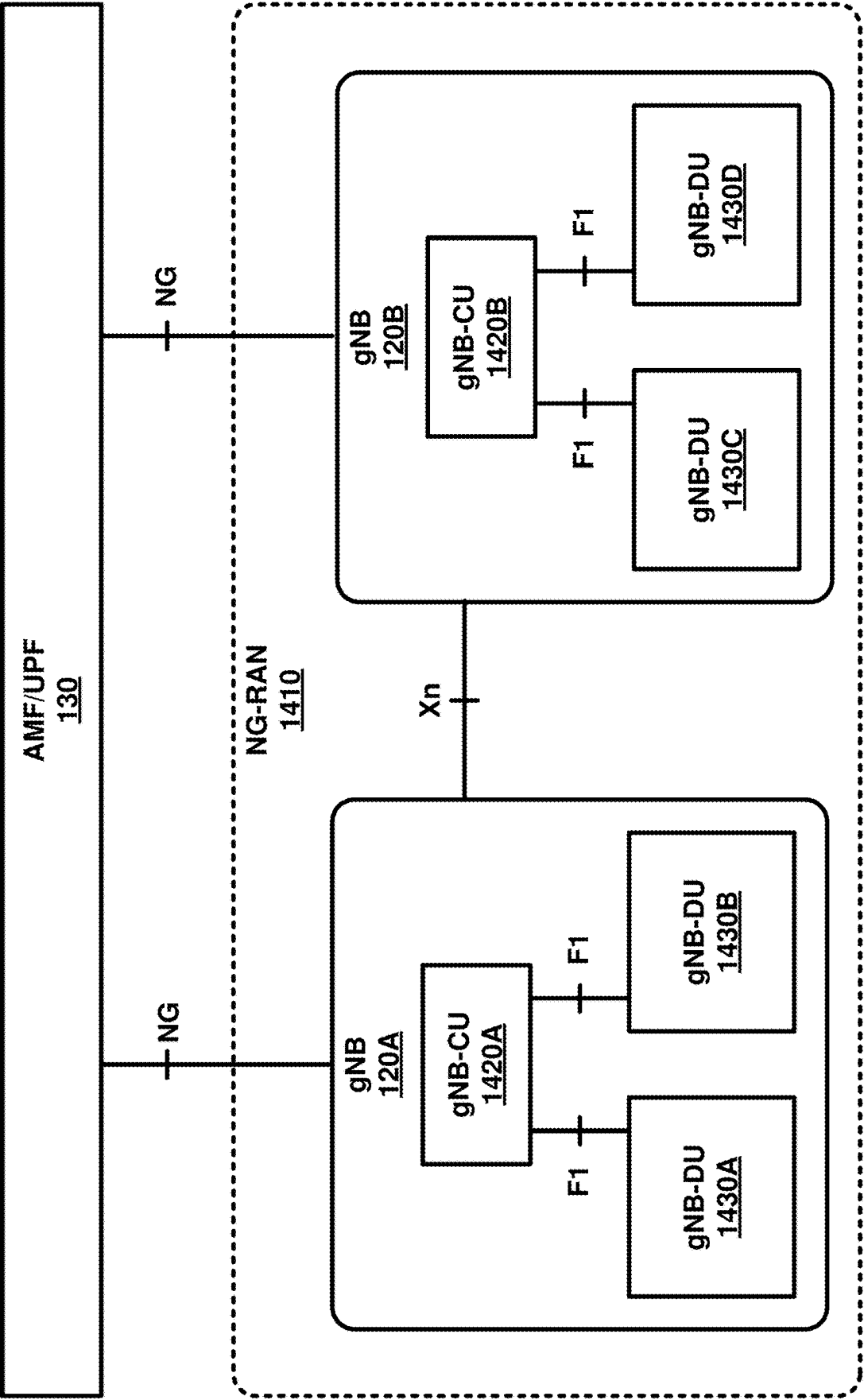


FIG. 14

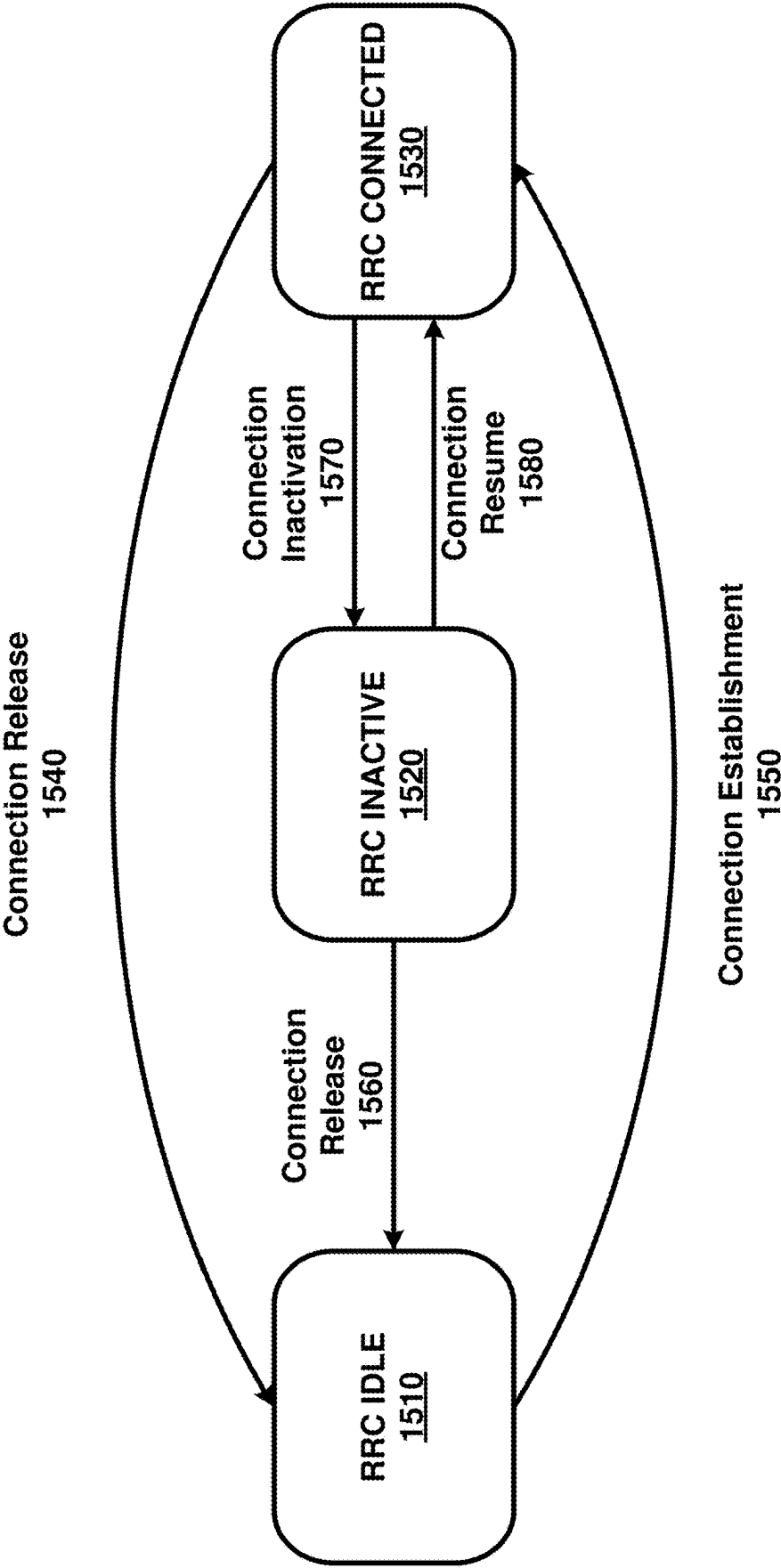


FIG. 15

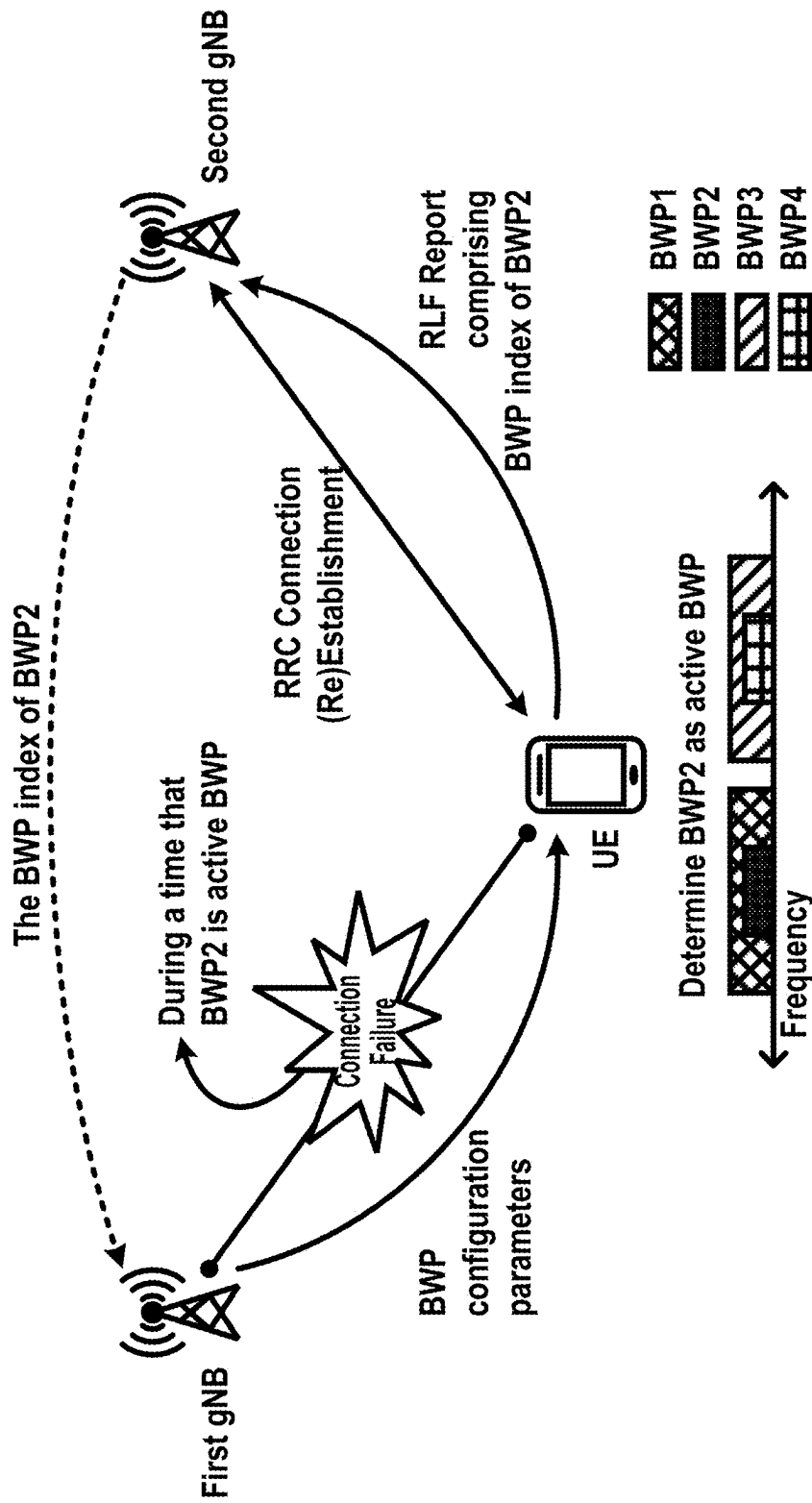


FIG. 16

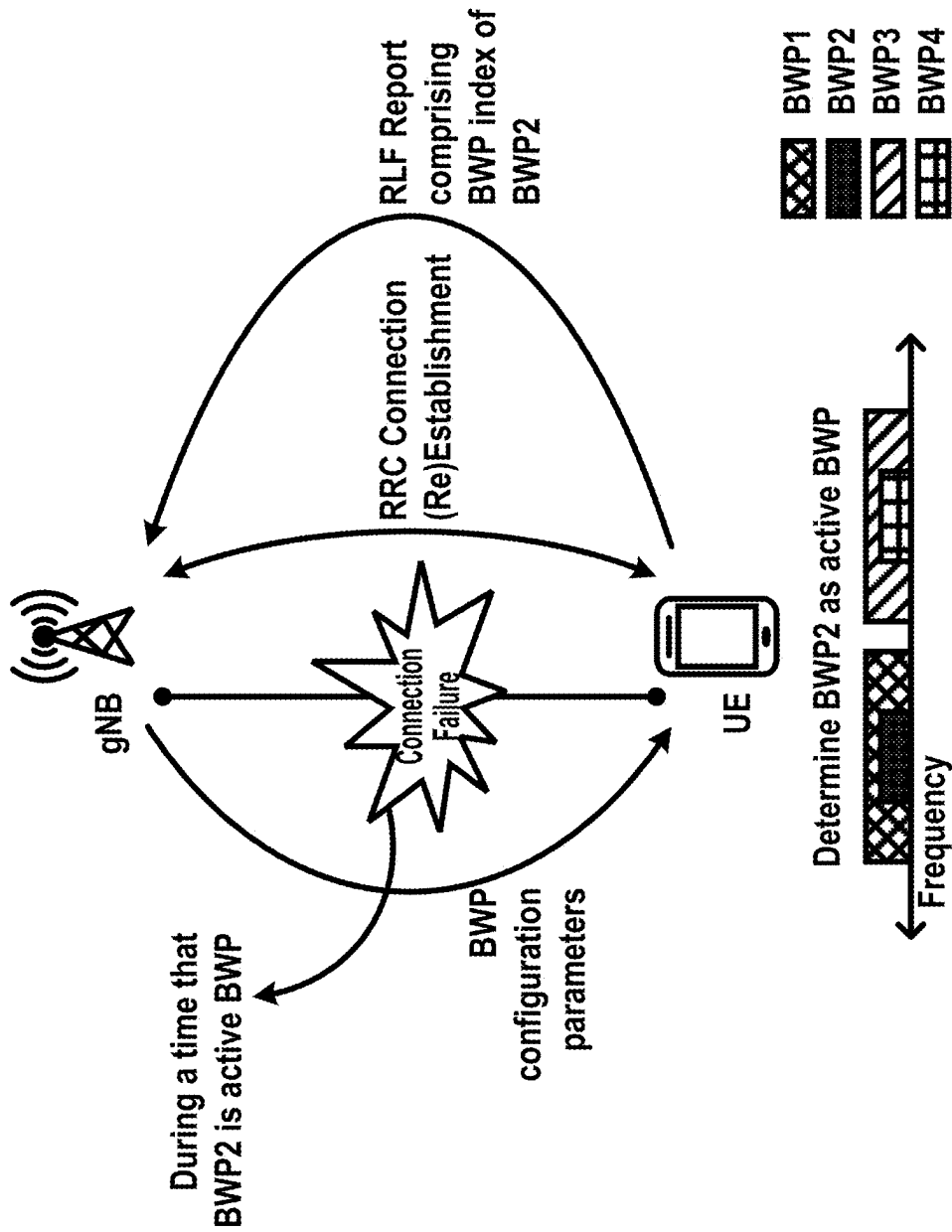


FIG. 17

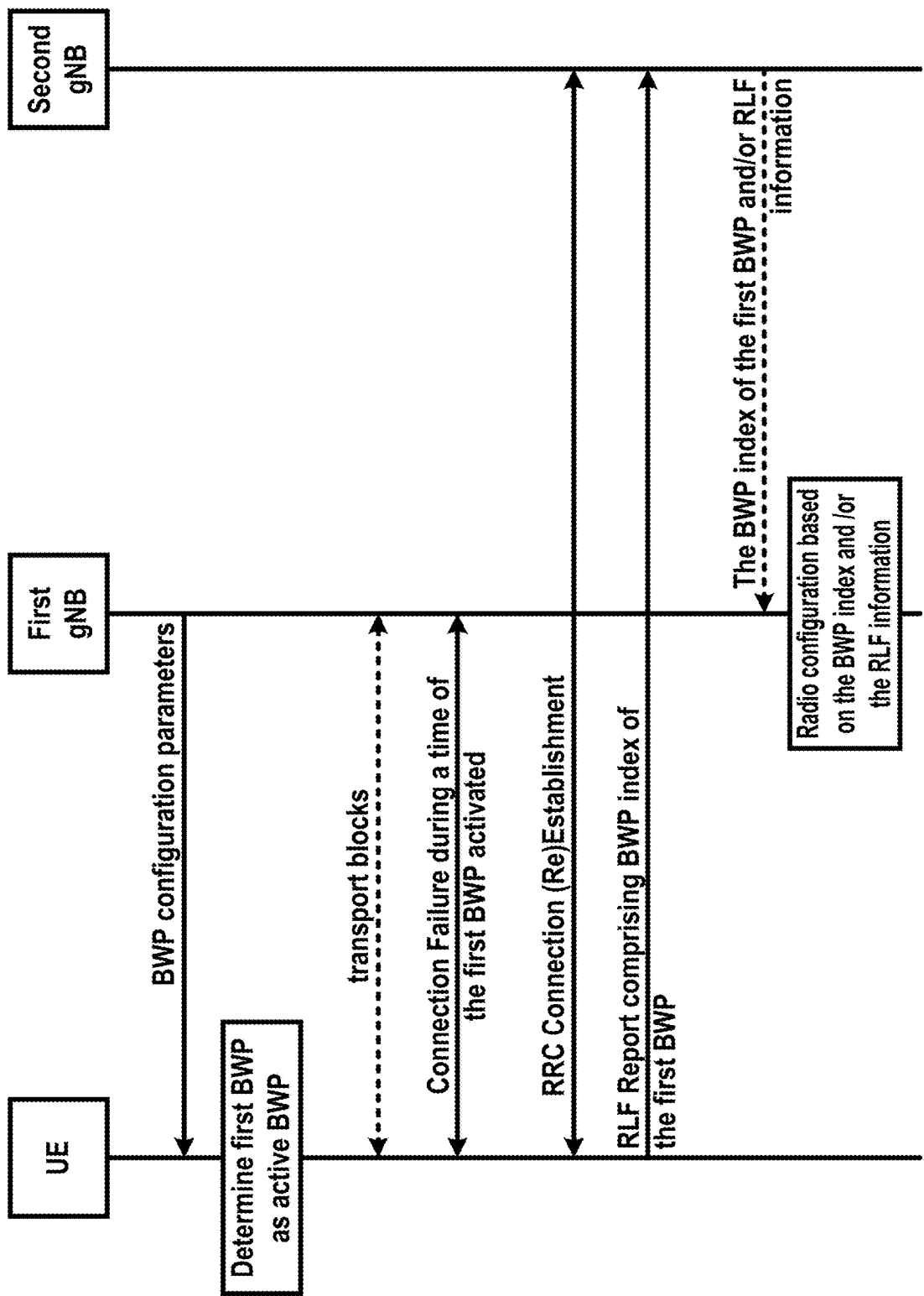


FIG. 18

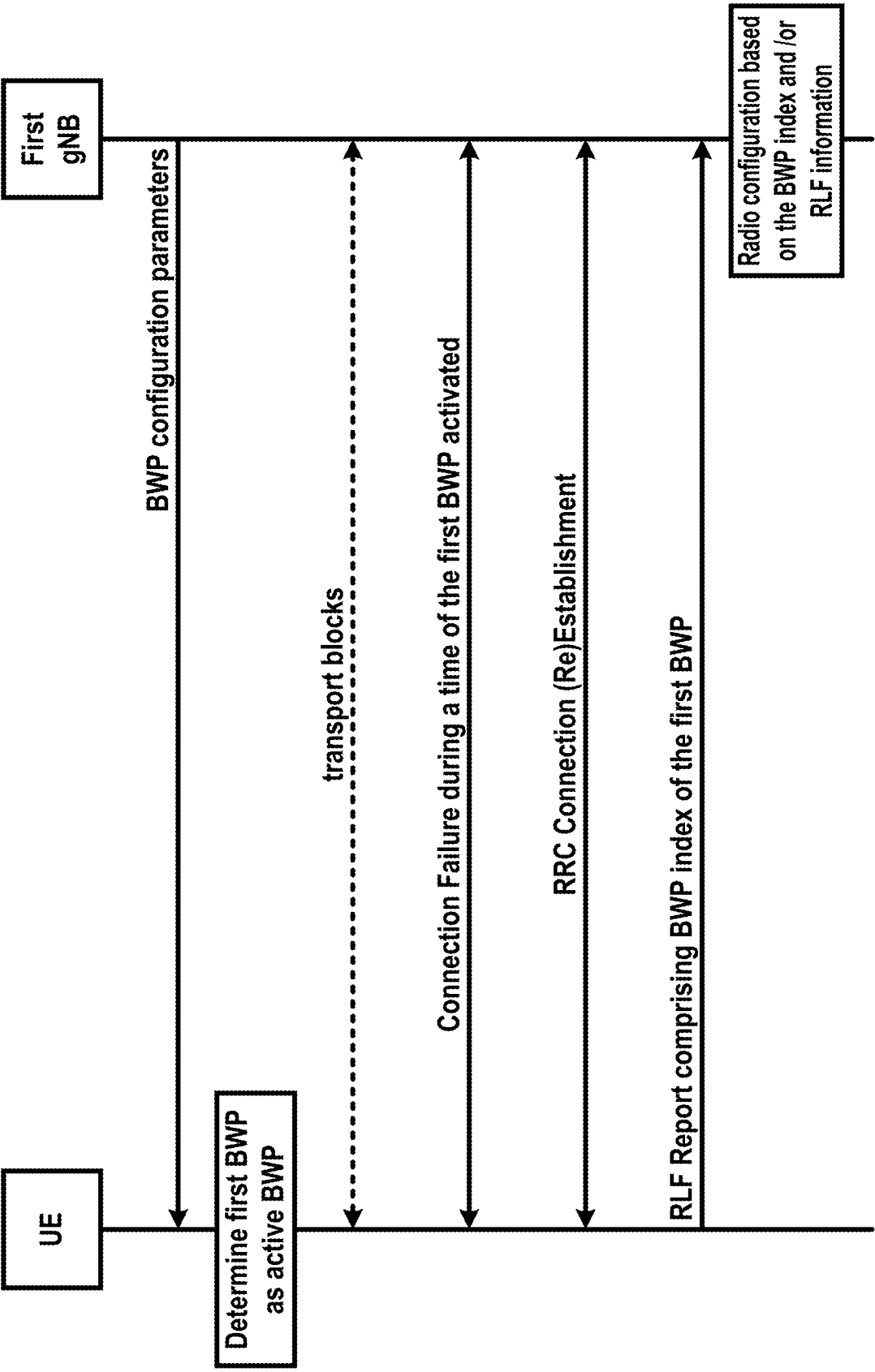


FIG. 19

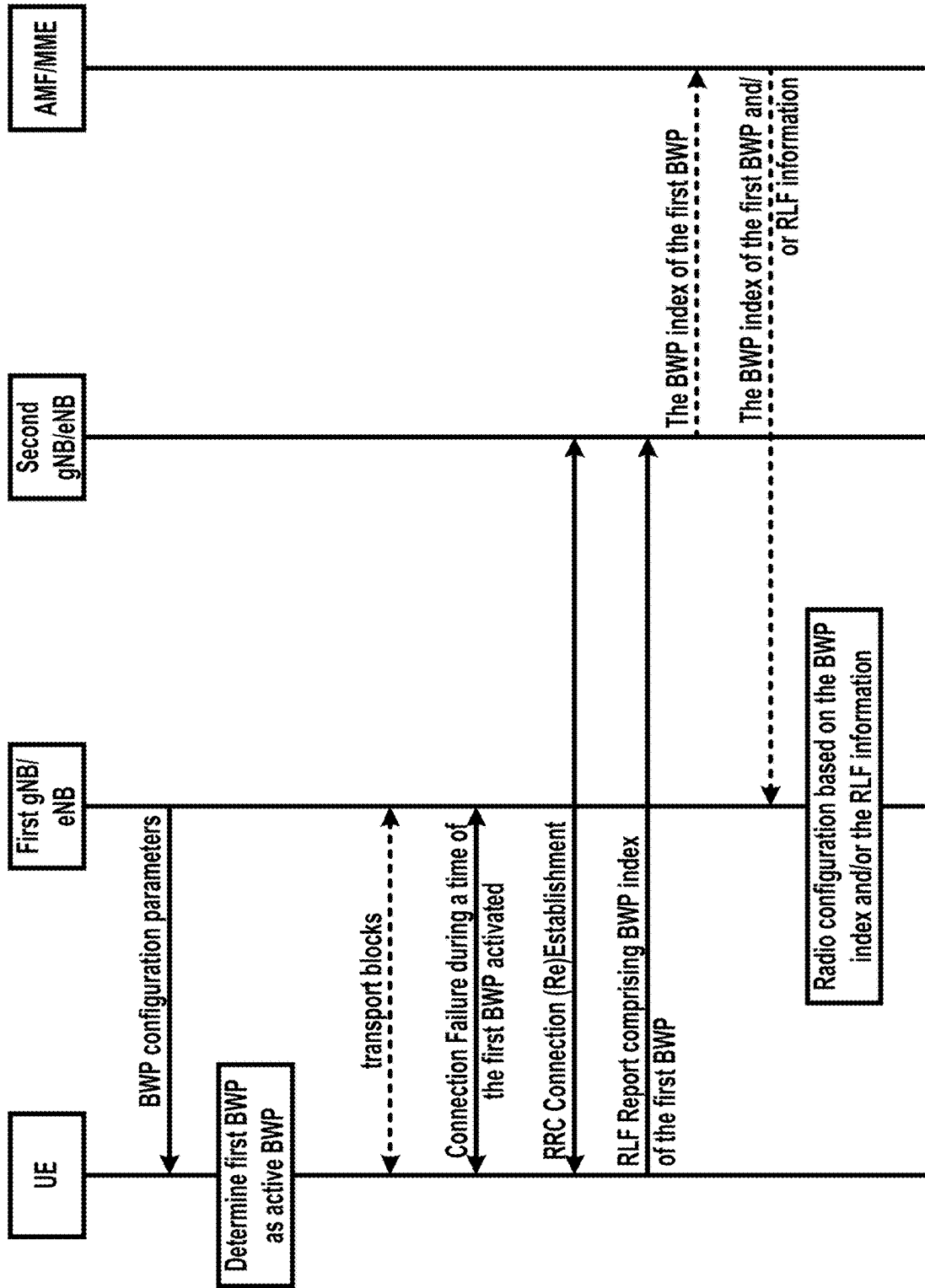


FIG. 20

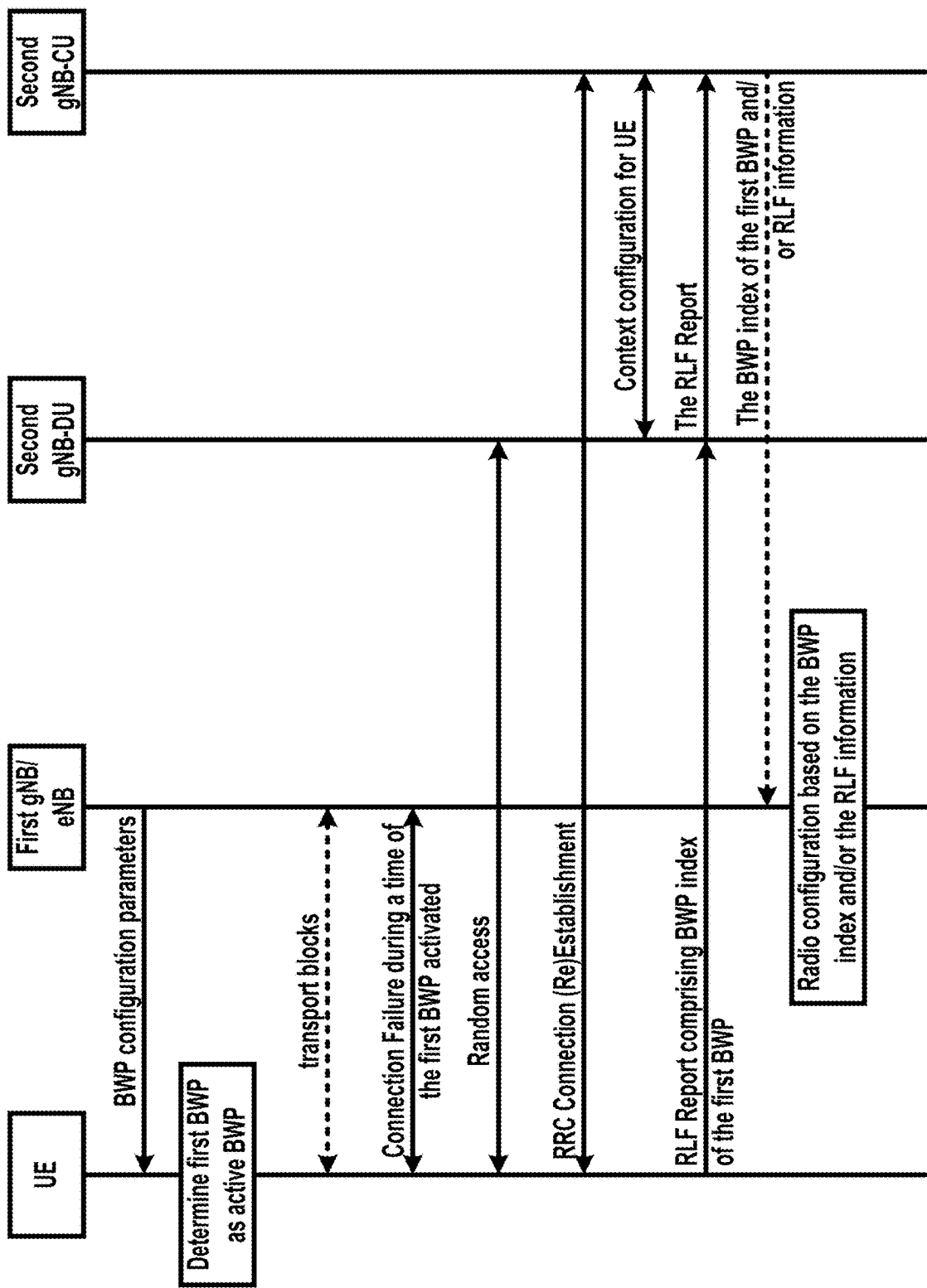


FIG. 21

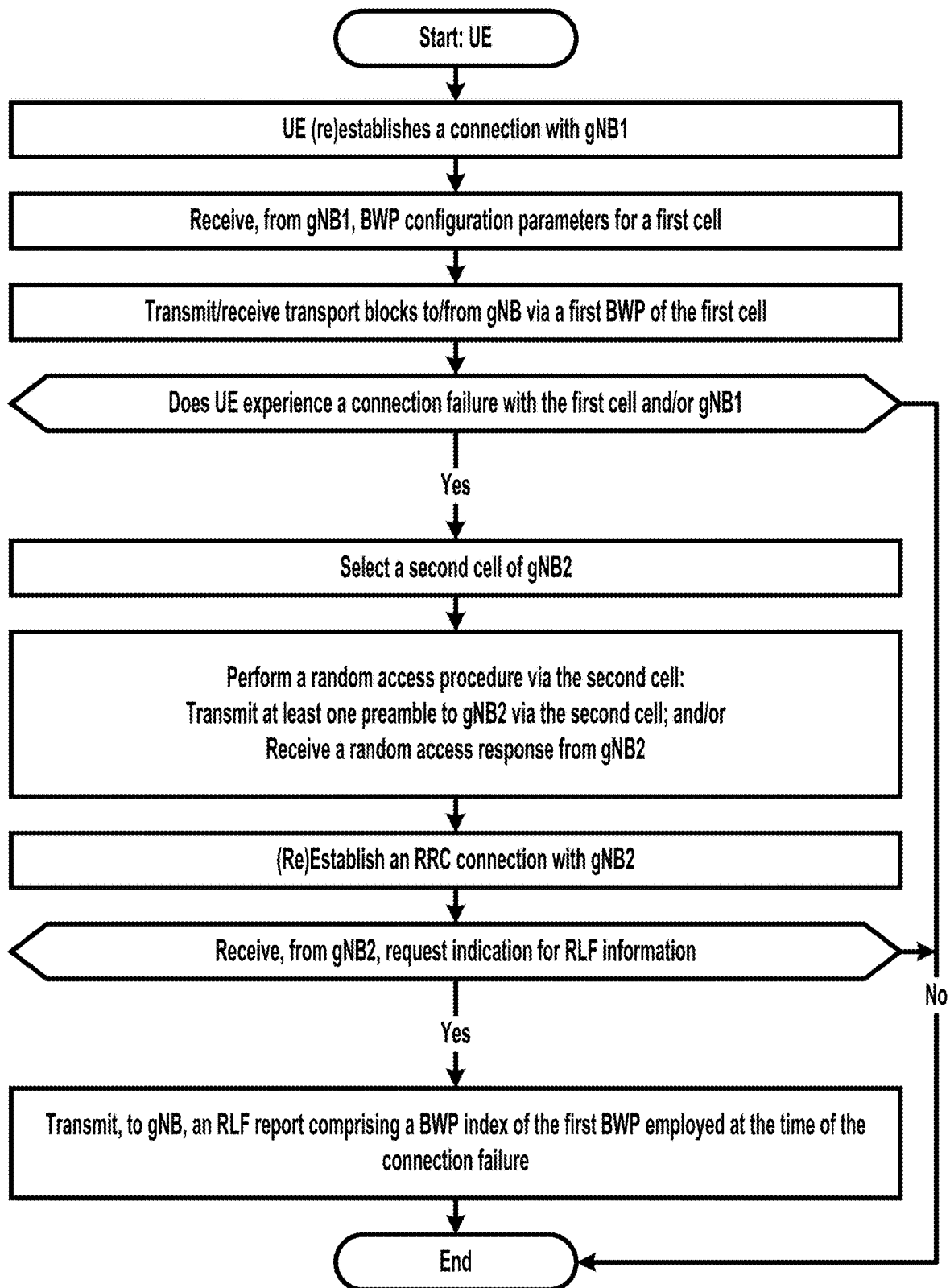


FIG. 22

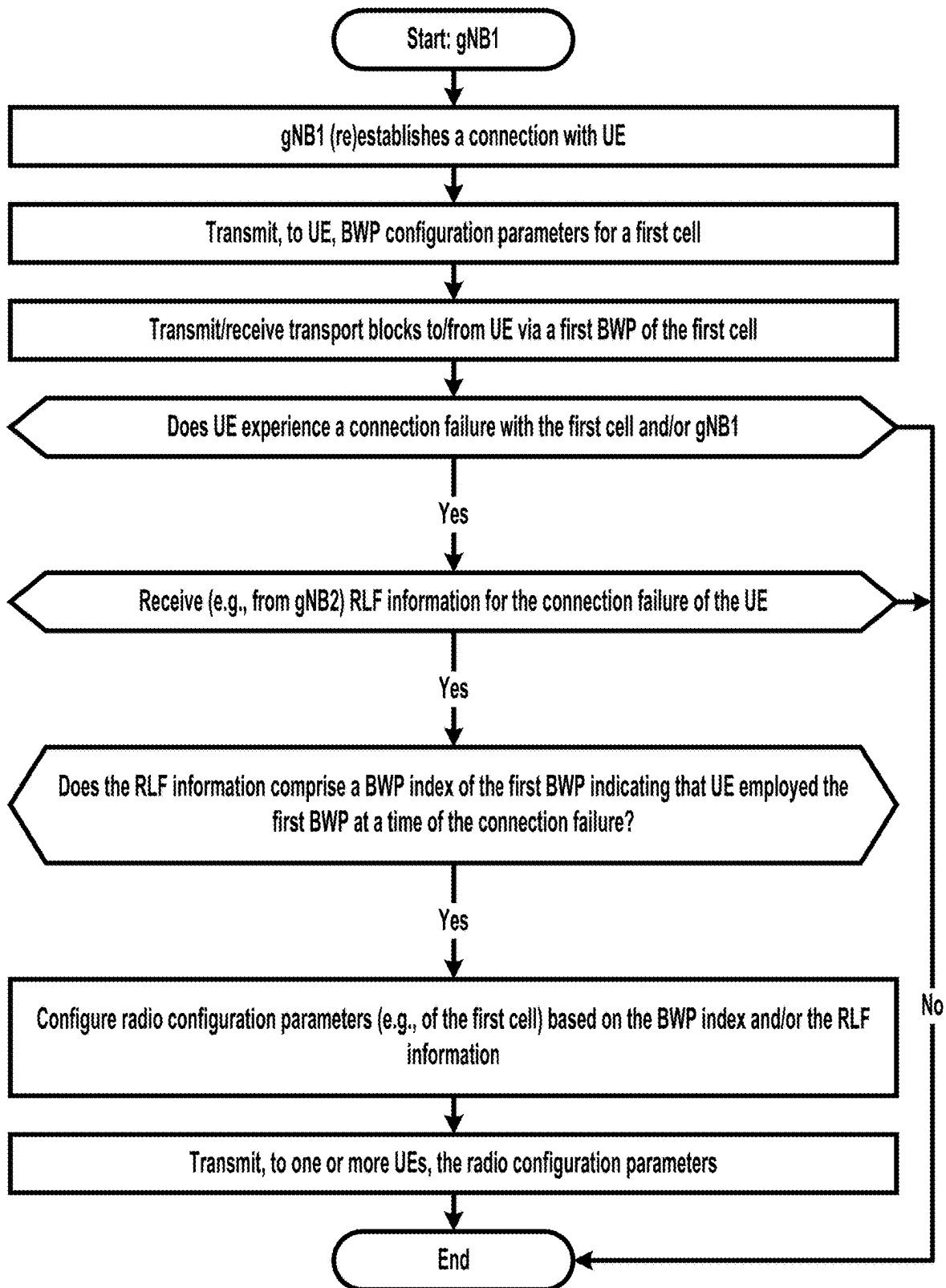


FIG. 23

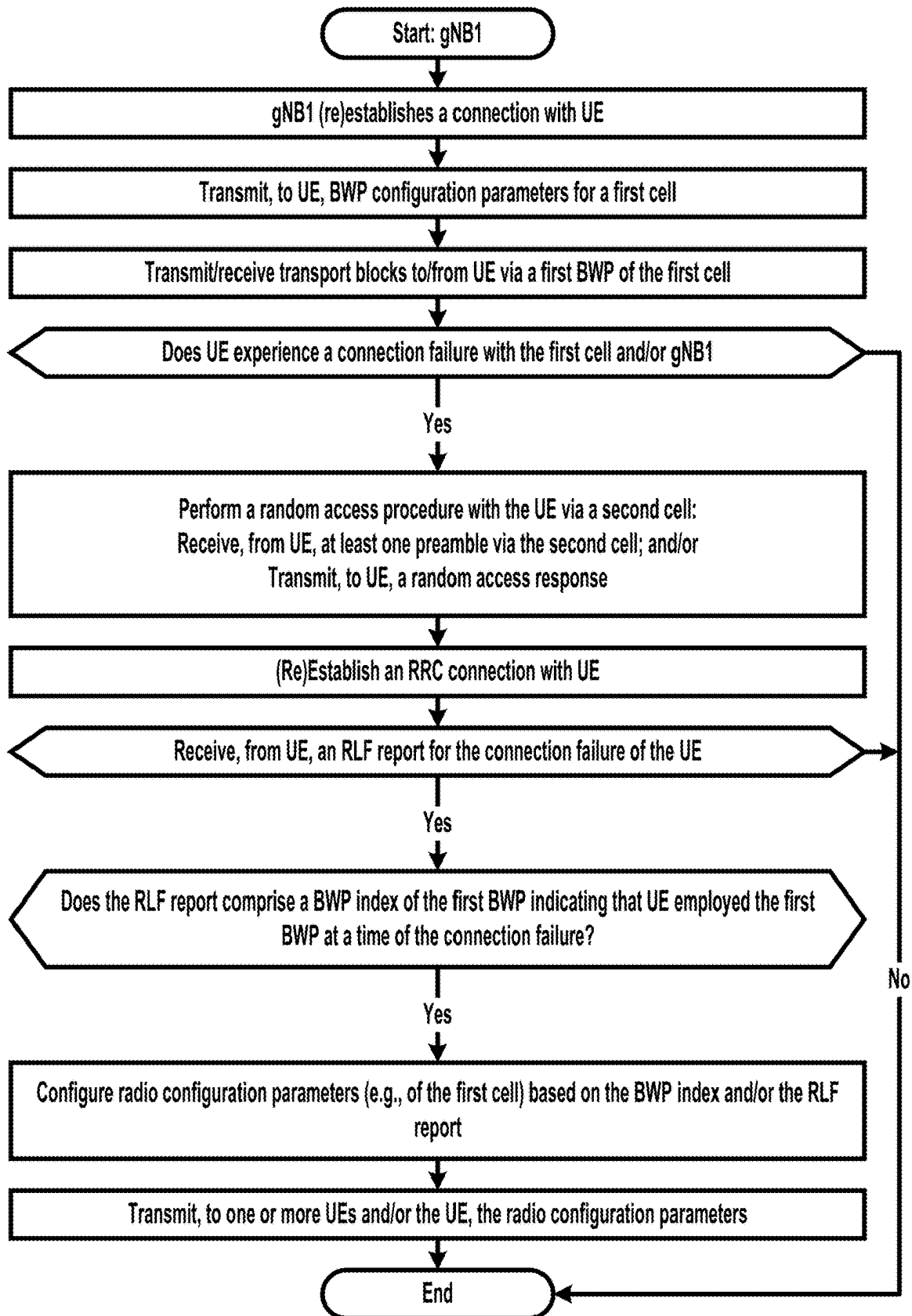


FIG. 24

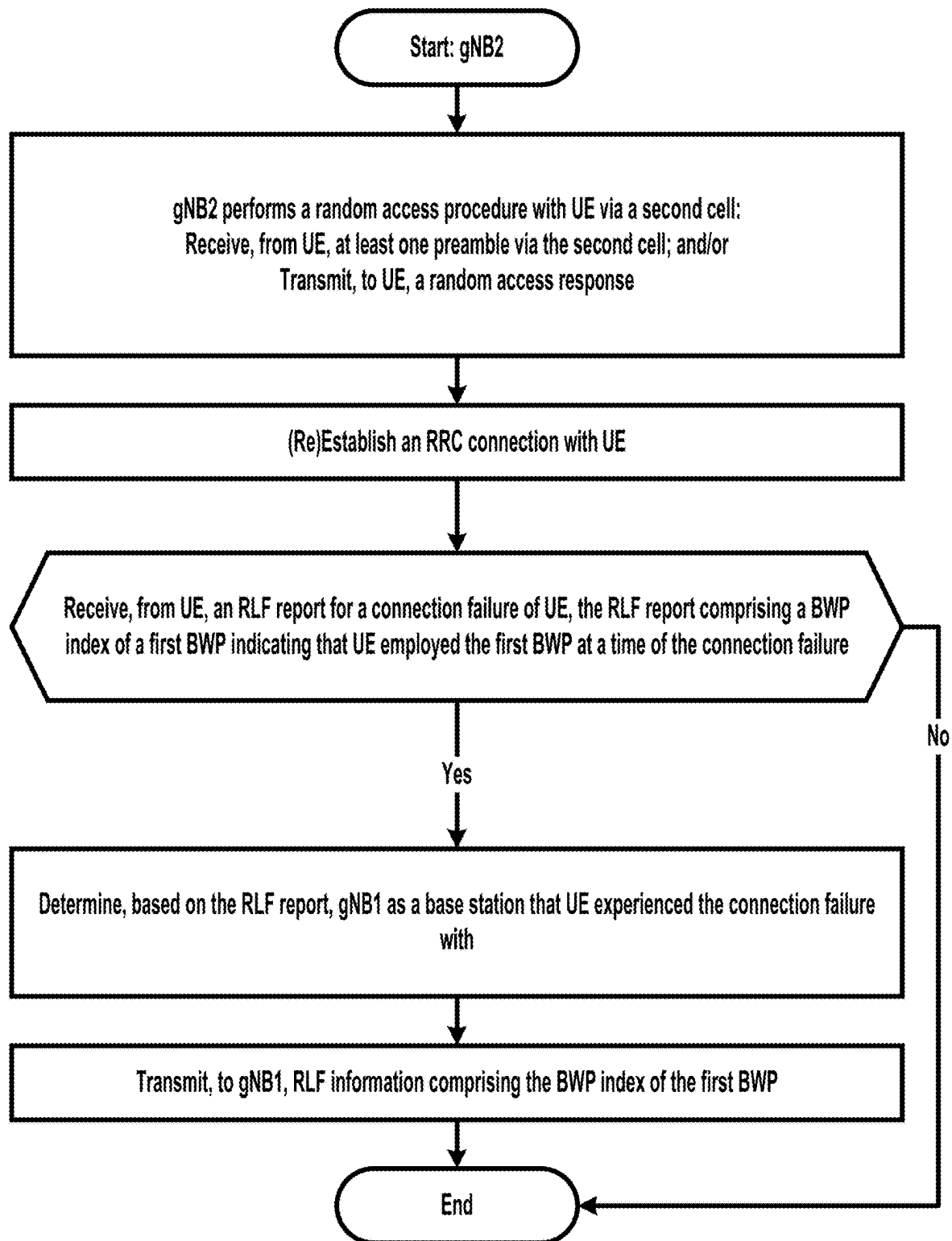


FIG. 25

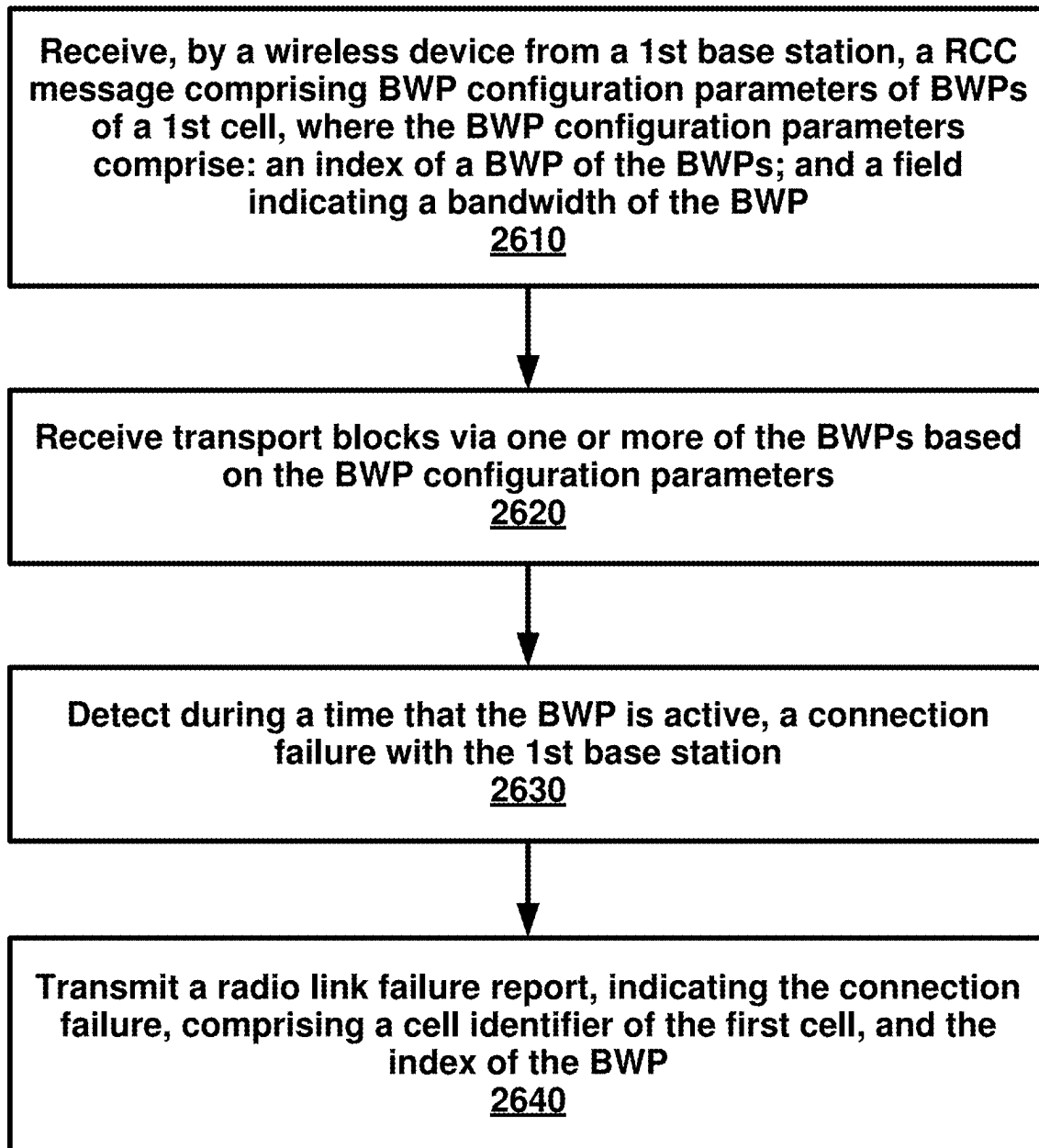


FIG. 26

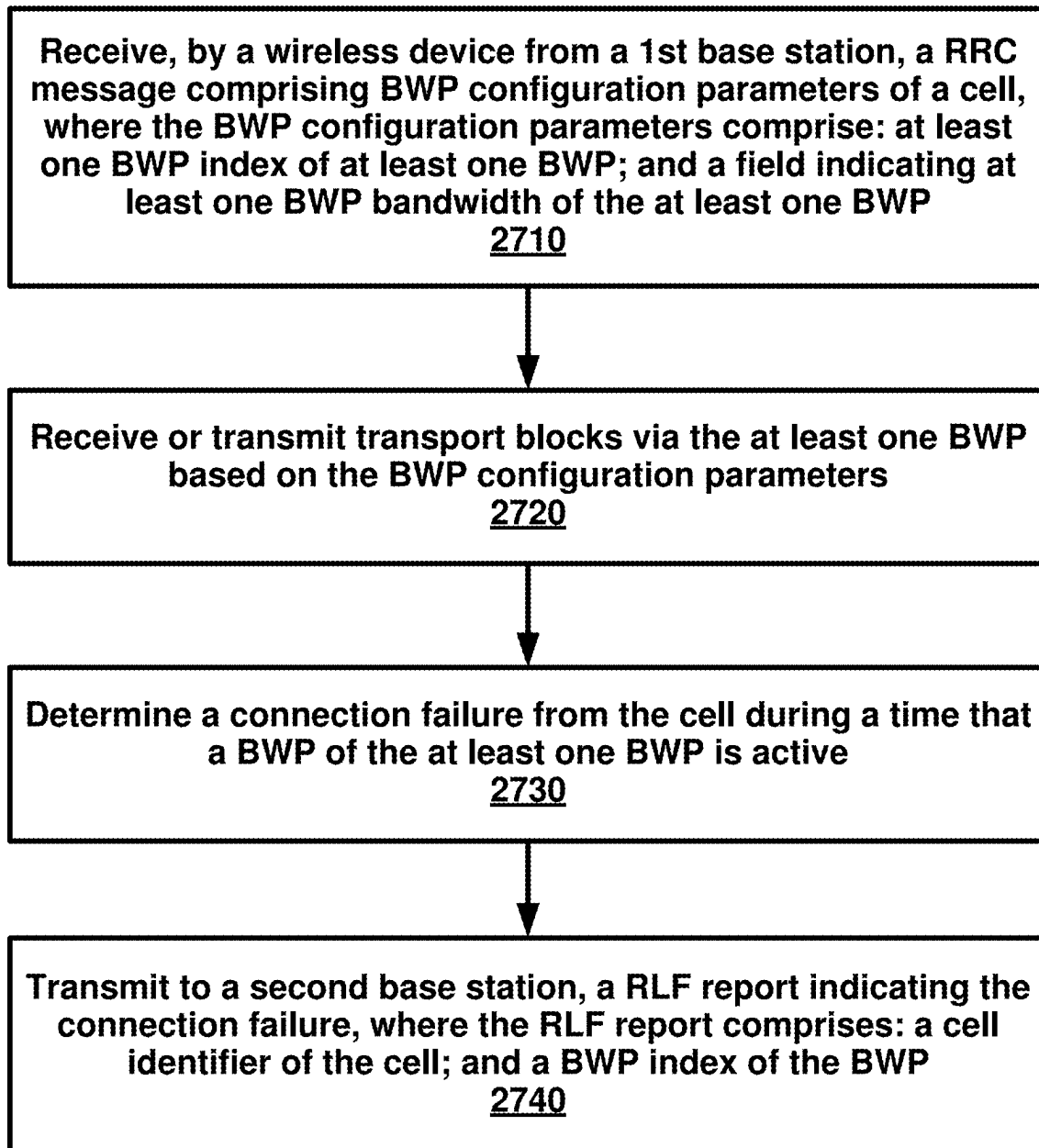


FIG. 27

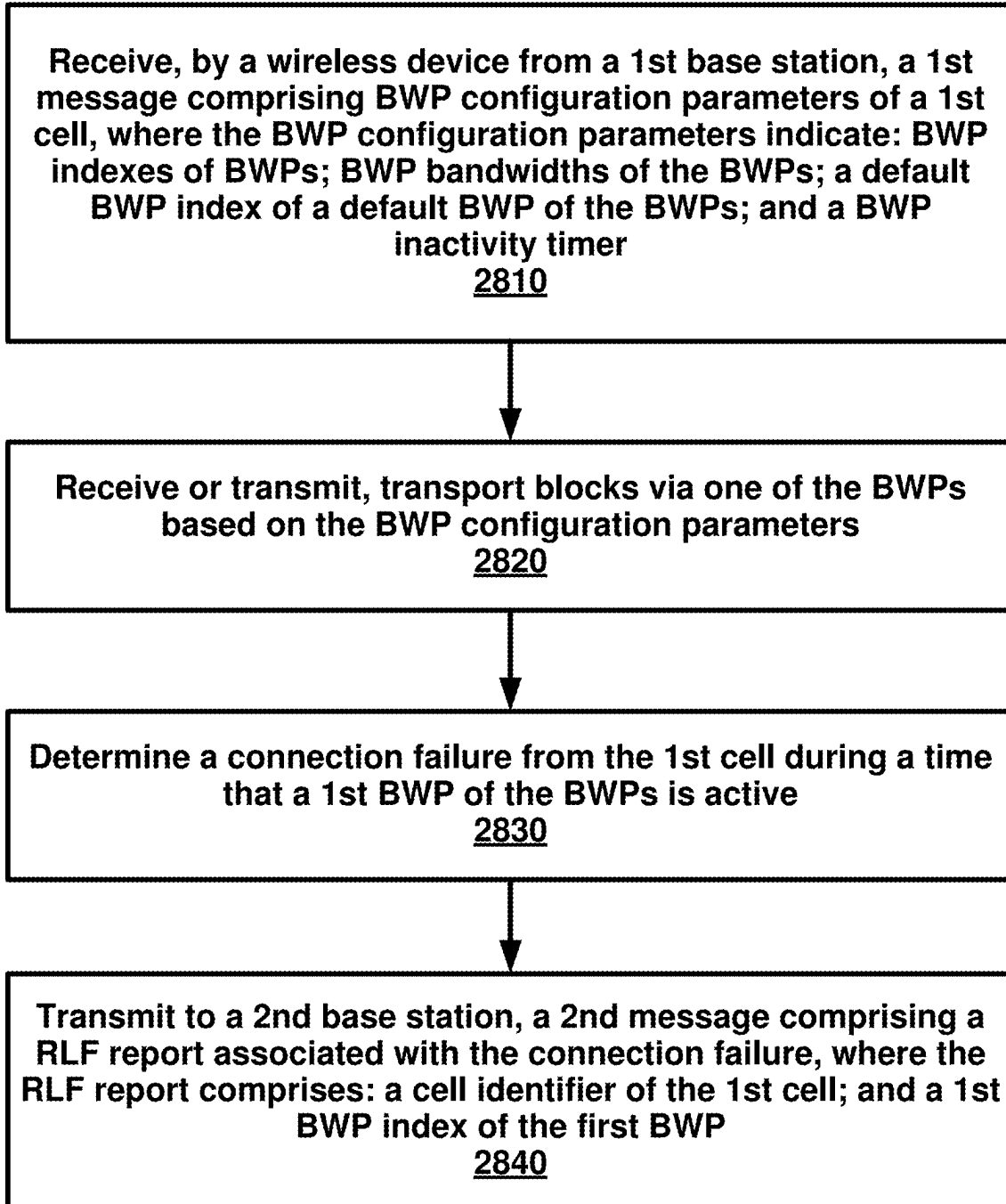


FIG. 28

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BANDWIDTH PART FAILURE INDICATION IN RADIO LINK FAILURE REPORT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/233,460, filed Aug. 14, 2023, which is a continuation of U.S. patent application Ser. No. 18/047,018, filed Oct. 17, 2022, which is a continuation of U.S. patent application Ser. No. 16/992,656, filed Aug. 13, 2020, which is a continuation of U.S. patent application Ser. No. 16/267,322, filed Feb. 4, 2019, which claims the benefit of U.S. Provisional Application No. 62/625,637, filed Feb. 2, 2018, all of which are hereby incorporated by reference in their entireties.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

Examples of several of the various embodiments of the present disclosure are described herein with reference to the drawings.

FIG. 1 is a diagram of an example RAN architecture as per an aspect of an embodiment of the present disclosure.

FIG. 2A is a diagram of an example user plane protocol stack as per an aspect of an embodiment of the present disclosure.

FIG. 2B is a diagram of an example control plane protocol stack as per an aspect of an embodiment of the present disclosure.

FIG. 3 is a diagram of an example wireless device and two base stations as per an aspect of an embodiment of the present disclosure.

FIG. 4A, FIG. 4B, FIG. 4C and FIG. 4D are example diagrams for uplink and downlink signal transmission as per an aspect of an embodiment of the present disclosure.

FIG. 5A is a diagram of an example uplink channel mapping and example uplink physical signals as per an aspect of an embodiment of the present disclosure.

FIG. 5B is a diagram of an example downlink channel mapping and example downlink physical signals as per an aspect of an embodiment of the present disclosure.

FIG. 6 is a diagram depicting an example transmission time or reception time for a carrier as per an aspect of an embodiment of the present disclosure.

FIG. 7A and FIG. 7B are diagrams depicting example sets of OFDM subcarriers as per an aspect of an embodiment of the present disclosure.

FIG. 8 is a diagram depicting example OFDM radio resources as per an aspect of an embodiment of the present disclosure.

FIG. 9A is a diagram depicting an example CSI-RS and/or SS block transmission in a multi-beam system.

FIG. 9B is a diagram depicting an example downlink beam management procedure as per an aspect of an embodiment of the present disclosure.

FIG. 10 is an example diagram of configured BWPs as per an aspect of an embodiment of the present disclosure.

FIG. 11A, and FIG. 11B are diagrams of an example multi-connectivity as per an aspect of an embodiment of the present disclosure.

FIG. 12 is a diagram of an example random access procedure as per an aspect of an embodiment of the present disclosure.

FIG. 13 is a structure of example MAC entities as per an aspect of an embodiment of the present disclosure.

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FIG. 14 is a diagram of an example RAN architecture as per an aspect of an embodiment of the present disclosure.

FIG. 15 is a diagram of example RRC states as per an aspect of an embodiment of the present disclosure.

FIG. 16 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 17 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 18 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 19 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 20 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 21 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 22 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 23 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 24 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 25 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 26 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 27 is an example diagram of an aspect of an embodiment of the present disclosure.

FIG. 28 is an example diagram of an aspect of an embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

Example embodiments of the present disclosure enable operation of wireless communication systems. Embodiments of the technology disclosed herein may be employed in the technical field of multicarrier communication systems. More particularly, the embodiments of the technology disclosed herein may relate to wireless communication systems in multicarrier communication systems.

The following Acronyms are used throughout the present disclosure:

3GPP 3rd Generation Partnership Project
 5GC 5G Core Network
 ACK Acknowledgement
 AMF Access and Mobility Management Function
 ARQ Automatic Repeat Request
 AS Access Stratum
 ASIC Application-Specific Integrated Circuit
 BA Bandwidth Adaptation
 BCCH Broadcast Control Channel
 BCH Broadcast Channel
 BPSK Binary Phase Shift Keying
 BWP Bandwidth Part
 CA Carrier Aggregation
 CC Component Carrier
 CCH Common Control Channel
 CDMA Code Division Multiple Access
 CN Core Network
 CP Cyclic Prefix
 CP-OFDM Cyclic Prefix-Orthogonal Frequency Division Multiplex
 C-RNTI Cell-Radio Network Temporary Identifier
 CS Configured Scheduling
 CSI Channel State Information
 CSI-RS Channel State Information-Reference Signal
 CQI Channel Quality Indicator

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CSS Common Search Space
 CU Central Unit
 DC Dual Connectivity
 DCCCH Dedicated Control Channel
 DCI Downlink Control Information
 DL Downlink
 DL-SCH Downlink Shared CHannel
 DM-RS DeModulation Reference Signal
 DRB Data Radio Bearer
 DRX Discontinuous Reception
 DTCH Dedicated Traffic Channel
 DU Distributed Unit
 EPC Evolved Packet Core
 E-UTRA Evolved UMTS Terrestrial Radio Access
 E-UTRAN Evolved-Universal Terrestrial Radio Access Network
 FDD Frequency Division Duplex
 FPGA Field Programmable Gate Arrays
 F1-C F1-Control plane
 F1-U F1-User plane
 gNB next generation Node B
 HARQ Hybrid Automatic Repeat reQuest
 HDL Hardware Description Languages
 IE Information Element
 IP Internet Protocol
 LCID Logical Channel Identifier
 LTE Long Term Evolution
 MAC Media Access Control
 MCG Master Cell Group
 MCS Modulation and Coding Scheme
 MeNB Master evolved Node B
 MIB Master Information Block
 MME Mobility Management Entity
 MN Master Node
 NACK Negative Acknowledgement
 NAS Non-Access Stratum
 NG CP Next Generation Control Plane
 NGC Next Generation Core
 NG-C NG-Control plane
 ng-eNB next generation evolved Node B
 NG-U NG-User plane
 NR New Radio
 NR MAC New Radio MAC
 NR PDCP New Radio PDCP
 NR PHY New Radio PHYSical
 NR RLC New Radio RLC
 NR RRC New Radio RRC
 NSSAI Network Slice Selection Assistance Information
 O&M Operation and Maintenance
 OFDM orthogonal Frequency Division Multiplexing
 PBCH Physical Broadcast Channel
 PCC Primary Component Carrier
 PCCH Paging Control CHannel
 PCell Primary Cell
 PCH Paging CHannel
 PDCCH Physical Downlink Control CHannel
 PDCP Packet Data Convergence Protocol
 PDSCH Physical Downlink Shared CHannel
 PDU Protocol Data Unit
 PHICH Physical HARQ Indicator CHannel
 PHY PHYSical
 PLMN Public Land Mobile Network
 PMI Precoding Matrix Indicator
 PRACH Physical Random Access CHannel
 PRB Physical Resource Block
 PSCell Primary Secondary Cell
 PSS Primary Synchronization Signal

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pTAG primary Timing Advance Group
 PT-RS Phase Tracking Reference Signal
 PUCCH Physical Uplink Control CHannel
 PUSCH Physical Uplink Shared CHannel
 5 QAM Quadrature Amplitude Modulation
 QFI Quality of Service Indicator
 QOS Quality of Service
 QPSK Quadrature Phase Shift Keying
 RA Random Access
 10 RACH Random Access CHannel
 RAN Radio Access Network
 RAT Radio Access Technology
 RA-RNTI Random Access-Radio Network Temporary Identifier
 RB Resource Blocks
 RBG Resource Block Groups
 RI Rank indicator
 RLC Radio Link Control
 RRC Radio Resource Control
 20 RS Reference Signal
 RSRP Reference Signal Received Power
 SCC Secondary Component Carrier
 SCell Secondary Cell
 SCG Secondary Cell Group
 25 SC-FDMA Single Carrier-Frequency Division Multiple Access
 SDAP Service Data Adaptation Protocol
 SDU Service Data Unit
 SeNB Secondary evolved Node B
 30 SFN System Frame Number
 S-GW Serving Gate Way
 SI System Information
 SIB System Information Block
 SMF Session Management Function
 35 SN Secondary Node
 SpCell Special Cell
 SRB Signaling Radio Bearer
 SRS Sounding Reference Signal
 SS Synchronization Signal
 40 SSS Secondary Synchronization Signal
 STAG secondary Timing Advance Group
 TA Timing Advance
 TAG Timing Advance Group
 TAI Tracking Area Identifier
 45 TAT Time Alignment Timer
 TB Transport Block
 TC-RNTI Temporary Cell-Radio Network Temporary Identifier
 TDD Time Division Duplex
 50 TDMA Time Division Multiple Access
 TTI Transmission Time Interval
 UCI Uplink Control Information
 UE User Equipment
 UL Uplink
 55 UL-SCH Uplink Shared CHannel
 UPF User Plane Function
 UPGW User Plane Gateway
 VHDL VHSIC Hardware Description Language
 Xn-C Xn-Control plane
 60 Xn-U Xn-User plane
 Example embodiments of the disclosure may be implemented using various physical layer modulation and transmission mechanisms. Example transmission mechanisms may include, but are not limited to: Code Division Multiple Access (CDMA), Orthogonal Frequency Division Multiple Access (OFDMA), Time Division Multiple Access (TDMA), Wavelet technologies, and/or the like. Hybrid

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transmission mechanisms such as TDMA/CDMA, and OFDM/CDMA may also be employed. Various modulation schemes may be applied for signal transmission in the physical layer. Examples of modulation schemes include, but are not limited to: phase, amplitude, code, a combination of these, and/or the like. An example radio transmission method may implement Quadrature Amplitude Modulation (QAM) using Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK), 16-QAM, 64-QAM, 256-QAM, and/or the like. Physical radio transmission may be enhanced by dynamically or semi-dynamically changing the modulation and coding scheme depending on transmission requirements and radio conditions.

FIG. 1 is an example Radio Access Network (RAN) architecture as per an aspect of an embodiment of the present disclosure. As illustrated in this example, a RAN node may be a next generation Node B (gNB) (e.g. 120A, 120B) providing New Radio (NR) user plane and control plane protocol terminations towards a first wireless device (e.g. 110A). In an example, a RAN node may be a next generation evolved Node B (ng-eNB) (e.g. 120C, 120D), providing Evolved UMTS Terrestrial Radio Access (E-UTRA) user plane and control plane protocol terminations towards a second wireless device (e.g. 110B). The first wireless device may communicate with a gNB over a Uu interface. The second wireless device may communicate with a ng-eNB over a Uu interface.

A gNB or an ng-eNB may host functions such as radio resource management and scheduling, IP header compression, encryption and integrity protection of data, selection of Access and Mobility Management Function (AMF) at User Equipment (UE) attachment, routing of user plane and control plane data, connection setup and release, scheduling and transmission of paging messages (originated from the AMF), scheduling and transmission of system broadcast information (originated from the AMF or Operation and Maintenance (O&M)), measurement and measurement reporting configuration, transport level packet marking in the uplink, session management, support of network slicing, Quality of Service (QoS) flow management and mapping to data radio bearers, support of UEs in RRC_INACTIVE state, distribution function for Non-Access Stratum (NAS) messages, RAN sharing, dual connectivity or tight interworking between NR and E-UTRA.

In an example, one or more gNBs and/or one or more ng-eNBs may be interconnected with each other by means of Xn interface. A gNB or an ng-eNB may be connected by means of NG interfaces to 5G Core Network (5GC). In an example, 5GC may comprise one or more AMF/User Plane Function (UPF) functions (e.g. 130A or 130B). A gNB or an ng-eNB may be connected to a UPF by means of an NG-User plane (NG-U) interface. The NG-U interface may provide delivery (e.g. non-guaranteed delivery) of user plane Protocol Data Units (PDUs) between a RAN node and the UPF. A gNB or an ng-eNB may be connected to an AMF by means of an NG-Control plane (NG-C) interface. The NG-C interface may provide functions such as NG interface management, UE context management, UE mobility management, transport of NAS messages, paging, PDU session management, configuration transfer or warning message transmission.

In an example, a UPF may host functions such as anchor point for intra-/inter-Radio Access Technology (RAT) mobility (when applicable), external PDU session point of interconnect to data network, packet routing and forwarding, packet inspection and user plane part of policy rule enforcement, traffic usage reporting, uplink classifier to support

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routing traffic flows to a data network, branching point to support multi-homed PDU session, QoS handling for user plane, e.g. packet filtering, gating, Uplink (UL)/Downlink (DL) rate enforcement, uplink traffic verification (e.g. Service Data Flow (SDF) to QoS flow mapping), downlink packet buffering and/or downlink data notification triggering.

In an example, an AMF may host functions such as NAS signaling termination, NAS signaling security, Access Stratum (AS) security control, inter Core Network (CN) node signaling for mobility between 3rd Generation Partnership Project (3GPP) access networks, idle mode UE reachability (e.g., control and execution of paging retransmission), registration area management, support of intra-system and inter-system mobility, access authentication, access authorization including check of roaming rights, mobility management control (subscription and policies), support of network slicing and/or Session Management Function (SMF) selection.

FIG. 2A is an example user plane protocol stack, where Service Data Adaptation Protocol (SDAP) (e.g. 211 and 221), Packet Data Convergence Protocol (PDCP) (e.g. 212 and 222), Radio Link Control (RLC) (e.g. 213 and 223) and Media Access Control (MAC) (e.g. 214 and 224) sublayers and Physical (PHY) (e.g. 215 and 225) layer may be terminated in wireless device (e.g. 110) and gNB (e.g. 120) on the network side. In an example, a PHY layer provides transport services to higher layers (e.g. MAC, RRC, etc.). In an example, services and functions of a MAC sublayer may comprise mapping between logical channels and transport channels, multiplexing/demultiplexing of MAC Service Data Units (SDUs) belonging to one or different logical channels into/from Transport Blocks (TBs) delivered to/from the PHY layer, scheduling information reporting, error correction through Hybrid Automatic Repeat request (HARQ) (e.g. one HARQ entity per carrier in case of Carrier Aggregation (CA)), priority handling between UEs by means of dynamic scheduling, priority handling between logical channels of one UE by means of logical channel prioritization, and/or padding. A MAC entity may support one or multiple numerologies and/or transmission timings. In an example, mapping restrictions in a logical channel prioritization may control which numerology and/or transmission timing a logical channel may use. In an example, an RLC sublayer may support transparent mode (TM), unacknowledged mode (UM) and acknowledged mode (AM) transmission modes. The RLC configuration may be per logical channel with no dependency on numerologies and/or Transmission Time Interval (TTI) durations. In an example, Automatic Repeat Request (ARQ) may operate on any of the numerologies and/or TTI durations the logical channel is configured with. In an example, services and functions of the PDCP layer for the user plane may comprise sequence numbering, header compression and decompression, transfer of user data, reordering and duplicate detection, PDCP PDU routing (e.g. in case of split bearers), retransmission of PDCP SDUs, ciphering, deciphering and integrity protection, PDCP SDU discard, PDCP re-establishment and data recovery for RLC AM, and/or duplication of PDCP PDUs. In an example, services and functions of SDAP may comprise mapping between a QoS flow and a data radio bearer. In an example, services and functions of SDAP may comprise mapping Quality of Service Indicator (QFI) in DL and UL packets. In an example, a protocol entity of SDAP may be configured for an individual PDU session.

FIG. 2B is an example control plane protocol stack where PDCP (e.g. 233 and 242), RLC (e.g. 234 and 243) and MAC

(e.g. **235** and **244**) sublayers and PHY (e.g. **236** and **245**) layer may be terminated in wireless device (e.g. **110**) and gNB (e.g. **120**) on a network side and perform service and functions described above. In an example, RRC (e.g. **232** and **241**) may be terminated in a wireless device and a gNB on a network side. In an example, services and functions of RRC may comprise broadcast of system information related to AS and NAS, paging initiated by 5GC or RAN, establishment, maintenance and release of an RRC connection between the UE and RAN, security functions including key management, establishment, configuration, maintenance and release of Signaling Radio Bearers (SRBs) and Data Radio Bearers (DRBs), mobility functions, QoS management functions, UE measurement reporting and control of the reporting, detection of and recovery from radio link failure, and/or NAS message transfer to/from NAS from/to a UE. In an example, NAS control protocol (e.g. **231** and **251**) may be terminated in the wireless device and AMF (e.g. **130**) on a network side and may perform functions such as authentication, mobility management between a UE and a AMF for 3GPP access and non-3GPP access, and session management between a UE and a SMF for 3GPP access and non-3GPP access.

In an example, a base station may configure a plurality of logical channels for a wireless device. A logical channel in the plurality of logical channels may correspond to a radio bearer and the radio bearer may be associated with a QoS requirement. In an example, a base station may configure a logical channel to be mapped to one or more TTIs/numerologies in a plurality of TTIs/numerologies. The wireless device may receive a Downlink Control Information (DCI) via Physical Downlink Control CHannel (PDCCH) indicating an uplink grant. In an example, the uplink grant may be for a first TTI/numerology and may indicate uplink resources for transmission of a transport block. The base station may configure each logical channel in the plurality of logical channels with one or more parameters to be used by a logical channel prioritization procedure at the MAC layer of the wireless device. The one or more parameters may comprise priority, prioritized bit rate, etc. A logical channel in the plurality of logical channels may correspond to one or more buffers comprising data associated with the logical channel. The logical channel prioritization procedure may allocate the uplink resources to one or more first logical channels in the plurality of logical channels and/or one or more MAC Control Elements (CEs). The one or more first logical channels may be mapped to the first TTI/numerology. The MAC layer at the wireless device may multiplex one or more MAC CEs and/or one or more MAC SDUs (e.g., logical channel) in a MAC PDU (e.g., transport block). In an example, the MAC PDU may comprise a MAC header comprising a plurality of MAC sub-headers. A MAC sub-header in the plurality of MAC sub-headers may correspond to a MAC CE or a MAC SUD (logical channel) in the one or more MAC CEs and/or one or more MAC SDUs. In an example, a MAC CE or a logical channel may be configured with a Logical Channel Identifier (LCID). In an example, LCID for a logical channel or a MAC CE may be fixed/pre-configured. In an example, LCID for a logical channel or MAC CE may be configured for the wireless device by the base station. The MAC sub-header corresponding to a MAC CE or a MAC SDU may comprise LCID associated with the MAC CE or the MAC SDU.

In an example, a base station may activate and/or deactivate and/or impact one or more processes (e.g., set values of one or more parameters of the one or more processes or start and/or stop one or more timers of the one or more

processes) at the wireless device by employing one or more MAC commands. The one or more MAC commands may comprise one or more MAC control elements. In an example, the one or more processes may comprise activation and/or deactivation of PDCP packet duplication for one or more radio bearers. The base station may transmit a MAC CE comprising one or more fields, the values of the fields indicating activation and/or deactivation of PDCP duplication for the one or more radio bearers. In an example, the one or more processes may comprise Channel State Information (CSI) transmission of on one or more cells. The base station may transmit one or more MAC CEs indicating activation and/or deactivation of the CSI transmission on the one or more cells. In an example, the one or more processes may comprise activation or deactivation of one or more secondary cells. In an example, the base station may transmit a MAC CE indicating activation or deactivation of one or more secondary cells. In an example, the base station may transmit one or more MAC CEs indicating starting and/or stopping one or more Discontinuous Reception (DRX) timers at the wireless device. In an example, the base station may transmit one or more MAC CEs indicating one or more timing advance values for one or more Timing Advance Groups (TAGs).

FIG. 3 is a block diagram of base stations (base station 1, **120A**, and base station 2, **120B**) and a wireless device **110**. A wireless device may be called an UE. A base station may be called a NB, cNB, gNB, and/or ng-eNB. In an example, a wireless device and/or a base station may act as a relay node. The base station 1, **120A**, may comprise at least one communication interface **320A** (e.g. a wireless modem, an antenna, a wired modem, and/or the like), at least one processor **321A**, and at least one set of program code instructions **323A** stored in non-transitory memory **322A** and executable by the at least one processor **321A**. The base station 2, **120B**, may comprise at least one communication interface **320B**, at least one processor **321B**, and at least one set of program code instructions **323B** stored in non-transitory memory **322B** and executable by the at least one processor **321B**.

A base station may comprise many sectors for example: 1, 2, 3, 4, or 6 sectors. A base station may comprise many cells, for example, ranging from 1 to 50 cells or more. A cell may be categorized, for example, as a primary cell or secondary cell. At Radio Resource Control (RRC) connection establishment/re-establishment/handover, one serving cell may provide the NAS (non-access stratum) mobility information (e.g. Tracking Area Identifier (TAI)). At RRC connection re-establishment/handover, one serving cell may provide the security input. This cell may be referred to as the Primary Cell (PCell). In the downlink, a carrier corresponding to the PCell may be a DL Primary Component Carrier (PCC), while in the uplink, a carrier may be an UL PCC. Depending on wireless device capabilities, Secondary Cells (SCells) may be configured to form together with a PCell a set of serving cells. In a downlink, a carrier corresponding to an SCell may be a downlink secondary component carrier (DL SCC), while in an uplink, a carrier may be an uplink secondary component carrier (UL SCC). An SCell may or may not have an uplink carrier.

A cell, comprising a downlink carrier and optionally an uplink carrier, may be assigned a physical cell ID and a cell index. A carrier (downlink or uplink) may belong to one cell. The cell ID or cell index may also identify the downlink carrier or uplink carrier of the cell (depending on the context it is used). In the disclosure, a cell ID may be equally referred to a carrier ID, and a cell index may be referred to

a carrier index. In an implementation, a physical cell ID or a cell index may be assigned to a cell. A cell ID may be determined using a synchronization signal transmitted on a downlink carrier. A cell index may be determined using RRC messages. For example, when the disclosure refers to a first physical cell ID for a first downlink carrier, the disclosure may mean the first physical cell ID is for a cell comprising the first downlink carrier. The same concept may apply to, for example, carrier activation. When the disclosure indicates that a first carrier is activated, the specification may equally mean that a cell comprising the first carrier is activated.

A base station may transmit to a wireless device one or more messages (e.g. RRC messages) comprising a plurality of configuration parameters for one or more cells. One or more cells may comprise at least one primary cell and at least one secondary cell. In an example, an RRC message may be broadcasted or unicasted to the wireless device. In an example, configuration parameters may comprise common parameters and dedicated parameters.

Services and/or functions of an RRC sublayer may comprise at least one of: broadcast of system information related to AS and NAS; paging initiated by 5GC and/or NG-RAN; establishment, maintenance, and/or release of an RRC connection between a wireless device and NG-RAN, which may comprise at least one of addition, modification and release of carrier aggregation; or addition, modification, and/or release of dual connectivity in NR or between E-UTRA and NR. Services and/or functions of an RRC sublayer may further comprise at least one of security functions comprising key management; establishment, configuration, maintenance, and/or release of Signaling Radio Bearers (SRBs) and/or Data Radio Bearers (DRBs); mobility functions which may comprise at least one of a handover (e.g. intra NR mobility or inter-RAT mobility) and a context transfer; or a wireless device cell selection and reselection and control of cell selection and reselection. Services and/or functions of an RRC sublayer may further comprise at least one of QoS management functions; a wireless device measurement configuration/reporting; detection of and/or recovery from radio link failure; or NAS message transfer to/from a core network entity (e.g. AMF, Mobility Management Entity (MME)) from/to the wireless device.

An RRC sublayer may support an RRC_Idle state, an RRC_Inactive state and/or an RRC_Connected state for a wireless device. In an RRC_Idle state, a wireless device may perform at least one of: Public Land Mobile Network (PLMN) selection; receiving broadcasted system information; cell selection/re-selection; monitoring/receiving a paging for mobile terminated data initiated by 5GC; paging for mobile terminated data area managed by 5GC; or DRX for CN paging configured via NAS. In an RRC_Inactive state, a wireless device may perform at least one of: receiving broadcasted system information; cell selection/re-selection; monitoring/receiving a RAN/CN paging initiated by NG-RAN/5GC; RAN-based notification area (RNA) managed by NG-RAN; or DRX for RAN/CN paging configured by NG-RAN/NAS. In an RRC_Idle state of a wireless device, a base station (e.g. NG-RAN) may keep a 5GC-NG-RAN connection (both C/U-planes) for the wireless device; and/or store a UE AS context for the wireless device. In an RRC_Connected state of a wireless device, a base station (e.g. NG-RAN) may perform at least one of: establishment of 5GC-NG-RAN connection (both C/U-planes) for the wireless device; storing a UE AS context for the wireless device; transmit/receive of unicast data to/from the wireless device; or network-controlled mobility based on measure-

ment results received from the wireless device. In an RRC_Connected state of a wireless device, an NG-RAN may know a cell that the wireless device belongs to.

System information (SI) may be divided into minimum SI and other SI. The minimum SI may be periodically broadcast. The minimum SI may comprise basic information required for initial access and information for acquiring any other SI broadcast periodically or provisioned on-demand, i.e. scheduling information. The other SI may either be broadcast, or be provisioned in a dedicated manner, either triggered by a network or upon request from a wireless device. A minimum SI may be transmitted via two different downlink channels using different messages (e.g. MasterInformationBlock and SystemInformationBlockType1). Another SI may be transmitted via SystemInformationBlockType2. For a wireless device in an RRC_Connected state, dedicated RRC signaling may be employed for the request and delivery of the other SI. For the wireless device in the RRC_Idle state and/or the RRC_Inactive state, the request may trigger a random-access procedure.

A wireless device may report its radio access capability information which may be static. A base station may request what capabilities for a wireless device to report based on band information. When allowed by a network, a temporary capability restriction request may be sent by the wireless device to signal the limited availability of some capabilities (e.g. due to hardware sharing, interference or overheating) to the base station. The base station may confirm or reject the request. The temporary capability restriction may be transparent to 5GC (e.g., only static capabilities may be stored in 5GC).

When CA is configured, a wireless device may have an RRC connection with a network. At RRC connection establishment/re-establishment/handover procedure, one serving cell may provide NAS mobility information, and at RRC connection re-establishment/handover, one serving cell may provide a security input. This cell may be referred to as the PCell. Depending on the capabilities of the wireless device, SCells may be configured to form together with the PCell a set of serving cells. The configured set of serving cells for the wireless device may comprise one PCell and one or more SCells.

The reconfiguration, addition and removal of SCells may be performed by RRC. At intra-NR handover, RRC may also add, remove, or reconfigure SCells for usage with the target PCell. When adding a new SCell, dedicated RRC signaling may be employed to send all required system information of the SCell i.e. while in connected mode, wireless devices may not need to acquire broadcasted system information directly from the SCells.

The purpose of an RRC connection reconfiguration procedure may be to modify an RRC connection, (e.g. to establish, modify and/or release RBs, to perform handover, to setup, modify, and/or release measurements, to add, modify, and/or release SCells and cell groups). As part of the RRC connection reconfiguration procedure, NAS dedicated information may be transferred from the network to the wireless device. The RRCConnectionReconfiguration message may be a command to modify an RRC connection. It may convey information for measurement configuration, mobility control, radio resource configuration (e.g. RBs, MAC main configuration and physical channel configuration) comprising any associated dedicated NAS information and security configuration. If the received RRC Connection Reconfiguration message includes the sCellToReleaseList, the wireless device may perform an SCell release. If the received RRC Connection Reconfiguration message

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includes the sCellToAddModList, the wireless device may perform SCell additions or modification.

An RRC connection establishment (or reestablishment, resume) procedure may be to establish (or reestablish, resume) an RRC connection, an RRC connection establishment procedure may comprise SRB1 establishment. The RRC connection establishment procedure may be used to transfer the initial NAS dedicated information/message from a wireless device to E-UTRAN. The RRCConnectionReestablishment message may be used to reestablish SRB1.

A measurement report procedure may be to transfer measurement results from a wireless device to NG-RAN. The wireless device may initiate a measurement report procedure after successful security activation. A measurement report message may be employed to transmit measurement results.

The wireless device 110 may comprise at least one communication interface 310 (e.g. a wireless modem, an antenna, and/or the like), at least one processor 314, and at least one set of program code instructions 316 stored in non-transitory memory 315 and executable by the at least one processor 314. The wireless device 110 may further comprise at least one of at least one speaker/microphone 311, at least one keypad 312, at least one display/touchpad 313, at least one power source 317, at least one global positioning system (GPS) chipset 318, and other peripherals 319.

The processor 314 of the wireless device 110, the processor 321A of the base station 1 120A, and/or the processor 321B of the base station 2 120B may comprise at least one of a general-purpose processor, a digital signal processor (DSP), a controller, a microcontroller, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) and/or other programmable logic device, discrete gate and/or transistor logic, discrete hardware components, and the like. The processor 314 of the wireless device 110, the processor 321A in base station 1 120A, and/or the processor 321B in base station 2 120B may perform at least one of signal coding/processing, data processing, power control, input/output processing, and/or any other functionality that may enable the wireless device 110, the base station 1 120A and/or the base station 2 120B to operate in a wireless environment.

The processor 314 of the wireless device 110 may be connected to the speaker/microphone 311, the keypad 312, and/or the display/touchpad 313. The processor 314 may receive user input data from and/or provide user output data to the speaker/microphone 311, the keypad 312, and/or the display/touchpad 313. The processor 314 in the wireless device 110 may receive power from the power source 317 and/or may be configured to distribute the power to the other components in the wireless device 110. The power source 317 may comprise at least one of one or more dry cell batteries, solar cells, fuel cells, and the like. The processor 314 may be connected to the GPS chipset 318. The GPS chipset 318 may be configured to provide geographic location information of the wireless device 110.

The processor 314 of the wireless device 110 may further be connected to other peripherals 319, which may comprise one or more software and/or hardware modules that provide additional features and/or functionalities. For example, the peripherals 319 may comprise at least one of an accelerometer, a satellite transceiver, a digital camera, a universal serial bus (USB) port, a hands-free headset, a frequency modulated (FM) radio unit, a media player, an Internet browser, and the like.

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The communication interface 320A of the base station 1, 120A, and/or the communication interface 320B of the base station 2, 120B, may be configured to communicate with the communication interface 310 of the wireless device 110 via a wireless link 330A and/or a wireless link 330B respectively. In an example, the communication interface 320A of the base station 1, 120A, may communicate with the communication interface 320B of the base station 2 and other RAN and core network nodes.

The wireless link 330A and/or the wireless link 330B may comprise at least one of a bi-directional link and/or a directional link. The communication interface 310 of the wireless device 110 may be configured to communicate with the communication interface 320A of the base station 1 120A and/or with the communication interface 320B of the base station 2 120B. The base station 1 120A and the wireless device 110 and/or the base station 2 120B and the wireless device 110 may be configured to send and receive transport blocks via the wireless link 330A and/or via the wireless link 330B, respectively. The wireless link 330A and/or the wireless link 330B may employ at least one frequency carrier. According to some of various aspects of embodiments, transceiver(s) may be employed. A transceiver may be a device that comprises both a transmitter and a receiver. Transceivers may be employed in devices such as wireless devices, base stations, relay nodes, and/or the like. Example embodiments for radio technology implemented in the communication interface 310, 320A, 320B and the wireless link 330A, 330B are illustrated in FIG. 4A, FIG. 4B, FIG. 4C, FIG. 4D, FIG. 6, FIG. 7A, FIG. 7B, FIG. 8, and associated text.

In an example, other nodes in a wireless network (e.g. AMF, UPF, SMF, etc.) may comprise one or more communication interfaces, one or more processors, and memory storing instructions.

A node (e.g. wireless device, base station, AMF, SMF, UPF, servers, switches, antennas, and/or the like) may comprise one or more processors, and memory storing instructions that when executed by the one or more processors causes the node to perform certain processes and/or functions. Example embodiments may enable operation of single-carrier and/or multi-carrier communications. Other example embodiments may comprise a non-transitory tangible computer readable media comprising instructions executable by one or more processors to cause operation of single-carrier and/or multi-carrier communications. Yet other example embodiments may comprise an article of manufacture that comprises a non-transitory tangible computer readable machine-accessible medium having instructions encoded thereon for enabling programmable hardware to cause a node to enable operation of single-carrier and/or multi-carrier communications. The node may include processors, memory, interfaces, and/or the like.

An interface may comprise at least one of a hardware interface, a firmware interface, a software interface, and/or a combination thereof. The hardware interface may comprise connectors, wires, electronic devices such as drivers, amplifiers, and/or the like. The software interface may comprise code stored in a memory device to implement protocol(s), protocol layers, communication drivers, device drivers, combinations thereof, and/or the like. The firmware interface may comprise a combination of embedded hardware and code stored in and/or in communication with a memory device to implement connections, electronic device operations, protocol(s), protocol layers, communication drivers, device drivers, hardware operations, combinations thereof, and/or the like.

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FIG. 4A, FIG. 4B, FIG. 4C and FIG. 4D are example diagrams for uplink and downlink signal transmission as per an aspect of an embodiment of the present disclosure. FIG. 4A shows an example uplink transmitter for at least one physical channel. A baseband signal representing a physical uplink shared channel may perform one or more functions. The one or more functions may comprise at least one of: scrambling; modulation of scrambled bits to generate complex-valued symbols; mapping of the complex-valued modulation symbols onto one or several transmission layers; transform precoding to generate complex-valued symbols; precoding of the complex-valued symbols; mapping of precoded complex-valued symbols to resource elements; generation of complex-valued time-domain Single Carrier-Frequency Division Multiple Access (SC-FDMA) or CP-OFDM signal for an antenna port; and/or the like. In an example, when transform precoding is enabled, a SC-FDMA signal for uplink transmission may be generated. In an example, when transform precoding is not enabled, an CP-OFDM signal for uplink transmission may be generated by FIG. 4A. These functions are illustrated as examples and it is anticipated that other mechanisms may be implemented in various embodiments.

An example structure for modulation and up-conversion to the carrier frequency of the complex-valued SC-FDMA or CP-OFDM baseband signal for an antenna port and/or the complex-valued Physical Random Access CHannel (PRACH) baseband signal is shown in FIG. 4B. Filtering may be employed prior to transmission.

An example structure for downlink transmissions is shown in FIG. 4C. The baseband signal representing a downlink physical channel may perform one or more functions. The one or more functions may comprise: scrambling of coded bits in a codeword to be transmitted on a physical channel; modulation of scrambled bits to generate complex-valued modulation symbols; mapping of the complex-valued modulation symbols onto one or several transmission layers; precoding of the complex-valued modulation symbols on a layer for transmission on the antenna ports; mapping of complex-valued modulation symbols for an antenna port to resource elements; generation of complex-valued time-domain OFDM signal for an antenna port; and/or the like. These functions are illustrated as examples and it is anticipated that other mechanisms may be implemented in various embodiments.

In an example, a gNB may transmit a first symbol and a second symbol on an antenna port, to a wireless device. The wireless device may infer the channel (e.g., fading gain, multipath delay, etc.) for conveying the second symbol on the antenna port, from the channel for conveying the first symbol on the antenna port. In an example, a first antenna port and a second antenna port may be quasi co-located if one or more large-scale properties of the channel over which a first symbol on the first antenna port is conveyed may be inferred from the channel over which a second symbol on a second antenna port is conveyed. The one or more large-scale properties may comprise at least one of: delay spread; doppler spread; doppler shift; average gain; average delay; and/or spatial Receiving (Rx) parameters.

An example modulation and up-conversion to the carrier frequency of the complex-valued OFDM baseband signal for an antenna port is shown in FIG. 4D. Filtering may be employed prior to transmission.

FIG. 5A is a diagram of an example uplink channel mapping and example uplink physical signals. FIG. 5B is a diagram of an example downlink channel mapping and a downlink physical signals. In an example, a physical layer

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may provide one or more information transfer services to a MAC and/or one or more higher layers. For example, the physical layer may provide the one or more information transfer services to the MAC via one or more transport channels. An information transfer service may indicate how and with what characteristics data are transferred over the radio interface.

In an example embodiment, a radio network may comprise one or more downlink and/or uplink transport channels. For example, a diagram in FIG. 5A shows example uplink transport channels comprising Uplink-Shared CHannel (UL-SCH) 501 and Random Access CHannel (RACH) 502. A diagram in FIG. 5B shows example downlink transport channels comprising Downlink-Shared CHannel (DL-SCH) 511, Paging CHannel (PCH) 512, and Broadcast CHannel (BCH) 513. A transport channel may be mapped to one or more corresponding physical channels. For example, UL-SCH 501 may be mapped to Physical Uplink Shared CHannel (PUSCH) 503. RACH 502 may be mapped to PRACH 505. DL-SCH 511 and PCH 512 may be mapped to Physical Downlink Shared CHannel (PDSCH) 514. BCH 513 may be mapped to Physical Broadcast CHannel (PBCH) 516.

There may be one or more physical channels without a corresponding transport channel. The one or more physical channels may be employed for Uplink Control Information (UCI) 509 and/or Downlink Control Information (DCI) 517. For example, Physical Uplink Control CHannel (PUCCH) 504 may carry UCI 509 from a UE to a base station. For example, Physical Downlink Control CHannel (PDCCH) 515 may carry DCI 517 from a base station to a UE. NR may support UCI 509 multiplexing in PUSCH 503 when UCI 509 and PUSCH 503 transmissions may coincide in a slot at least in part. The UCI 509 may comprise at least one of CSI, Acknowledgement (ACK)/Negative Acknowledgement (NACK), and/or scheduling request. The DCI 517 on PDCCH 515 may indicate at least one of following: one or more downlink assignments and/or one or more uplink scheduling grants

In uplink, a UE may transmit one or more Reference Signals (RSs) to a base station. For example, the one or more RSs may be at least one of Demodulation-RS (DM-RS) 506, Phase Tracking-RS (PT-RS) 507, and/or Sounding RS (SRS) 508. In downlink, a base station may transmit (e.g., unicast, multicast, and/or broadcast) one or more RSs to a UE. For example, the one or more RSs may be at least one of Primary Synchronization Signal (PSS)/Secondary Synchronization Signal (SSS) 521, CSI-RS 522, DM-RS 523, and/or PT-RS 524.

In an example, a UE may transmit one or more uplink DM-RSs 506 to a base station for channel estimation, for example, for coherent demodulation of one or more uplink physical channels (e.g., PUSCH 503 and/or PUCCH 504). For example, a UE may transmit a base station at least one uplink DM-RS 506 with PUSCH 503 and/or PUCCH 504, wherein the at least one uplink DM-RS 506 may be spanning a same frequency range as a corresponding physical channel. In an example, a base station may configure a UE with one or more uplink DM-RS configurations. At least one DM-RS configuration may support a front-loaded DM-RS pattern. A front-loaded DM-RS may be mapped over one or more OFDM symbols (e.g., 1 or 2 adjacent OFDM symbols). One or more additional uplink DM-RS may be configured to transmit at one or more symbols of a PUSCH and/or PUCCH. A base station may semi-statistically configure a UE with a maximum number of front-loaded DM-RS symbols for PUSCH and/or PUCCH. For example, a UE may

schedule a single-symbol DM-RS and/or double symbol DM-RS based on a maximum number of front-loaded DM-RS symbols, wherein a base station may configure the UE with one or more additional uplink DM-RS for PUSCH and/or PUCCH. A new radio network may support, e.g., at least for CP-OFDM, a common DM-RS structure for DL and UL, wherein a DM-RS location, DM-RS pattern, and/or scrambling sequence may be same or different.

In an example, whether uplink PT-RS 507 is present or not may depend on a RRC configuration. For example, a presence of uplink PT-RS may be UE-specifically configured. For example, a presence and/or a pattern of uplink PT-RS 507 in a scheduled resource may be UE-specifically configured by a combination of RRC signaling and/or association with one or more parameters employed for other purposes (e.g., Modulation and Coding Scheme (MCS)) which may be indicated by DCI. When configured, a dynamic presence of uplink PT-RS 507 may be associated with one or more DCI parameters comprising at least MCS. A radio network may support plurality of uplink PT-RS densities defined in time/frequency domain. When present, a frequency domain density may be associated with at least one configuration of a scheduled bandwidth. A UE may assume a same precoding for a DMRS port and a PT-RS port. A number of PT-RS ports may be fewer than a number of DM-RS ports in a scheduled resource. For example, uplink PT-RS 507 may be confined in the scheduled time/frequency duration for a UE.

In an example, a UE may transmit SRS 508 to a base station for channel state estimation to support uplink channel dependent scheduling and/or link adaptation. For example, SRS 508 transmitted by a UE may allow for a base station to estimate an uplink channel state at one or more different frequencies. A base station scheduler may employ an uplink channel state to assign one or more resource blocks of good quality for an uplink PUSCH transmission from a UE. A base station may semi-statistically configure a UE with one or more SRS resource sets. For an SRS resource set, a base station may configure a UE with one or more SRS resources. An SRS resource set applicability may be configured by a higher layer (e.g., RRC) parameter. For example, when a higher layer parameter indicates beam management, a SRS resource in each of one or more SRS resource sets may be transmitted at a time instant. A UE may transmit one or more SRS resources in different SRS resource sets simultaneously. A new radio network may support aperiodic, periodic and/or semi-persistent SRS transmissions. A UE may transmit SRS resources based on one or more trigger types, wherein the one or more trigger types may comprise higher layer signaling (e.g., RRC) and/or one or more DCI formats (e.g., at least one DCI format may be employed for a UE to select at least one of one or more configured SRS resource sets. An SRS trigger type 0 may refer to an SRS triggered based on a higher layer signaling. An SRS trigger type 1 may refer to an SRS triggered based on one or more DCI formats. In an example, when PUSCH 503 and SRS 508 are transmitted in a same slot, a UE may be configured to transmit SRS 508 after a transmission of PUSCH 503 and corresponding uplink DM-RS 506.

In an example, a base station may semi-statistically configure a UE with one or more SRS configuration parameters indicating at least one of following: a SRS resource configuration identifier, a number of SRS ports, time domain behavior of SRS resource configuration (e.g., an indication of periodic, semi-persistent, or aperiodic SRS), slot (mini-slot, and/or subframe) level periodicity and/or offset for a periodic and/or aperiodic SRS resource, a number of OFDM symbols in a SRS resource, starting OFDM symbol of a SRS

resource, a SRS bandwidth, a frequency hopping bandwidth, a cyclic shift, and/or a SRS sequence ID.

In an example, in a time domain, an SS/PBCH block may comprise one or more OFDM symbols (e.g., 4 OFDM symbols numbered in increasing order from 0 to 3) within the SS/PBCH block. An SS/PBCH block may comprise PSS/SSS 521 and PBCH 516. In an example, in the frequency domain, an SS/PBCH block may comprise one or more contiguous subcarriers (e.g., 240 contiguous subcarriers with the subcarriers numbered in increasing order from 0 to 239) within the SS/PBCH block. For example, a PSS/SSS 521 may occupy 1 OFDM symbol and 127 subcarriers. For example, PBCH 516 may span across 3 OFDM symbols and 240 subcarriers. A UE may assume that one or more SS/PBCH blocks transmitted with a same block index may be quasi co-located, e.g., with respect to Doppler spread, Doppler shift, average gain, average delay, and spatial Rx parameters. A UE may not assume quasi co-location for other SS/PBCH block transmissions. A periodicity of an SS/PBCH block may be configured by a radio network (e.g., by an RRC signaling) and one or more time locations where the SS/PBCH block may be sent may be determined by sub-carrier spacing. In an example, a UE may assume a band-specific sub-carrier spacing for an SS/PBCH block unless a radio network has configured a UE to assume a different sub-carrier spacing.

In an example, downlink CSI-RS 522 may be employed for a UE to acquire channel state information. A radio network may support periodic, aperiodic, and/or semi-persistent transmission of downlink CSI-RS 522. For example, a base station may semi-statistically configure and/or reconfigure a UE with periodic transmission of downlink CSI-RS 522. A configured CSI-RS resources may be activated and/or deactivated. For semi-persistent transmission, an activation and/or deactivation of CSI-RS resource may be triggered dynamically. In an example, CSI-RS configuration may comprise one or more parameters indicating at least a number of antenna ports. For example, a base station may configure a UE with 32 ports. A base station may semi-statistically configure a UE with one or more CSI-RS resource sets. One or more CSI-RS resources may be allocated from one or more CSI-RS resource sets to one or more UEs. For example, a base station may semi-statistically configure one or more parameters indicating CSI-RS resource mapping, for example, time-domain location of one or more CSI-RS resources, a bandwidth of a CSI-RS resource, and/or a periodicity. In an example, a UE may be configured to employ a same OFDM symbols for downlink CSI-RS 522 and Control Resource Set (CORESET) when the downlink CSI-RS 522 and CORESET are spatially quasi co-located and resource elements associated with the downlink CSI-RS 522 are the outside of PRBs configured for CORESET. In an example, a UE may be configured to employ a same OFDM symbols for downlink CSI-RS 522 and SSB/PBCH when the downlink CSI-RS 522 and SSB/PBCH are spatially quasi co-located and resource elements associated with the downlink CSI-RS 522 are the outside of PRBs configured for SSB/PBCH.

In an example, a UE may transmit one or more downlink DM-RSs 523 to a base station for channel estimation, for example, for coherent demodulation of one or more downlink physical channels (e.g., PDSCH 514). For example, a radio network may support one or more variable and/or configurable DM-RS patterns for data demodulation. At least one downlink DM-RS configuration may support a front-loaded DM-RS pattern. A front-loaded DM-RS may be mapped over one or more OFDM symbols (e.g., 1 or 2

adjacent OFDM symbols). A base station may semi-statistically configure a UE with a maximum number of front-loaded DM-RS symbols for PDSCH **514**. For example, a DM-RS configuration may support one or more DM-RS ports. For example, for single user-MIMO, a DM-RS configuration may support at least 8 orthogonal downlink DM-RS ports. For example, for multiuser-MIMO, a DM-RS configuration may support 12 orthogonal downlink DM-RS ports. A radio network may support, e.g., at least for CP-OFDM, a common DM-RS structure for DL and UL, wherein a DM-RS location, DM-RS pattern, and/or scrambling sequence may be same or different.

In an example, whether downlink PT-RS **524** is present or not may depend on a RRC configuration. For example, a presence of downlink PT-RS **524** may be UE-specifically configured. For example, a presence and/or a pattern of downlink PT-RS **524** in a scheduled resource may be UE-specifically configured by a combination of RRC signaling and/or association with one or more parameters employed for other purposes (e.g., MCS) which may be indicated by DCI. When configured, a dynamic presence of downlink PT-RS **524** may be associated with one or more DCI parameters comprising at least MCS. A radio network may support plurality of PT-RS densities defined in time/frequency domain. When present, a frequency domain density may be associated with at least one configuration of a scheduled bandwidth. A UE may assume a same precoding for a DMRS port and a PT-RS port. A number of PT-RS ports may be fewer than a number of DM-RS ports in a scheduled resource. For example, downlink PT-RS **524** may be confined in the scheduled time/frequency duration for a UE.

FIG. **6** is a diagram depicting an example transmission time and reception time for a carrier as per an aspect of an embodiment of the present disclosure. A multicarrier OFDM communication system may include one or more carriers, for example, ranging from 1 to 32 carriers, in case of carrier aggregation, or ranging from 1 to 64 carriers, in case of dual connectivity. Different radio frame structures may be supported (e.g., for FDD and for TDD duplex mechanisms). FIG. **6** shows an example frame timing. Downlink and uplink transmissions may be organized into radio frames **601**. In this example, radio frame duration is 10 ms. In this example, a 10 ms radio frame **601** may be divided into ten equally sized subframes **602** with 1 ms duration. Subframe(s) may comprise one or more slots (e.g. slots **603** and **605**) depending on subcarrier spacing and/or CP length. For example, a subframe with 15 kHz, 30 kHz, 60 kHz, 120 kHz, 240 kHz and 480 kHz subcarrier spacing may comprise one, two, four, eight, sixteen and thirty-two slots, respectively. In FIG. **6**, a subframe may be divided into two equally sized slots **603** with 0.5 ms duration. For example, 10 subframes may be available for downlink transmission and 10 subframes may be available for uplink transmissions in a 10 ms interval. Uplink and downlink transmissions may be separated in the frequency domain. Slot(s) may include a plurality of OFDM symbols **604**. The number of OFDM symbols **604** in a slot **605** may depend on the cyclic prefix length. For example, a slot may be 14 OFDM symbols for the same subcarrier spacing of up to 480 kHz with normal CP. A slot may be 12 OFDM symbols for the same subcarrier spacing of 60 KHz with extended CP. A slot may contain downlink, uplink, or a downlink part and an uplink part and/or alike.

FIG. **7A** is a diagram depicting example sets of OFDM subcarriers as per an aspect of an embodiment of the present disclosure. In the example, a gNB may communicate with a wireless device with a carrier with an example channel

bandwidth **700**. Arrow(s) in the diagram may depict a subcarrier in a multicarrier OFDM system. The OFDM system may use technology such as OFDM technology, SC-FDMA technology, and/or the like. In an example, an arrow **701** shows a subcarrier transmitting information symbols. In an example, a subcarrier spacing **702**, between two contiguous subcarriers in a carrier, may be any one of 15 KHz, 30 KHz, 60 KHz, 120 KHz, 240 KHz etc. In an example, different subcarrier spacing may correspond to different transmission numerologies. In an example, a transmission numerology may comprise at least: a numerology index; a value of subcarrier spacing; a type of cyclic prefix (CP). In an example, a gNB may transmit to/receive from a UE on a number of subcarriers **703** in a carrier. In an example, a bandwidth occupied by a number of subcarriers **703** (transmission bandwidth) may be smaller than the channel bandwidth **700** of a carrier, due to guard band **704** and **705**. In an example, a guard band **704** and **705** may be used to reduce interference to and from one or more neighbor carriers. A number of subcarriers (transmission bandwidth) in a carrier may depend on the channel bandwidth of the carrier and the subcarrier spacing. For example, a transmission bandwidth, for a carrier with 20 MHz channel bandwidth and 15 KHz subcarrier spacing, may be in number of 1024 subcarriers.

In an example, a gNB and a wireless device may communicate with multiple CCs when configured with CA. In an example, different component carriers may have different bandwidth and/or subcarrier spacing, if CA is supported. In an example, a gNB may transmit a first type of service to a UE on a first component carrier. The gNB may transmit a second type of service to the UE on a second component carrier. Different type of services may have different service requirement (e.g., data rate, latency, reliability), which may be suitable for transmission via different component carrier having different subcarrier spacing and/or bandwidth. FIG. **7B** shows an example embodiment. A first component carrier may comprise a first number of subcarriers **706** with a first subcarrier spacing **709**. A second component carrier may comprise a second number of subcarriers **707** with a second subcarrier spacing **710**. A third component carrier may comprise a third number of subcarriers **708** with a third subcarrier spacing **711**. Carriers in a multicarrier OFDM communication system may be contiguous carriers, non-contiguous carriers, or a combination of both contiguous and non-contiguous carriers.

FIG. **8** is a diagram depicting OFDM radio resources as per an aspect of an embodiment of the present disclosure. In an example, a carrier may have a transmission bandwidth **801**. In an example, a resource grid may be in a structure of frequency domain **802** and time domain **803**. In an example, a resource grid may comprise a first number of OFDM symbols in a subframe and a second number of resource blocks, starting from a common resource block indicated by higher-layer signaling (e.g. RRC signaling), for a transmission numerology and a carrier. In an example, in a resource grid, a resource unit identified by a subcarrier index and a symbol index may be a resource element **805**. In an example, a subframe may comprise a first number of OFDM symbols **807** depending on a numerology associated with a carrier. For example, when a subcarrier spacing of a numerology of a carrier is 15 KHz, a subframe may have 14 OFDM symbols for a carrier. When a subcarrier spacing of a numerology is 30 KHz, a subframe may have 28 OFDM symbols. When a subcarrier spacing of a numerology is 60 KHz, a subframe may have 56 OFDM symbols, etc. In an

example, a second number of resource blocks comprised in a resource grid of a carrier may depend on a bandwidth and a numerology of the carrier.

As shown in FIG. 8, a resource block 806 may comprise 12 subcarriers. In an example, multiple resource blocks may be grouped into a Resource Block Group (RBG) 804. In an example, a size of a RBG may depend on at least one of: a RRC message indicating a RBG size configuration; a size of a carrier bandwidth; or a size of a bandwidth part of a carrier. In an example, a carrier may comprise multiple bandwidth parts. A first bandwidth part of a carrier may have different frequency location and/or bandwidth from a second bandwidth part of the carrier.

In an example, a gNB may transmit a downlink control information comprising a downlink or uplink resource block assignment to a wireless device. A base station may transmit to or receive from, a wireless device, data packets (e.g. transport blocks) scheduled and transmitted via one or more resource blocks and one or more slots according to parameters in a downlink control information and/or RRC message(s). In an example, a starting symbol relative to a first slot of the one or more slots may be indicated to the wireless device. In an example, a gNB may transmit to or receive from, a wireless device, data packets scheduled on one or more RBGs and one or more slots.

In an example, a gNB may transmit a downlink control information comprising a downlink assignment to a wireless device via one or more PDCCHs. The downlink assignment may comprise parameters indicating at least modulation and coding format; resource allocation; and/or HARQ information related to DL-SCH. In an example, a resource allocation may comprise parameters of resource block allocation; and/or slot allocation. In an example, a gNB may dynamically allocate resources to a wireless device via a Cell-Radio Network Temporary Identifier (C-RNTI) on one or more PDCCHs. The wireless device may monitor the one or more PDCCHs in order to find possible allocation when its downlink reception is enabled. The wireless device may receive one or more downlink data package on one or more PDSCH scheduled by the one or more PDCCHs, when successfully detecting the one or more PDCCHs.

In an example, a gNB may allocate Configured Scheduling (CS) resources for down link transmission to a wireless device. The gNB may transmit one or more RRC messages indicating a periodicity of the CS grant. The gNB may transmit a DCI via a PDCCH addressed to a Configured Scheduling-RNTI (CS-RNTI) activating the CS resources. The DCI may comprise parameters indicating that the downlink grant is a CS grant. The CS grant may be implicitly reused according to the periodicity defined by the one or more RRC messages, until deactivated.

In an example, a gNB may transmit a downlink control information comprising an uplink grant to a wireless device via one or more PDCCHs. The uplink grant may comprise parameters indicating at least modulation and coding format; resource allocation; and/or HARQ information related to UL-SCH. In an example, a resource allocation may comprise parameters of resource block allocation; and/or slot allocation. In an example, a gNB may dynamically allocate resources to a wireless device via a C-RNTI on one or more PDCCHs. The wireless device may monitor the one or more PDCCHs in order to find possible resource allocation. The wireless device may transmit one or more uplink data package via one or more PUSCH scheduled by the one or more PDCCHs, when successfully detecting the one or more PDCCHs.

In an example, a gNB may allocate CS resources for uplink data transmission to a wireless device. The gNB may transmit one or more RRC messages indicating a periodicity of the CS grant. The gNB may transmit a DCI via a PDCCH addressed to a CS-RNTI activating the CS resources. The DCI may comprise parameters indicating that the uplink grant is a CS grant. The CS grant may be implicitly reused according to the periodicity defined by the one or more RRC message, until deactivated.

In an example, a base station may transmit DCI/control signaling via PDCCH. The DCI may take a format in a plurality of formats. A DCI may comprise downlink and/or uplink scheduling information (e.g., resource allocation information, HARQ related parameters, MCS), request for CSI (e.g., aperiodic CQI reports), request for SRS, uplink power control commands for one or more cells, one or more timing information (e.g., TB transmission/reception timing, HARQ feedback timing, etc.), etc. In an example, a DCI may indicate an uplink grant comprising transmission parameters for one or more transport blocks. In an example, a DCI may indicate downlink assignment indicating parameters for receiving one or more transport blocks. In an example, a DCI may be used by base station to initiate a contention-free random access at the wireless device. In an example, the base station may transmit a DCI comprising slot format indicator (SFI) notifying a slot format. In an example, the base station may transmit a DCI comprising pre-emption indication notifying the PRB(s) and/or OFDM symbol(s) where a UE may assume no transmission is intended for the UE. In an example, the base station may transmit a DCI for group power control of PUCCH or PUSCH or SRS. In an example, a DCI may correspond to an RNTI. In an example, the wireless device may obtain an RNTI in response to completing the initial access (e.g., C-RNTI). In an example, the base station may configure an RNTI for the wireless (e.g., CS-RNTI, TPC-CS-RNTI, TPC-PUCCH-RNTI, TPC-PUSCH-RNTI, TPC-SRS-RNTI). In an example, the wireless device may compute an RNTI (e.g., the wireless device may compute RA-RNTI based on resources used for transmission of a preamble). In an example, an RNTI may have a pre-configured value (e.g., P-RNTI or SI-RNTI). In an example, a wireless device may monitor a group common search space which may be used by base station for transmitting DCIs that are intended for a group of UEs. In an example, a group common DCI may correspond to an RNTI which is commonly configured for a group of UEs. In an example, a wireless device may monitor a UE-specific search space. In an example, a UE specific DCI may correspond to an RNTI configured for the wireless device.

An NR system may support a single beam operation and/or a multi-beam operation. In a multi-beam operation, a base station may perform a downlink beam sweeping to provide coverage for common control channels and/or downlink SS blocks, which may comprise at least a PSS, a SSS, and/or PBCH. A wireless device may measure quality of a beam pair link using one or more RSs. One or more SS blocks, or one or more CSI-RS resources, associated with a CSI-RS resource index (CRI), or one or more DM-RSs of PBCH, may be used as RS for measuring quality of a beam pair link. Quality of a beam pair link may be defined as a reference signal received power (RSRP) value, or a reference signal received quality (RSRQ) value, and/or a CSI value measured on RS resources. The base station may indicate whether an RS resource, used for measuring a beam pair link quality, is quasi-co-located (QCLed) with DM-RSs of a control channel. A RS resource and DM-RSs of a control

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channel may be called QCLed when a channel characteristics from a transmission on an RS to a wireless device, and that from a transmission on a control channel to a wireless device, are similar or same under a configured criterion. In a multi-beam operation, a wireless device may perform an uplink beam sweeping to access a cell.

In an example, a wireless device may be configured to monitor PDCCH on one or more beam pair links simultaneously depending on a capability of a wireless device. This may increase robustness against beam pair link blocking. A base station may transmit one or more messages to configure a wireless device to monitor PDCCH on one or more beam pair links in different PDCCH OFDM symbols. For example, a base station may transmit higher layer signaling (e.g. RRC signaling) or MAC CE comprising parameters related to the Rx beam setting of a wireless device for monitoring PDCCH on one or more beam pair links. A base station may transmit indication of spatial QCL assumption between an DL RS antenna port(s) (for example, cell-specific CSI-RS, or wireless device-specific CSI-RS, or SS block, or PBCH with or without DM-RSs of PBCH), and DL RS antenna port(s) for demodulation of DL control channel. Signaling for beam indication for a PDCCH may be MAC CE signaling, or RRC signaling, or DCI signaling, or specification-transparent and/or implicit method, and combination of these signaling methods.

For reception of unicast DL data channel, a base station may indicate spatial QCL parameters between DL RS antenna port(s) and DM-RS antenna port(s) of DL data channel. The base station may transmit DCI (e.g. downlink grants) comprising information indicating the RS antenna port(s). The information may indicate RS antenna port(s) which may be QCLed with the DM-RS antenna port(s). Different set of DM-RS antenna port(s) for a DL data channel may be indicated as QCL with different set of the RS antenna port(s).

FIG. 9A is an example of beam sweeping in a DL channel. In an RRC_INACTIVE state or RRC_IDLE state, a wireless device may assume that SS blocks form an SS burst 940, and an SS burst set 950. The SS burst set 950 may have a given periodicity. For example, in a multi-beam operation, a base station 120 may transmit SS blocks in multiple beams, together forming a SS burst 940. One or more SS blocks may be transmitted on one beam. If multiple SS bursts 940 are transmitted with multiple beams, SS bursts together may form SS burst set 950.

A wireless device may further use CSI-RS in the multi-beam operation for estimating a beam quality of a links between a wireless device and a base station. A beam may be associated with a CSI-RS. For example, a wireless device may, based on a RSRP measurement on CSI-RS, report a beam index, as indicated in a CRI for downlink beam selection, and associated with a RSRP value of a beam. A CSI-RS may be transmitted on a CSI-RS resource including at least one of one or more antenna ports, one or more time or frequency radio resources. A CSI-RS resource may be configured in a cell-specific way by common RRC signaling, or in a wireless device-specific way by dedicated RRC signaling, and/or L1/L2 signaling. Multiple wireless devices covered by a cell may measure a cell-specific CSI-RS resource. A dedicated subset of wireless devices covered by a cell may measure a wireless device-specific CSI-RS resource.

A CSI-RS resource may be transmitted periodically, or using aperiodic transmission, or using a multi-shot or semi-persistent transmission. For example, in a periodic transmission in FIG. 9A, a base station 120 may transmit configured

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CSI-RS resources 940 periodically using a configured periodicity in a time domain. In an aperiodic transmission, a configured CSI-RS resource may be transmitted in a dedicated time slot. In a multi-shot or semi-persistent transmission, a configured CSI-RS resource may be transmitted within a configured period. Beams used for CSI-RS transmission may have different beam width than beams used for SS-blocks transmission.

FIG. 9B is an example of a beam management procedure in an example new radio network. A base station 120 and/or a wireless device 110 may perform a downlink L1/L2 beam management procedure. One or more of the following downlink L1/L2 beam management procedures may be performed within one or more wireless devices 110 and one or more base stations 120. In an example, a P-1 procedure 910 may be used to enable the wireless device 110 to measure one or more Transmission (Tx) beams associated with the base station 120 to support a selection of a first set of Tx beams associated with the base station 120 and a first set of Rx beam(s) associated with a wireless device 110. For beamforming at a base station 120, a base station 120 may sweep a set of different TX beams. For beamforming at a wireless device 110, a wireless device 110 may sweep a set of different Rx beams. In an example, a P-2 procedure 920 may be used to enable a wireless device 110 to measure one or more Tx beams associated with a base station 120 to possibly change a first set of Tx beams associated with a base station 120. A P-2 procedure 920 may be performed on a possibly smaller set of beams for beam refinement than in the P-1 procedure 910. A P-2 procedure 920 may be a special case of a P-1 procedure 910. In an example, a P-3 procedure 930 may be used to enable a wireless device 110 to measure at least one Tx beam associated with a base station 120 to change a first set of Rx beams associated with a wireless device 110.

A wireless device 110 may transmit one or more beam management reports to a base station 120. In one or more beam management reports, a wireless device 110 may indicate some beam pair quality parameters, comprising at least, one or more beam identifications; RSRP; Precoding Matrix Indicator (PMI)/Channel Quality Indicator (CQI)/Rank Indicator (RI) of a subset of configured beams. Based on one or more beam management reports, a base station 120 may transmit to a wireless device 110 a signal indicating that one or more beam pair links are one or more serving beams. A base station 120 may transmit PDCCH and PDSCH for a wireless device 110 using one or more serving beams.

In an example embodiment, new radio network may support a Bandwidth Adaptation (BA). In an example, receive and/or transmit bandwidths configured by an UE employing a BA may not be large. For example, a receive and/or transmit bandwidths may not be as large as a bandwidth of a cell. Receive and/or transmit bandwidths may be adjustable. For example, a UE may change receive and/or transmit bandwidths, e.g., to shrink during period of low activity to save power. For example, a UE may change a location of receive and/or transmit bandwidths in a frequency domain, e.g. to increase scheduling flexibility. For example, a UE may change a subcarrier spacing, e.g. to allow different services.

In an example embodiment, a subset of a total cell bandwidth of a cell may be referred to as a Bandwidth Part (BWP). A base station may configure a UE with one or more BWPs to achieve a BA. For example, a base station may indicate, to a UE, which of the one or more (configured) BWPs is an active BWP.

FIG. 10 is an example diagram of 3 BWPs configured: BWP1 (1010 and 1050) with a width of 40 MHz and subcarrier spacing of 15 kHz; BWP2 (1020 and 1040) with a width of 10 MHz and subcarrier spacing of 15 kHz; BWP3 1030 with a width of 20 MHz and subcarrier spacing of 60 KHz.

In an example, a UE, configured for operation in one or more BWPs of a cell, may be configured by one or more higher layers (e.g. RRC layer) for a cell a set of one or more BWPs (e.g., at most four BWPs) for receptions by the UE (DL BWP set) in a DL bandwidth by at least one parameter DL-BWP and a set of one or more BWPs (e.g., at most four BWPs) for transmissions by a UE (UL BWP set) in a UL bandwidth by at least one parameter UL-BWP for a cell.

To enable BA on the PCell, a base station may configure a UE with one or more UL and DL BWP pairs. To enable BA on SCCells (e.g., in case of CA), a base station may configure a UE at least with one or more DL BWPs (e.g., there may be none in an UL).

In an example, an initial active DL BWP may be defined by at least one of a location and number of contiguous PRBs, a subcarrier spacing, or a cyclic prefix, for a control resource set for at least one common search space. For operation on the PCell, one or more higher layer parameters may indicate at least one initial UL BWP for a random access procedure. If a UE is configured with a secondary carrier on a primary cell, the UE may be configured with an initial BWP for random access procedure on a secondary carrier.

In an example, for unpaired spectrum operation, a UE may expect that a center frequency for a DL BWP may be same as a center frequency for a UL BWP.

For example, for a DL BWP or an UL BWP in a set of one or more DL BWPs or one or more UL BWPs, respectively, a base station may semi-statistically configure a UE for a cell with one or more parameters indicating at least one of the following: a subcarrier spacing; a cyclic prefix; a number of contiguous PRBs; an index in the set of one or more DL BWPs and/or one or more UL BWPs; a link between a DL BWP and an UL BWP from a set of configured DL BWPs and UL BWPs; a DCI detection to a PDSCH reception timing; a PDSCH reception to a HARQ-ACK transmission timing value; a DCI detection to a PUSCH transmission timing value; an offset of a first PRB of a DL bandwidth or an UL bandwidth, respectively, relative to a first PRB of a bandwidth.

In an example, for a DL BWP in a set of one or more DL BWPs on a PCell, a base station may configure a UE with one or more control resource sets for at least one type of common search space and/or one UE-specific search space. For example, a base station may not configure a UE without a common search space on a PCell, or on a PSCell, in an active DL BWP.

For an UL BWP in a set of one or more UL BWPs, a base station may configure a UE with one or more resource sets for one or more PUCCH transmissions.

In an example, if a DCI comprises a BWP indicator field, a BWP indicator field value may indicate an active DL BWP, from a configured DL BWP set, for one or more DL receptions. If a DCI comprises a BWP indicator field, a BWP indicator field value may indicate an active UL BWP, from a configured UL BWP set, for one or more UL transmissions.

In an example, for a PCell, a base station may semi-statistically configure a UE with a default DL BWP among configured DL BWPs. If a UE is not provided a default DL BWP, a default BWP may be an initial active DL BWP.

In an example, a base station may configure a UE with a timer value for a PCell. For example, a UE may start a timer, referred to as BWP inactivity timer, when a UE detects a DCI indicating an active DL BWP, other than a default DL BWP, for a paired spectrum operation or when a UE detects a DCI indicating an active DL BWP or UL BWP, other than a default DL BWP or UL BWP, for an unpaired spectrum operation. The UE may increment the timer by an interval of a first value (e.g., the first value may be 1 millisecond or 0.5 milliseconds) if the UE does not detect a DCI during the interval for a paired spectrum operation or for an unpaired spectrum operation. In an example, the timer may expire when the timer is equal to the timer value. A UE may switch to the default DL BWP from an active DL BWP when the timer expires.

In an example, a base station may semi-statistically configure a UE with one or more BWPs. A UE may switch an active BWP from a first BWP to a second BWP in response to receiving a DCI indicating the second BWP as an active BWP and/or in response to an expiry of BWP inactivity timer (for example, the second BWP may be a default BWP). For example, FIG. 10 is an example diagram of 3 BWPs configured, BWP1 (1010 and 1050), BWP2 (1020 and 1040), and BWP3 (1030). BWP2 (1020 and 1040) may be a default BWP. BWP1 (1010) may be an initial active BWP. In an example, a UE may switch an active BWP from BWP1 1010 to BWP2 1020 in response to an expiry of BWP inactivity timer. For example, a UE may switch an active BWP from BWP2 1020 to BWP3 1030 in response to receiving a DCI indicating BWP3 1030 as an active BWP. Switching an active BWP from BWP3 1030 to BWP2 1040 and/or from BWP2 1040 to BWP1 1050 may be in response to receiving a DCI indicating an active BWP and/or in response to an expiry of BWP inactivity timer.

In an example, if a UE is configured for a secondary cell with a default DL BWP among configured DL BWPs and a timer value, UE procedures on a secondary cell may be same as on a primary cell using the timer value for the secondary cell and the default DL BWP for the secondary cell.

In an example, if a base station configures a UE with a first active DL BWP and a first active UL BWP on a secondary cell or carrier, a UE may employ an indicated DL BWP and an indicated UL BWP on a secondary cell as a respective first active DL BWP and first active UL BWP on a secondary cell or carrier.

FIG. 11A and FIG. 11B show packet flows employing a multi connectivity (e.g. dual connectivity, multi connectivity, tight interworking, and/or the like). FIG. 11A is an example diagram of a protocol structure of a wireless device 110 (e.g. UE) with CA and/or multi connectivity as per an aspect of an embodiment. FIG. 11B is an example diagram of a protocol structure of multiple base stations with CA and/or multi connectivity as per an aspect of an embodiment. The multiple base stations may comprise a master node, MN 1130 (e.g. a master node, a master base station, a master gNB, a master eNB, and/or the like) and a secondary node, SN 1150 (e.g. a secondary node, a secondary base station, a secondary gNB, a secondary eNB, and/or the like). A master node 1130 and a secondary node 1150 may co-work to communicate with a wireless device 110.

When multi connectivity is configured for a wireless device 110, the wireless device 110, which may support multiple reception/transmission functions in an RRC connected state, may be configured to utilize radio resources provided by multiple schedulers of a multiple base stations. Multiple base stations may be inter-connected via a non-ideal or ideal backhaul (e.g. Xn interface, X2 interface,

and/or the like). A base station involved in multi connectivity for a certain wireless device may perform at least one of two different roles: a base station may either act as a master base station or as a secondary base station. In multi connectivity, a wireless device may be connected to one master base station and one or more secondary base stations. In an example, a master base station (e.g. the MN 1130) may provide a master cell group (MCG) comprising a primary cell and/or one or more secondary cells for a wireless device (e.g. the wireless device 110). A secondary base station (e.g. the SN 1150) may provide a secondary cell group (SCG) comprising a primary secondary cell (PSCell) and/or one or more secondary cells for a wireless device (e.g. the wireless device 110).

In multi connectivity, a radio protocol architecture that a bearer employs may depend on how a bearer is setup. In an example, three different type of bearer setup options may be supported: an MCG bearer, an SCG bearer, and/or a split bearer. A wireless device may receive/transmit packets of an MCG bearer via one or more cells of the MCG, and/or may receive/transmits packets of an SCG bearer via one or more cells of an SCG. Multi-connectivity may also be described as having at least one bearer configured to use radio resources provided by the secondary base station. Multi-connectivity may or may not be configured/implemented in some of the example embodiments.

In an example, a wireless device (e.g. Wireless Device 110) may transmit and/or receive: packets of an MCG bearer via an SDAP layer (e.g. SDAP 1110), a PDCP layer (e.g. NR PDCP 1111), an RLC layer (e.g. MN RLC 1114), and a MAC layer (e.g. MN MAC 1118); packets of a split bearer via an SDAP layer (e.g. SDAP 1110), a PDCP layer (e.g. NR PDCP 1112), one of a master or secondary RLC layer (e.g. MN RLC 1115, SN RLC 1116), and one of a master or secondary MAC layer (e.g. MN MAC 1118, SN MAC 1119); and/or packets of an SCG bearer via an SDAP layer (e.g. SDAP 1110), a PDCP layer (e.g. NR PDCP 1113), an RLC layer (e.g. SN RLC 1117), and a MAC layer (e.g. MN MAC 1119).

In an example, a master base station (e.g. MN 1130) and/or a secondary base station (e.g. SN 1150) may transmit/receive: packets of an MCG bearer via a master or secondary node SDAP layer (e.g. SDAP 1120, SDAP 1140), a master or secondary node PDCP layer (e.g. NR PDCP 1121, NR PDCP 1142), a master node RLC layer (e.g. MN RLC 1124, MN RLC 1125), and a master node MAC layer (e.g. MN MAC 1128); packets of an SCG bearer via a master or secondary node SDAP layer (e.g. SDAP 1120, SDAP 1140), a master or secondary node PDCP layer (e.g. NR PDCP 1122, NR PDCP 1143), a secondary node RLC layer (e.g. SN RLC 1146, SN RLC 1147), and a secondary node MAC layer (e.g. SN MAC 1148); packets of a split bearer via a master or secondary node SDAP layer (e.g. SDAP 1120, SDAP 1140), a master or secondary node PDCP layer (e.g. NR PDCP 1123, NR PDCP 1141), a master or secondary node RLC layer (e.g. MN RLC 1126, SN RLC 1144, SN RLC 1145, MN RLC 1127), and a master or secondary node MAC layer (e.g. MN MAC 1128, SN MAC 1148).

In multi connectivity, a wireless device may configure multiple MAC entities: one MAC entity (e.g. MN MAC 1118) for a master base station, and other MAC entities (e.g. SN MAC 1119) for a secondary base station. In multi-connectivity, a configured set of serving cells for a wireless device may comprise two subsets: an MCG comprising serving cells of a master base station, and SCGs comprising serving cells of a secondary base station. For an SCG, one or more of following configurations may be applied: at least one cell of an SCG has a configured UL CC and at least one

cell of a SCG, named as primary secondary cell (PSCell, PCell of SCG, or sometimes called PCell), is configured with PUCCH resources; when an SCG is configured, there may be at least one SCG bearer or one Split bearer; upon detection of a physical layer problem or a random access problem on a PSCell, or a number of NR RLC retransmissions has been reached associated with the SCG, or upon detection of an access problem on a PSCell during a SCG addition or a SCG change: an RRC connection re-establishment procedure may not be triggered, UL transmissions towards cells of an SCG may be stopped, a master base station may be informed by a wireless device of a SCG failure type, for split bearer, a DL data transfer over a master base station may be maintained; an NR RLC acknowledged mode (AM) bearer may be configured for a split bearer; PCell and/or PSCell may not be de-activated; PSCell may be changed with a SCG change procedure (e.g. with security key change and a RACH procedure); and/or a bearer type change between a split bearer and a SCG bearer or simultaneous configuration of a SCG and a split bearer may or may not be supported.

With respect to interaction between a master base station and a secondary base stations for multi-connectivity, one or more of the following may be applied: a master base station and/or a secondary base station may maintain RRM measurement configurations of a wireless device; a master base station may (e.g. based on received measurement reports, traffic conditions, and/or bearer types) may decide to request a secondary base station to provide additional resources (e.g. serving cells) for a wireless device; upon receiving a request from a master base station, a secondary base station may create/modify a container that may result in configuration of additional serving cells for a wireless device (or decide that the secondary base station has no resource available to do so); for a UE capability coordination, a master base station may provide (a part of) an AS configuration and UE capabilities to a secondary base station; a master base station and a secondary base station may exchange information about a UE configuration by employing of RRC containers (inter-node messages) carried via Xn messages; a secondary base station may initiate a reconfiguration of the secondary base station existing serving cells (e.g. PUCCH towards the secondary base station); a secondary base station may decide which cell is a PSCell within a SCG; a master base station may or may not change content of RRC configurations provided by a secondary base station; in case of a SCG addition and/or a SCG SCell addition, a master base station may provide recent (or the latest) measurement results for SCG cell(s); a master base station and secondary base stations may receive information of SFN and/or subframe offset of each other from OAM and/or via an Xn interface, (e.g. for a purpose of DRX alignment and/or identification of a measurement gap). In an example, when adding a new SCG SCell, dedicated RRC signaling may be used for sending required system information of a cell as for CA, except for a SFN acquired from a MIB of a PSCell of a SCG.

FIG. 12 is an example diagram of a random access procedure. One or more events may trigger a random access procedure. For example, one or more events may be at least one of following: initial access from RRC_IDLE, RRC connection re-establishment procedure, handover, DL or UL data arrival during RRC_CONNECTED when UL synchronization status is non-synchronized, transition from RRC_Inactive, and/or request for other system information. For example, a PDCCH order, a MAC entity, and/or a beam failure indication may initiate a random access procedure.

In an example embodiment, a random access procedure may be at least one of a contention based random access procedure and a contention free random access procedure. For example, a contention based random access procedure may comprise, one or more Msg 1 1220 transmissions, one or more Msg2 1230 transmissions, one or more Msg3 1240 transmissions, and contention resolution 1250. For example, a contention free random access procedure may comprise one or more Msg 1 1220 transmissions and one or more Msg2 1230 transmissions.

In an example, a base station may transmit (e.g., unicast, multicast, or broadcast), to a UE, a RACH configuration 1210 via one or more beams. The RACH configuration 1210 may comprise one or more parameters indicating at least one of the following: available set of PRACH resources for a transmission of a random access preamble, initial preamble power (e.g., random access preamble initial received target power), an RSRP threshold for a selection of a SS block and corresponding PRACH resource, a power-ramping factor (e.g., random access preamble power ramping step), random access preamble index, a maximum number of preamble transmission, preamble group A and group B, a threshold (e.g., message size) to determine the groups of random access preambles, a set of one or more random access preambles for system information request and corresponding PRACH resource(s), if any, a set of one or more random access preambles for beam failure recovery request and corresponding PRACH resource(s), if any, a time window to monitor RA response(s), a time window to monitor response(s) on beam failure recovery request, and/or a contention resolution timer.

In an example, the Msg1 1220 may be one or more transmissions of a random access preamble. For a contention based random access procedure, a UE may select a SS block with a RSRP above the RSRP threshold. If random access preambles group B exists, a UE may select one or more random access preambles from a group A or a group B depending on a potential Msg3 1240 size. If a random access preambles group B does not exist, a UE may select the one or more random access preambles from a group A. A UE may select a random access preamble index randomly (e.g. with equal probability or a normal distribution) from one or more random access preambles associated with a selected group. If a base station semi-statistically configures a UE with an association between random access preambles and SS blocks, the UE may select a random access preamble index randomly with equal probability from one or more random access preambles associated with a selected SS block and a selected group.

For example, a UE may initiate a contention free random access procedure based on a beam failure indication from a lower layer. For example, a base station may semi-statistically configure a UE with one or more contention free PRACH resources for beam failure recovery request associated with at least one of SS blocks and/or CSI-RSs. If at least one of SS blocks with a RSRP above a first RSRP threshold amongst associated SS blocks or at least one of CSI-RSs with a RSRP above a second RSRP threshold amongst associated CSI-RSs is available, a UE may select a random access preamble index corresponding to a selected SS block or CSI-RS from a set of one or more random access preambles for beam failure recovery request.

For example, a UE may receive, from a base station, a random access preamble index via PDCCH or RRC for a contention free random access procedure. If a base station does not configure a UE with at least one contention free PRACH resource associated with SS blocks or CSI-RS, the

UE may select a random access preamble index. If a base station configures a UE with one or more contention free PRACH resources associated with SS blocks and at least one SS block with a RSRP above a first RSRP threshold amongst associated SS blocks is available, the UE may select the at least one SS block and select a random access preamble corresponding to the at least one SS block. If a base station configures a UE with one or more contention free PRACH resources associated with CSI-RSs and at least one CSI-RS with a RSRP above a second RSRP threshold amongst the associated CSI-RSs is available, the UE may select the at least one CSI-RS and select a random access preamble corresponding to the at least one CSI-RS.

A UE may perform one or more Msg1 1220 transmissions by transmitting the selected random access preamble. For example, if a UE selects an SS block and is configured with an association between one or more PRACH occasions and one or more SS blocks, the UE may determine an PRACH occasion from one or more PRACH occasions corresponding to a selected SS block. For example, if a UE selects a CSI-RS and is configured with an association between one or more PRACH occasions and one or more CSI-RSs, the UE may determine a PRACH occasion from one or more PRACH occasions corresponding to a selected CSI-RS. A UE may transmit, to a base station, a selected random access preamble via a selected PRACH occasions. A UE may determine a transmit power for a transmission of a selected random access preamble at least based on an initial preamble power and a power-ramping factor. A UE may determine a RA-RNTI associated with a selected PRACH occasions in which a selected random access preamble is transmitted. For example, a UE may not determine a RA-RNTI for a beam failure recovery request. A UE may determine an RA-RNTI at least based on an index of a first OFDM symbol and an index of a first slot of a selected PRACH occasions, and/or an uplink carrier index for a transmission of Msg1 1220.

In an example, a UE may receive, from a base station, a random access response, Msg 2 1230. A UE may start a time window (e.g., ra-ResponseWindow) to monitor a random access response. For beam failure recovery request, a base station may configure a UE with a different time window (e.g., bfr-ResponseWindow) to monitor response on beam failure recovery request. For example, a UE may start a time window (e.g., ra-Response Window or bfr-Response Window) at a start of a first PDCCH occasion after a fixed duration of one or more symbols from an end of a preamble transmission. If a UE transmits multiple preambles, the UE may start a time window at a start of a first PDCCH occasion after a fixed duration of one or more symbols from an end of a first preamble transmission. A UE may monitor a PDCCH of a cell for at least one random access response identified by a RA-RNTI or for at least one response to beam failure recovery request identified by a C-RNTI while a timer for a time window is running.

In an example, a UE may consider a reception of random access response successful if at least one random access response comprises a random access preamble identifier corresponding to a random access preamble transmitted by the UE. A UE may consider the contention free random access procedure successfully completed if a reception of random access response is successful. If a contention free random access procedure is triggered for a beam failure recovery request, a UE may consider a contention free random access procedure successfully complete if a PDCCH transmission is addressed to a C-RNTI. In an example, if at least one random access response comprises only a random access preamble identifier, a UE may consider the random

access procedure successfully completed and may indicate a reception of an acknowledgement for a system information request to upper layers. If a UE has signaled multiple preamble transmissions, the UE may stop transmitting remaining preambles (if any) in response to a successful reception of a corresponding random access response.

In an example, a UE may perform one or more Msg 3 **1240** transmissions in response to a successful reception of random access response (e.g., for a contention based random access procedure). A UE may adjust an uplink transmission timing based on a timing advanced command indicated by a random access response and may transmit one or more transport blocks based on an uplink grant indicated by a random access response. Subcarrier spacing for PUSCH transmission for Msg3 **1240** may be provided by at least one higher layer (e.g. RRC) parameter. A UE may transmit a random access preamble via PRACH and Msg3 **1240** via PUSCH on a same cell. A base station may indicate an UL BWP for a PUSCH transmission of Msg3 **1240** via system information block. A UE may employ HARQ for a retransmission of Msg 3 **1240**.

In an example, multiple UEs may perform Msg 1 **1220** by transmitting a same preamble to a base station and receive, from the base station, a same random access response comprising an identity (e.g., TC-RNTI). Contention resolution **1250** may ensure that a UE does not incorrectly use an identity of another UE. For example, contention resolution **1250** may be based on C-RNTI on PDCCH or a UE contention resolution identity on DL-SCH. For example, if a base station assigns a C-RNTI to a UE, the UE may perform contention resolution **1250** based on a reception of a PDCCH transmission that is addressed to the C-RNTI. In response to detection of a C-RNTI on a PDCCH, a UE may consider contention resolution **1250** successful and may consider a random access procedure successfully completed. If a UE has no valid C-RNTI, a contention resolution may be addressed by employing a TC-RNTI. For example, if a MAC PDU is successfully decoded and a MAC PDU comprises a UE contention resolution identity MAC CE that matches the CCCH SDU transmitted in Msg3 **1250**, a UE may consider the contention resolution **1250** successful and may consider the random access procedure successfully completed.

FIG. **13** is an example structure for MAC entities as per an aspect of an embodiment. In an example, a wireless device may be configured to operate in a multi-connectivity mode. A wireless device in RRC_CONNECTED with multiple RX/TX may be configured to utilize radio resources provided by multiple schedulers located in a plurality of base stations. The plurality of base stations may be connected via a non-ideal or ideal backhaul over the Xn interface. In an example, a base station in a plurality of base stations may act as a master base station or as a secondary base station. A wireless device may be connected to one master base station and one or more secondary base stations. A wireless device may be configured with multiple MAC entities, e.g. one MAC entity for master base station, and one or more other MAC entities for secondary base station(s). In an example, a configured set of serving cells for a wireless device may comprise two subsets: an MCG comprising serving cells of a master base station, and one or more SCGs comprising serving cells of a secondary base station(s). FIG. **13** illustrates an example structure for MAC entities when MCG and SCG are configured for a wireless device.

In an example, at least one cell in a SCG may have a configured UL CC, wherein a cell of at least one cell may be called PSCell or PCell of SCG, or sometimes may be simply

called PCell. A PSCell may be configured with PUCCH resources. In an example, when a SCG is configured, there may be at least one SCG bearer or one split bearer. In an example, upon detection of a physical layer problem or a random access problem on a PSCell, or upon reaching a number of RLC retransmissions associated with the SCG, or upon detection of an access problem on a PSCell during a SCG addition or a SCG change: an RRC connection re-establishment procedure may not be triggered, UL transmissions towards cells of an SCG may be stopped, a master base station may be informed by a UE of a SCG failure type and DL data transfer over a master base station may be maintained.

In an example, a MAC sublayer may provide services such as data transfer and radio resource allocation to upper layers (e.g. **1310** or **1320**). A MAC sublayer may comprise a plurality of MAC entities (e.g. **1350** and **1360**). A MAC sublayer may provide data transfer services on logical channels. To accommodate different kinds of data transfer services, multiple types of logical channels may be defined. A logical channel may support transfer of a particular type of information. A logical channel type may be defined by what type of information (e.g., control or data) is transferred. For example, BCCH, PCCH, CCCH and DCCH may be control channels and DTCH may be a traffic channel. In an example, a first MAC entity (e.g. **1310**) may provide services on PCCH, BCCH, CCCH, DCCH, DTCH and MAC control elements. In an example, a second MAC entity (e.g. **1320**) may provide services on BCCH, DCCH, DTCH and MAC control elements.

A MAC sublayer may expect from a physical layer (e.g. **1330** or **1340**) services such as data transfer services, signaling of HARQ feedback, signaling of scheduling request or measurements (e.g. CQI). In an example, in dual connectivity, two MAC entities may be configured for a wireless device: one for MCG and one for SCG. A MAC entity of wireless device may handle a plurality of transport channels. In an example, a first MAC entity may handle first transport channels comprising a PCCH of MCG, a first BCH of MCG, one or more first DL-SCHs of MCG, one or more first UL-SCHs of MCG and one or more first RACHs of MCG. In an example, a second MAC entity may handle second transport channels comprising a second BCH of SCG, one or more second DL-SCHs of SCG, one or more second UL-SCHs of SCG and one or more second RACHs of SCG.

In an example, if a MAC entity is configured with one or more SCGs, there may be multiple DL-SCHs and there may be multiple UL-SCHs as well as multiple RACHs per MAC entity. In an example, there may be one DL-SCH and UL-SCH on a SpCell. In an example, there may be one DL-SCH, zero or one UL-SCH and zero or one RACH for an SCG. A DL-SCH may support receptions using different numerologies and/or TTI duration within a MAC entity. A UL-SCH may also support transmissions using different numerologies and/or TTI duration within the MAC entity.

In an example, a MAC sublayer may support different functions and may control these functions with a control (e.g. **1355** or **1365**) element. Functions performed by a MAC entity may comprise mapping between logical channels and transport channels (e.g., in uplink or downlink), multiplexing (e.g. **1352** or **1362**) of MAC SDUs from one or different logical channels onto transport blocks (TB) to be delivered to the physical layer on transport channels (e.g., in uplink), demultiplexing (e.g. **1352** or **1362**) of MAC SDUs to one or different logical channels from transport blocks (TB) delivered from the physical layer on transport channels (e.g., in

downlink), scheduling information reporting (e.g., in uplink), error correction through HARQ in uplink or downlink (e.g. **1363**), and logical channel prioritization in uplink (e.g. **1351** or **1361**). A MAC entity may handle a random access process (e.g. **1354** or **1364**).

FIG. **14** is an example diagram of a RAN architecture comprising one or more base stations. In an example, a protocol stack (e.g. RRC, SDAP, PDCP, RLC, MAC, and PHY) may be supported at a node. A base station (e.g. gNB **120A** or **120B**) may comprise a base station central unit (CU) (e.g. gNB-CU **1420A** or **1420B**) and at least one base station distributed unit (DU) (e.g. gNB-DU **1430A**, **1430B**, **1430C**, or **1430D**) if a functional split is configured. Upper protocol layers of a base station may be located in a base station CU, and lower layers of the base station may be located in the base station DUs. An F1 interface (e.g. CU-DU interface) connecting a base station CU and base station DUs may be an ideal or non-ideal backhaul. F1-C may provide a control plane connection over an F1 interface, and F1-U may provide a user plane connection over the F1 interface. In an example, an Xn interface may be configured between base station CUs.

In an example, a base station CU may comprise an RRC function, an SDAP layer, and a PDCP layer, and base station DUs may comprise an RLC layer, a MAC layer, and a PHY layer. In an example, various functional split options between a base station CU and base station DUs may be possible by locating different combinations of upper protocol layers (RAN functions) in a base station CU and different combinations of lower protocol layers (RAN functions) in base station DUs. A functional split may support flexibility to move protocol layers between a base station CU and base station DUs depending on service requirements and/or network environments.

In an example, functional split options may be configured per base station, per base station CU, per base station DU, per UE, per bearer, per slice, or with other granularities. In per base station CU split, a base station CU may have a fixed split option, and base station DUs may be configured to match a split option of a base station CU. In per base station DU split, a base station DU may be configured with a different split option, and a base station CU may provide different split options for different base station DUs. In per UE split, a base station (base station CU and at least one base station DUs) may provide different split options for different wireless devices. In per bearer split, different split options may be utilized for different bearers. In per slice split, different split options may be applied for different slices.

FIG. **15** is an example diagram showing RRC state transitions of a wireless device. In an example, a wireless device may be in at least one RRC state among an RRC connected state (e.g. RRC Connected **1530**, RRC_Connected), an RRC idle state (e.g. RRC Idle **1510**, RRC_Idle), and/or an RRC inactive state (e.g. RRC Inactive **1520**, RRC_Inactive). In an example, in an RRC connected state, a wireless device may have at least one RRC connection with at least one base station (e.g. gNB and/or eNB), which may have a UE context of the wireless device. A UE context (e.g. a wireless device context) may comprise at least one of an access stratum context, one or more radio link configuration parameters, bearer (e.g. data radio bearer (DRB)), signaling radio bearer (SRB), logical channel, QoS flow, PDU session, and/or the like configuration information, security information, PHY/MAC/RLC/PDCP/SDAP layer configuration information, and/or the like configuration information for a wireless device. In an example, in an RRC idle state, a wireless device may not have an RRC connection

with a base station, and a UE context of a wireless device may not be stored in a base station. In an example, in an RRC inactive state, a wireless device may not have an RRC connection with a base station. A UE context of a wireless device may be stored in a base station, which may be called as an anchor base station (e.g. last serving base station).

In an example, a wireless device may transition a UE RRC state between an RRC idle state and an RRC connected state in both ways (e.g. connection release **1540** or connection establishment **1550**; or connection reestablishment) and/or between an RRC inactive state and an RRC connected state in both ways (e.g. connection inactivation **1570** or connection resume **1580**). In an example, a wireless device may transition its RRC state from an RRC inactive state to an RRC idle state (e.g. connection release **1560**).

In an example, an anchor base station may be a base station that may keep a UE context (a wireless device context) of a wireless device at least during a time period that a wireless device stays in a RAN notification area (RNA) of an anchor base station, and/or that a wireless device stays in an RRC inactive state. In an example, an anchor base station may be a base station that a wireless device in an RRC inactive state was lastly connected to in a latest RRC connected state or that a wireless device lastly performed an RNA update procedure in. In an example, an RNA may comprise one or more cells operated by one or more base stations. In an example, a base station may belong to one or more RNAs. In an example, a cell may belong to one or more RNAs.

In an example, a wireless device may transition a UE RRC state from an RRC connected state to an RRC inactive state in a base station. A wireless device may receive RNA information from the base station. RNA information may comprise at least one of an RNA identifier, one or more cell identifiers of one or more cells of an RNA, a base station identifier, an IP address of the base station, an AS context identifier of the wireless device, a resume identifier, and/or the like.

In an example, an anchor base station may broadcast a message (e.g. RAN paging message) to base stations of an RNA to reach to a wireless device in an RRC inactive state, and/or the base stations receiving the message from the anchor base station may broadcast and/or multicast another message (e.g. paging message) to wireless devices in their coverage area, cell coverage area, and/or beam coverage area associated with the RNA through an air interface.

In an example, when a wireless device in an RRC inactive state moves into a new RNA, the wireless device may perform an RNA update (RNAU) procedure, which may comprise a random access procedure by the wireless device and/or a UE context retrieve procedure. A UE context retrieve may comprise: receiving, by a base station from a wireless device, a random access preamble; and fetching, by a base station, a UE context of the wireless device from an old anchor base station. Fetching may comprise: sending a retrieve UE context request message comprising a resume identifier to the old anchor base station and receiving a retrieve UE context response message comprising the UE context of the wireless device from the old anchor base station.

In an example embodiment, a wireless device in an RRC inactive state may select a cell to camp on based on at least a measurement results for one or more cells, a cell where a wireless device may monitor an RNA paging message and/or a core network paging message from a base station. In an example, a wireless device in an RRC inactive state

may select a cell to perform a random access procedure to resume an RRC connection and/or to transmit one or more packets to a base station (e.g. to a network). In an example, if a cell selected belongs to a different RNA from an RNA for a wireless device in an RRC inactive state, the wireless device may initiate a random access procedure to perform an RNA update procedure. In an example, if a wireless device in an RRC inactive state has one or more packets, in a buffer, to transmit to a network, the wireless device may initiate a random access procedure to transmit one or more packets to a base station of a cell that the wireless device selects. A random access procedure may be performed with two messages (e.g. 2 stage random access) and/or four messages (e.g. 4 stage random access) between the wireless device and the base station.

In an example embodiment, a base station receiving one or more uplink packets from a wireless device in an RRC inactive state may fetch a UE context of a wireless device by transmitting a retrieve UE context request message for the wireless device to an anchor base station of the wireless device based on at least one of an AS context identifier, an RNA identifier, a base station identifier, a resume identifier, and/or a cell identifier received from the wireless device. In response to fetching a UE context, a base station may transmit a path switch request for a wireless device to a core network entity (e.g. AMF, MME, and/or the like). A core network entity may update a downlink tunnel endpoint identifier for one or more bearers established for the wireless device between a user plane core network entity (e.g. UPF, S-GW, and/or the like) and a RAN node (e.g. the base station), e.g. changing a downlink tunnel endpoint identifier from an address of the anchor base station to an address of the base station.

A channel status of a wireless device may vary depending on a bandwidth part (BWP) that the wireless device is employing. A radio link connectivity of a wireless device may depend on a channel status, and further may depend on a BWP that the wireless device is employing. In an implementation of existing technologies, a wireless device may provide information of a failed cell when reporting a connection failure to a base station. A base station and/or a failed base station may configure cell configuration parameters based on the information of the failed cell. The failed base station may configure, based on the information of the failed cell, whole BWP configuration parameters of the failed cell including parameters of BWPs that are reliable. Configuring whole BWP configuration parameters when a certain BWP has a problem is inefficient. The inefficient BWP configuration may decrease resource utilization efficiency and radio connection reliability of wireless devices.

Example embodiments may enhance a radio link failure report by providing BWP information (for example, information of a failed BWP), so that a base station is able to identify a cause (for example, a failed BWP) of a connection failure. A base station may employ BWP information of a radio link failure report to update radio configuration parameters to improve uplink/downlink connections.

In an example embodiment, a wireless device (e.g. UE) may experience a connection failure (e.g. radio link failure (RLF) and/or handover failure (HOF)) at a first cell of a first base station (e.g. gNB, eNB) during a time that the wireless device is employing a certain bandwidth part (BWP) of a plurality of BWPs configured by the first base station. The wireless device may report the connection failure to a second base station to which the wireless device (re) establishes a radio resource control (RRC) connection. The wireless device may send, to the second base station, infor-

mation of the certain BWP, the plurality of BWPs, and/or BWP configuration parameters of the first cell when reporting the connection failure. In an example, the second base station may be the first base station. If the second base station is different from the first base station, the first base station may receive the information from the second base station. The first base station may determine one or more radio resource configuration parameters for one or more wireless devices based on the information.

In an example, a channel status of a wireless device may vary according to which BWP the wireless device is employing for transmission and/or reception. A radio link status of a wireless device may depend on a channel status, and further may depend on BWP that the wireless device is employing. In a legacy system, a wireless device does not provide a BWP information when reporting a connection failure to a base station. A base station may not know what BWP was employed when a wireless device experienced a connection failure, and what BWP the wireless device had a problem with. A base station may not be able to identify a cause of wireless device's connection failure without information of BWP configuration employed when a wireless device experiences a connection failure. An example embodiment may enhance an RLF report by providing BWP information, so that a base station is able to identify a cause of wireless device's connection failure. A base station may employ BWP information of an RLF report to update one or more radio link configuration parameters to improve uplink/downlink connections.

Connection Failure

In an example, a wireless device in an RRC connected state may experience (and/or determine) a connection failure from at least one cell of a base station. A connection failure may comprise a radio link failure (RLF) and/or a handover failure (HOF).

In an example, a wireless device may determine an RLF when one of following criteria are met: expiry of a timer started after indication of radio problems (e.g. out of sync) from a physical layer (if radio problems are recovered before the timer is expired, the UE may stop the timer); random access procedure failure; RLC failure (e.g. a number of retransmission is over a threshold); and/or the like. In an example, after the RLF is declared, the wireless device may: stay in an RRC_CONNECTED state (e.g. RRC connected state); select a suitable cell and initiate RRC re-establishment; enter an RRC_IDLE state (e.g. RRC idle state) if a suitable cell wasn't found within a certain time after the RLF was declared; stay at an RRC_INACTIVE state (e.g. RRC inactive state) if a suitable cell wasn't found within a certain time after the RLF was declared; and/or the like.

In an example, a behavior of a wireless device associated with the RLF may comprise two phases. A first phase may be started upon a radio problem detection; may lead to the RLF detection; may not initiate a UE based mobility; may be based on a timer (e.g. T1) or other criteria (e.g. counting); and/or the like. A second phase may be started upon the RLF or the HOF; may leads to an RRC idle state of the wireless device; may initiate a UE based mobility; may be based on a timer (e.g. T2); and/or the like. In an example, the wireless device in a normal operation of an RRC connected state may experience the first phase comprising the radio problem detection and no recovery during the T1, and/or may experience the second phase comprising no recovery during the T2. The second phase may be followed by an RRC idle state (e.g. the wireless device may go to an RRC idle state). During the normal operation, the first phase, and/or the second phase, the wireless device may be considered as

being in an RRC connected state. In an example, the RLF may be considered as occurring between the first phase and the second phase (e.g. at an ending time of the first phase and/or at the starting time of the second phase).

In an example, upon detecting physical layer problems for a PCell, i.e. upon receiving N310 consecutive out-of-sync indications from lower layers, a wireless device may start a timer, T310. Upon receiving N311 consecutive in-sync indications from lower layers for the PCell, and/or upon triggering a handover procedure and upon initiating a connection re-establishment procedure, a wireless device may stop T310. In an example, when a timer, T310, is expired, if security is not activated, a wireless device may go to RRC_IDLE state; and/or if security is activated, a wireless device may initiate a connection re-establishment procedure.

In an example, upon initiating an RRC connection re-establishment procedure, a wireless device may start a timer, T311. Upon selecting a suitable cell of LTE/5G/another RAT, a wireless device may stop T311. If T311 is expired, a wireless device may enter an RRC_IDLE state.

In an example, a wireless device may detect the HOF that may occur due to a too early handover, a too late handover, a handover to wrong cell, and/or the like. In case of the too early handover, a radio link failure may occur shortly after a successful handover from a source cell to a target cell, and/or a handover failure may occur during a handover procedure. In case of the too late handover, the wireless device may attempt to reestablish a radio link connection in the source cell. In case of the too late handover, a radio link failure may occur after the wireless device may stay for a long period of time in a cell (e.g. a radio link failure may occur because a serving base station may not initiate a handover for the wireless device when a radio link quality is not good enough to serve the wireless device in a current cell), and/or the wireless device may attempt to reestablish a radio link connection in a different cell. In case of the handover to wrong cell, a radio link failure may occur shortly after a successful handover from a source cell to a target cell, and/or a handover failure may occur during a handover procedure. In case of the handover to wrong cell, the UE may attempt to reestablish a radio link connection in a cell other than the source cell and the target cell. In an example, the "successful handover" may refer to wireless device's state of a successful completion of a random access (RA) procedure.

In an example, to resolve a problem of a connection failure (e.g. RLF and/or HOF), a wireless device and/or a network (e.g. a base station, a base station CU, a base station DU, a gNB, an eNB, a core network, and/or the like) may trigger one or more of following functions: a detection of the connection failure after an RRC reestablishment attempt; a detection of the connection failure after an RRC connection setup; a retrieval of information needed for problem analysis; and/or the like. Triggering of each of the functions may be optional, and/or may depend on situation and/or implementation.

Bandwidth Part

FIG. 10 is an example diagram of 3 BWPs configured: BWP1 (1010 and 1050) with a width of 40 MHz and subcarrier spacing of 15 kHz; BWP2 (1020 and 1040) with a width of 10 MHz and subcarrier spacing of 15 kHz; BWP3 1030 with a width of 20 MHz and subcarrier spacing of 60 KHz.

In an example, a UE, configured for operation in one or more BWPs of a cell, may be configured by one or more higher layers (e.g. RRC layer) for a cell a set of one or more BWPs (e.g., at most four BWPs) for receptions by the UE

(DL BWP set) in a DL bandwidth by at least one parameter DL-BWP and a set of one or more BWPs (e.g., at most four BWPs) for transmissions by a UE (UL BWP set) in an UL bandwidth by at least one parameter UL-BWP for a cell.

To enable BA on the PCell, a base station may configure a UE with one or more UL and DL BWP pairs. To enable BA on SCells (e.g., in case of CA), a base station may configure a UE at least with one or more DL BWPs (e.g., there may be none in an UL).

In an example, an initial active DL BWP may be defined by at least one of a location and number of contiguous PRBs, a subcarrier spacing, or a cyclic prefix, for a control resource set for at least one common search space. For operation on the PCell, one or more higher layer parameters may indicate at least one initial UL BWP for a random access procedure. If a UE is configured with a secondary carrier on a primary cell, the UE may be configured with an initial BWP for random access procedure on a secondary carrier.

In an example, for unpaired spectrum operation, a UE may expect that a center frequency for a DL BWP may be same as a center frequency for a UL BWP.

For example, for a DL BWP or an UL BWP in a set of one or more DL BWPs or one or more UL BWPs, respectively, a base station may semi-statistically configure a UE for a cell with one or more parameters indicating at least one of following: a subcarrier spacing; a cyclic prefix; a number of contiguous PRBs; an index in the set of one or more DL BWPs and/or one or more UL BWPs; a link between a DL BWP and an UL BWP from a set of configured DL BWPs and UL BWPs; a DCI detection to a PDSCH reception timing; a PDSCH reception to a HARQ-ACK transmission timing value; a DCI detection to a PUSCH transmission timing value; an offset of a first PRB of a DL bandwidth or an UL bandwidth, respectively, relative to a first PRB of a bandwidth.

In an example, for a DL BWP in a set of one or more DL BWPs on a PCell, a base station may configure a UE with one or more control resource sets for at least one type of common search space and/or one UE-specific search space. For example, a base station may not configure a UE without a common search space on a PCell, or on a PSCell, in an active DL BWP.

For an UL BWP in a set of one or more UL BWPs, a base station may configure a UE with one or more resource sets for one or more PUCCH transmissions.

In an example, if a DCI comprises a BWP indicator field, a BWP indicator field value may indicate an active DL BWP, from a configured DL BWP set, for one or more DL receptions. If a DCI comprises a BWP indicator field, a BWP indicator field value may indicate an active UL BWP, from a configured UL BWP set, for one or more UL transmissions.

In an example, for a PCell, a base station may semi-statistically configure a UE with a default DL BWP among configured DL BWPs. If a UE is not provided a default DL BWP, a default BWP may be an initial active DL BWP.

In an example, a base station may configure a UE with a timer value for a PCell. For example, a UE may start a timer, referred to as BWP inactivity timer, when a UE detects a DCI indicating an active DL BWP, other than a default DL BWP, for a paired spectrum operation or when a UE detects a DCI indicating an active DL BWP or UL BWP, other than a default DL BWP or UL BWP, for an unpaired spectrum operation. The UE may increment the timer by an interval of a first value (e.g., the first value may be 1 millisecond or 0.5 milliseconds) if the UE does not detect a DCI during the interval for a paired spectrum operation or for an unpaired

spectrum operation. In an example, the timer may expire when the timer is equal to the timer value. A UE may switch to the default DL BWP from an active DL BWP when the timer expires.

In an example, a base station may semi-statistically configure a UE with one or more BWPs. A UE may switch an active BWP from a first BWP to a second BWP in response to receiving a DCI indicating the second BWP as an active BWP and/or in response to an expiry of BWP inactivity timer (for example, the second BWP may be a default BWP). For example, FIG. 10 is an example diagram of 3 BWPs configured, BWP1 (1010 and 1050), BWP2 (1020 and 1040), and BWP3 (1030). BWP2 (1020 and 1040) may be a default BWP. BWP1 (1010) may be an initial active BWP. In an example, a UE may switch an active BWP from BWP1 1010 to BWP2 1020 in response to an expiry of BWP inactivity timer. For example, a UE may switch an active BWP from BWP2 1020 to BWP3 1030 in response to receiving a DCI indicating BWP3 1030 as an active BWP. Switching an active BWP from BWP3 1030 to BWP2 1040 and/or from BWP2 1040 to BWP1 1050 may be in response to receiving a DCI indicating an active BWP and/or in response to an expiry of BWP inactivity timer.

In an example, if a UE is configured for a secondary cell with a default DL BWP among configured DL BWPs and a timer value, UE procedures on a secondary cell may be same as on a primary cell using the timer value for the secondary cell and the default DL BWP for the secondary cell.

In an example, if a base station configures a UE with a first active DL BWP and a first active UL BWP on a secondary cell or carrier, a UE may employ an indicated DL BWP and an indicated UL BWP on a secondary cell as a respective first active DL BWP and first active UL BWP on a secondary cell or carrier.

Example Embodiments

In an example, as shown in FIG. 16, FIG. 17, FIG. 18 and/or FIG. 19, a first base station (e.g. gNB, cNB, gNB-CU and gNB-DU, and/or the like) may serve a wireless device at least via a first cell. The first cell may be a cell of the first base station and/or a primary cell of the wireless device. In an example, the first base station may configure one or more secondary cells for the wireless device. In an example, the first base station may transmit a first message to the wireless device. The first message may be an RRC message, e.g. RRC connection reconfiguration message, RRC connection reestablishment message, RRC connection resume message, RRC connection setup message, and/or the like. The first message may be transmitted via the first cell. In an example, the first message may comprise bandwidth part (BWP) configuration parameters for an uplink, a downlink, and/or a sidelink of the first cell. The BWP configuration parameters may indicate at least one of: a plurality of BWP indexes of a plurality of BWPs; a plurality of BWP bandwidths of the plurality of BWPs; a default BWP index of a default BWP of the plurality of BWPs; a BWP inactivity timer; and/or the like.

In an example, the plurality of BWP indexes may comprise an integer value, e.g. from 0 to 3. The plurality of BWP bandwidths may indicate bandwidths and/or frequencies (e.g. subcarriers, PRB, actual bandwidth and/or frequency values, and/or the like) for a plurality of BWPs, which may be identified based on the plurality of BWP indexes. In an example, the default BWP may be a BWP that the wireless device employs in response to expiration of the BWP inactivity timer. The wireless device may start/restart the

BWP inactivity timer when an associated BWP is activated and/or when the wireless device receives/transmits one or more transport blocks via an associated BWP.

In an example, the BWP configuration parameters may indicate at least one of: an initial BWP index of an initial BWP (e.g. initial active BWP) of the plurality of BWPs; a subcarrier spacing for the plurality of BWPs; a cyclic prefix; a number of contiguous PRBs; an index in the set of one or more DL BWPs and/or one or more UL BWPs; a link between a DL BWP and an UL BWP from a set of configured DL BWPs and UL BWPs; a DCI detection to a PDSCH reception timing value; a PDSCH reception to a HARQ-ACK transmission timing value; a DCI detection to a PUSCH transmission timing value; an offset of a first PRB of a DL bandwidth or an UL bandwidth, respectively, relative to a first PRB of a bandwidth.

In an example, the initial BWP may be a BWP that the wireless device starts employing when BWPs associated with the BWP configuration parameters are configured. In an example, the subcarrier spacing may be spacing information of BWPs (e.g. between BWPs, of OFDM subcarriers). In an example, the number of contiguous PRBs may be a number of PRBs employed/configured for a BWP. In an example, the DCI detection to a PDSCH reception timing value may indicate a time duration between detection of a DCI indicating an active BWP switching and starting of receiving (monitoring) a PDSCH associated with a BWP indicated via the DCI. In an example, the PDSCH reception to a HARQ-ACK transmission timing value may indicate a time duration between transport block reception via a PDSCH of a BWP and a HARQ-ACK transmission for a BWP. In an example, the DCI detection to a PUSCH transmission timing value may indicate a time duration between detection of a DCI indicating an active BWP switching and starting of transmitting a transport block via a PUSCH associated with a BWP indicated via the DCI.

In an example, the wireless device may transmit/receive transport blocks via at least one BWP of the plurality of BWPs configured via the BWP configuration parameters of the first message. In an example, the wireless device may transmit/receive transport blocks via the initial BWP in response to receiving the first message. In an example, the base station may transmit one or more DCIs indicating an active BWP switching. The wireless device may switch an active BWP based on the one or more DCIs and/or may transmit/receive transport blocks via an active BWP switched. In an example, upon expiration of the BWP inactivity timer, the wireless device may switch an active BWP to the default BWP, and/or transmit/receive transport blocks via the default BWP.

In an example, the wireless device may determine a connection failure in a first cell (e.g. a failed cell) of a first base station. In an example, the connection failure may comprise at least one of a radio link failure (RLF) or a handover failure (HOF). In an example the wireless device may determine the connection failure during a time duration that a first BWP of the plurality of BWPs is an active BWP. In an example, the first BWP may be an uplink BWP and/or a downlink BWP. In an example, a BWP index of the first BWP may be an integer value 2.

In an example, after determining the connection failure, the wireless device may determine (select/reselect) a second cell of a second base station based on a measurement result of the wireless device (e.g. RSRP, RSRQ). In an example, the second base station may be the first base station, or a base station different from the first base station. In an example, the second cell may be the first cell, or a cell

different from the first cell. In response to determining the second cell, the wireless device may initiate a random access procedure to make an RRC connection via the second cell. In an example, the wireless device may send, to the second base station, an RRC connection reestablishment/setup/resume request message for an RRC connection reestablishment/setup/resume request, which may comprise an indication parameter indicating that the wireless device experienced a connection failure (e.g. RLF, HOF, reconfiguration failure, other failure, and/or the like). The indication parameter may indicate a cause of the RRC connection reestablishment/setup/resume request. The indication parameter may indicate that the wireless device has a report associated with a connection failure to transmit to a base station (e.g. the second base station).

In an example, the wireless device may receive, from the second base station, an UE information request message indicating a request to transmit an RLF report to the second base station. In an example, the wireless device may transmit, to the second base station, a second message. In an example, the second message may be transmitted in response to the UE information request message. In an example, the second message may be an UE information response message (e.g. UEInformationResponse message). The second message may be an RRC message. In an example the second message may comprise a radio link failure (RLF) report for the connection failure of the wireless device.

In an example, the RLF report may comprise at least one of a cell identifier of the first cell, a first BWP index of the first BWP (e.g. a BWP index of the active BWP when the connection failure is determined, e.g. a BWP index of 2), and/or the like. In an example, the RLF report may comprise one or more elements of the BWP configuration parameters of the first message. The RLF report may comprise parameters (of the BWP configuration parameters) indicating at least one of: the plurality of BWP indexes of the plurality of BWPs; the plurality of BWP bandwidths of the plurality of BWPs; the default BWP index of a default BWP of the plurality of BWPs; the BWP inactivity timer; the initial BWP index of the initial BWP (e.g. initial active BWP) of the plurality of BWPs; the subcarrier spacing for the plurality of BWPs; the cyclic prefix; the number of contiguous PRBs; the index in the set of one or more DL BWPs and/or one or more UL BWPs; the link between a DL BWP and an UL BWP from a set of configured DL BWPs and UL BWPs; the DCI detection to a PDSCH reception timing value; the PDSCH reception to a HARQ-ACK transmission timing value; the DCI detection to a PUSCH transmission timing value; the offset of a first PRB of a DL bandwidth or an UL bandwidth, respectively, relative to a first PRB of a bandwidth; and/or the like.

In an example, the RLF report may comprise one or more report IEs indicating information associated with the connection failure of the wireless device. The one or more report IEs may indicate at least one of: a failed cell identifier of the failed cell (e.g. the first cell), wherein the wireless device experienced the connection failure at the failed cell; an RLF cause of the connection failure, wherein the RLF cause indicating at least one of a t310 expiry, a random access problem, a radio link control (RLC) maximum number of retransmissions, or t312 expiry; a wireless device identifier of the wireless device at the failed cell; a carrier frequency value of the failed cell; a first time value (e.g. timeConn-Failure) indicating time that elapsed since a last handover initialization for the wireless device until the connection failure; a second time value (e.g. timeSinceFailure) indicat-

ing time that elapsed since the connection failure (or an establishment failure); a connection failure type indicating whether the connection failure is due to an RLF or a handover failure; a measurement result, the measurement result comprising at least one of a reference signal received power result or a reference signal received quality; a quality classification indication 1 bearer indicating the connection failure occurred while a bearer with QCI value equal to 1 was configured; a C-RNTI of the wireless device at the first cell; and/or the like.

In an example, the failed cell identifier of the failed cell (e.g. the first cell) may comprise a global cell identifier (e.g. NCGI, ECGI, CGI, and/or the like), a physical cell identifier (e.g. PCI), and/or the like. In an example, the failed cell may be a primary cell of the wireless device when the connection failure occurred.

In an example, the RLF cause of the connection failure may indicate that a cause of determining the connection failure of the wireless device comprises at least one of a t310 expiry, a random access problem (e.g. failure), a radio link control (RLC) maximum number of retransmissions, t312 expiry, and/or the like.

In an example, the t310 expiry may indicate that a timer t310 of the wireless device expired at the failed cell when the connection failure occurred. The timer t310 may start when the wireless device detects physical layer related problems (e.g. for a primary cell) (e.g. when the wireless device receives a certain number (e.g. N310) consecutive out-of-sync indications from lower layers). In an example, the timer t310 may stop: when the wireless device receives a certain number (e.g. N311) consecutive in-sync indications from lower layers (e.g. for a primary cell); upon triggering a handover procedure; upon initiating an RRC connection reestablishment procedure; and/or the like. In an example, at an expiry of the timer t310, if security is not activated, the wireless device may go to an RRC idle state. In an example, at an expiry of the timer t310, if security is activated, the wireless device may initiate an RRC connection reestablishment procedure.

In an example, the t312 expiry may indicate that a timer t312 of the wireless device expired at the failed cell when the connection failure occurred. The timer t312 may start upon triggering a measurement report for a measurement identity for which the timer t312 has been configured, while the timer t310 is running. In an example, the timer t312 may stop: upon receiving a certain number (e.g. N311) of consecutive in-sync indications from lower layers; upon triggering a handover procedure; upon initiating a connection re-establishment procedure; upon an expiry of the timer t310; and/or the like. In an example, at an expiry of the timer t312, if security is not activated, the wireless device may go to an idle state. In an example, at an expiry of the timer t312, if security is activated, the wireless device may initiate an RRC connection reestablishment procedure.

In an example, the random access problem (e.g. failure) may indicate that the wireless device experienced one or more random access problems (e.g. random access failures) at the failed cell when the connection failure occurred. In an example, the RLC maximum number of retransmissions may indicate that an RLC layer of the wireless device tried a maximum number of packet retransmissions when connection failure occurred.

In an example, the wireless device identifier of the wireless device may comprise a C-RNTI at the failed cell, a temporary mobile subscriber identity (TMSI), an International Mobile Subscriber Identity (IMSI), Globally Unique Temporary Identifier (GUTI), and/or the like. In an example,

the carrier frequency value may indicate a carrier frequency value of the failed cell. In an example, the carrier frequency value may comprise at least one of an EARFCN, an ARFCN, and/or the like, for example with a maximum value of a maxEARFCN. In an example, the carrier frequency value may be determined according to a band used when obtaining a concerned measurement results.

In an example, the measurement result may comprise at least one of a reference signal received power result (RSRP) or a reference signal received quality (RSRQ) of at least one of the failed cell, a last serving cell, one or more serving cells, one or more neighboring cells of the failed cell, one or more secondary cells of the wireless device when the failed cell was a primary cell, and/or the like. In an example, the wireless device may determine one or more elements of the measurement result by measuring when (or before/after) the wireless device experiences the connection failure. In an example, the quality classification indication 1 bearer (e.g. drb-EstablishedWithQCI-1) may indicate that the connection failure occurred while a bearer with QCI value 1 was configured.

In an example, as shown in FIG. 16, FIG. 18, FIG. 20, and/or FIG. 21, the first base station may receive, from the second base station, the RLF report and/or one or more elements (e.g. the BWP index of the active BWP and/or the BWP configuration parameters) of the RLF report via an Xn interface, via an X2 interface, via an Xx interface, and/or the like (e.g. a direct interface between the first base station and the second base station). In an example, as shown in FIG. 20, the first base station may receive, from the second base station, the RLF report and/or one or more elements (e.g. the BWP index of the active BWP and/or the BWP configuration parameters) of the RLF report via one or more core network entities (e.g. one or more AMFs, one or more MMEs, one or more SGSNs, one or more GGSNs, and/or the like), employing logical direct interfaces between the first base station CU and the one or more core network entities and/or between the second base station and the one or more core network entities (e.g. an NG interface, an N2 interface, an SI interface, and/or the like). In an example, as shown in FIG. 17 and/or FIG. 19, if the first base station is the second base station, the first base station may receive the RLF report via the second message.

In an example, the first base station may determine at least one cell configuration parameter of one or more cells of the first base station for one or more wireless devices based on the RLF report. The one or more cells may comprise the failed cell (e.g. the first cell) of the wireless device. In an example, the at least one cell configuration parameter may comprise at least one of: at least one BWP configuration parameters; at least one transmission power configuration parameter; at least one frequency configuration parameter; at least one beamforming configuration parameter; at least one physical control channel scheduling parameter; at least one antenna configuration parameter; at least one cell selection or reselection configuration parameter for one or more wireless devices; at least one system information; at least one interference control parameter; and/or the like.

In an example, the at least one BWP configuration parameters may be for one or more wireless devices. The at least one BWP configuration parameters may comprise one or more parameters indicating at least one of: a plurality of BWP indexes of a plurality of BWPs; a plurality of BWP bandwidths of a plurality of BWPs; a default BWP index of a default BWP of the plurality of BWPs; a BWP inactivity timer; an initial BWP index of an initial BWP (e.g. initial active BWP) of a plurality of BWPs; a subcarrier spacing for

a plurality of BWPs; a cyclic prefix; a number of contiguous PRBs; an index in the set of one or more DL BWPs and/or one or more UL BWPs; a link between a DL BWP and an UL BWP from a set of configured DL BWPs and UL BWPs; a DCI detection to a PDSCH reception timing value; a PDSCH reception to a HARQ-ACK transmission timing value; a DCI detection to a PUSCH transmission timing value; an offset of a first PRB of a DL bandwidth or an UL bandwidth, respectively, relative to a first PRB of a bandwidth; and/or the like.

In an example, if a wireless device experiences a connection failure during a time that a first BWP of a plurality of BWPs of a cell is an active BWP, a base station may not configure the first BWP as a default BWP (and/or as an initial BWP) for one or more wireless devices.

In an example, if a channel quality (e.g. RSRP, RSRQ) of a second BWP of a plurality of BWPs of a cell is good (e.g. better than a channel quality of an active BWP) when a wireless device experiences a connection failure during a time that a first BWP of the plurality of BWPs is an active BWP, a base station may configure the second BWP as a default BWP (and/or as an initial BWP) for one or more wireless devices.

In an example, the at least one transmission power configuration parameter may comprise a maximum downlink/uplink cell transmission power, a physical downlink control channel (PDCCH) transmission power, one or more power control parameters for uplink and/or downlink, a TPC configuration parameter, an SRS configuration parameter, and/or the like for one or more wireless device and/or for the first base station. In an example, if the first base station determines that the connection failure occurred because of a low transmission power of a PDCCH (e.g. based on the measurement result of the RLF report), the first base station may increase a transmission power of the PDCCH. In an example, if the connection failure occurred because of large interferences on a PDCCH, the first base station may reschedule the PDCCH to be located at other subframes.

In an example, if a wireless device experiences a connection failure during a time that a first BWP of a plurality of BWPs of a cell is an active BWP, a base station may increase an uplink/downlink power level (e.g. 0.1 dB increase) for one or more wireless devices (e.g. UEs served in the first cell) when the one or more wireless devices employ the first BWP as an active BWP.

In an example, if a cause of a connection failure of a wireless device is a random access problem and the wireless device experiences the connection failure during a time that a first BWP of a plurality of BWPs of a cell is an active BWP, a base station may not configure the first BWP for a random access preamble transmission of one or more wireless devices.

In an example, if a cause of a connection failure of a wireless device is an RLC maximum number of retransmissions (e.g. uplink transmission problem; a number of RLC retransmissions is over a threshold value) and the wireless device experiences the connection failure during a time that a first BWP of a plurality of BWPs of a cell is an active BWP, a base station may increase an uplink power level for one or more wireless devices when the one or more wireless devices employ the first BWP as an active BWP.

In an example, the at least one frequency configuration parameter may comprise a carrier frequency, a bandwidth, one or more bandwidth part configuration parameters, and/or the like. In an example, if a cell of the first base station experiences large interferences from neighboring cells, the first base station may change an operation frequency to other

frequency. In an example, if a certain bandwidth part of a served cell of the first base station experiences large interferences from neighboring cells or other technologies, the first base station may change a default bandwidth part and/or an active bandwidth part of one or more wireless devices to other bandwidth part.

In an example, the at least one beamforming configuration parameter may comprise one or more beamforming direction configuration parameters, one or more beam sweeping configuration parameters, one or more synchronization signal (SS)/reference signal (e.g. CSI-RS) configuration parameters, one or more beam recovery related parameters, one or more BRACH parameter, one or more preamble configuration parameters for beam recovery, one or more random access configuration parameters of one or more beams, and/or the like. In an example, if the connection failure occurred because of a random access failure or a failure of beam recovery procedure (e.g. out-of-sync), the base station DU may reschedule random access resources and/or BRACH resources, and/or may reconfigure preambles to reduce random access contentions.

In an example, the at least one physical control channel scheduling parameter may comprise a subframe pattern configuration parameter, a measurement subframe pattern configuration parameter, a transmission type parameter indicating a localized transmission and/or distributed transmission, a resource block assignment configuration parameter, a CSI-RS configuration parameter, and/or the like. In an example, the at least one antenna configuration parameter may comprise default antenna configuration parameters, an antenna port configuration parameter, a number of CRS antenna port parameter, and/or the like. In an example, the at least one cell selection or reselection configuration parameter for one or more wireless devices may comprise one or more power/time threshold parameters for cell selection/reselection of at least one wireless device of the first base station, one or more cell priority configuration parameters for cell selection/reselection, and/or the like. In an example, the connection failure occurred because of a random access failure of the wireless device, the first base station may increase values of the one or more power/time threshold parameters to make wireless devices avoid the failed cell if the wireless devices do not satisfy increased thresholds.

In an example, the first base station may reconfigure one or more IEs of the at least one system information comprising at least one of system information type block type 1 to 21 based on the RLF report. In an example, the at least one interference control parameter may comprise one or more almost blank subframe configuration parameters, one or more CoMP interference management related parameters, and/or the like. In an example, if the connection failure occurred because of interferences from a neighboring cell of the failed cell, the first base station may schedule resource blocks for the neighboring cell and the failed cell not to use the resource blocks simultaneously.

In an example, the first base station may transmit at least one system information blocks comprising the at least one cell configuration parameter. The at least one system information blocks may be at least one of the system information block type 1 to 21. The first base station may transmit at least one of the at least one cell configuration parameter to one or more wireless devices, e.g. via an RRC message. The one or more wireless device may comprise the wireless device.

In an example, as shown in FIG. 22, FIG. 23, FIG. 24, and/or FIG. 25, a wireless device may receive, from a first base station, a first message comprising bandwidth part (BWP) configuration parameters of a first cell. The BWP

configuration parameters may indicate at least one of: a plurality of BWP indexes of a plurality of BWPs; a plurality of BWP bandwidths of the plurality of BWPs; a default BWP index of a default BWP of the plurality of BWPs; a BWP inactivity timer; and/or the like. The wireless device may receive/transmit transport blocks via one of the plurality of BWPs based on the BWP configuration parameters. The wireless device may determine a connection failure from the first cell during a time that a first BWP of the plurality of BWPs is active. The wireless device may transmit, to a second base station, a second message comprising a radio link failure (RLF) report associated with the connection failure. The RLF report may comprise at least one of: a cell identifier of the first cell; and/or a first BWP index of the first BWP. In an example, the BWP configuration parameters may further comprise at least one of: an initial BWP index of an initial BWP of the plurality of BWPs; a subcarrier spacing parameter indicating subcarrier spacing for the plurality of BWPs; a frequency band of the first BWP; a cyclic prefix; a number of contiguous PRBs; an index in the set of one or more DL BWPs and/or one or more UL BWPs; a link between a DL BWP and an UL BWP from a set of configured DL BWPs and UL BWPs; a DCI detection to a PDSCH reception timing value; a PDSCH reception to a HARQ-ACK transmission timing value; a DCI detection to a PUSCH transmission timing value; an offset of a first PRB of a DL bandwidth or an UL bandwidth; and/or the like.

In an example, the BWP configuration parameters may be associated with at least one of uplink BWP configurations or downlink BWP configurations. The RLF report may further comprise one or more elements of the BWP configuration parameters. In an example, the first base station may receive, from the second base station, at least one element of the RLF report. The first base station may determine one or more cell configuration parameters of the first cell based on the at least one element of the RLF report. In an example, the second base station may be the first base station, and/or the first base station may determine one or more cell configuration parameters of the first cell based on the RLF report.

Embodiments may be configured to operate as needed. The disclosed mechanism may be performed when certain criteria are met, for example, in a wireless device, a base station, a radio environment, a network, a combination of the above, and/or the like. Example criteria may be based, at least in part, on for example, wireless device or network node configurations, traffic load, initial system set up, packet sizes, traffic characteristics, a combination of the above, and/or the like. When the one or more criteria are met, various example embodiments may be applied. Therefore, it may be possible to implement example embodiments that selectively implement disclosed protocols.

A base station may communicate with a mix of wireless devices. Wireless devices and/or base stations may support multiple technologies, and/or multiple releases of the same technology. Wireless devices may have some specific capability(ies) depending on wireless device category and/or capability(ies). A base station may comprise multiple sectors. When this disclosure refers to a base station communicating with a plurality of wireless devices, this disclosure may refer to a subset of the total wireless devices in a coverage area. This disclosure may refer to, for example, a plurality of wireless devices of a given LTE or 5G release with a given capability and in a given sector of the base station. The plurality of wireless devices in this disclosure may refer to a selected plurality of wireless devices, and/or a subset of total wireless devices in a coverage area which

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perform according to disclosed methods, and/or the like. There may be a plurality of base stations or a plurality of wireless devices in a coverage area that may not comply with the disclosed methods, for example, because those wireless devices or base stations perform based on older releases of LTE or 5G technology.

According to various embodiments, a device such as, for example, a wireless device, off-network wireless device, a base station, and/or the like, may comprise one or more processors and memory. The memory may store instructions that, when executed by the one or more processors, cause the device to perform a series of actions. Embodiments of example actions are illustrated in the accompanying figures and specification. Features from various embodiments may be combined to create yet further embodiments.

FIG. 26 is an example flow diagram as per an aspect of an embodiment of the present disclosure. At 2610, a wireless device may receive a radio resource control message from a first base station. The radio resource control message may comprise bandwidth part (BWP) configuration parameters of a plurality of BWPs of a first cell. The BWP configuration parameters may comprise an index of a BWP of the plurality of BWPs. The BWP configuration parameters may comprise a field indicating a bandwidth of the BWP. At 2620, the wireless device may receive transport blocks via one or more of the plurality of BWPs based on the BWP configuration parameters. At 2630, the wireless device may detect a connection failure with the first base station during a time that the BWP is active. At 2640, the wireless device may transmit a radio link failure report indicating the connection failure. The radio link failure report may comprise a cell identifier of the first cell. The radio link failure report may comprise the index of the BWP.

According to an example embodiment, the first cell may be a primary cell. According to an example embodiment, the BWP configuration parameters may further comprise a default BWP index of a default BWP of the plurality of BWPs. According to an example embodiment, the radio link failure report may further comprise one or more elements of the BWP configuration parameters. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise a frequency band of the BWP. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise an initial BWP index of an initial BWP of the plurality of BWPs. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise a parameter indicating subcarrier spacing for the plurality of BWPs. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise a cyclic prefix. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise a number of contiguous physical resource blocks. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise an index in a set of one or more downlink BWPs. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise an index in a set of one or more uplink BWPs. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise a link between a downlink BWP and an uplink BWP. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise a downlink control information detection parameter for a physical downlink shared channel reception timing. According to an example embodiment, the

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one or more elements of the BWP configuration parameters may comprise a physical downlink shared channel reception parameter for a hybrid automatic repeat request-acknowledgment transmission timing. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise a downlink control information detection parameter for a physical uplink shared channel transmission timing. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise an offset of a physical resource block of a downlink BWP. According to an example embodiment, the one or more elements of the BWP configuration parameters may comprise an offset of a physical resource block of an uplink BWP.

According to an example embodiment, the plurality of BWPs may comprise one or more downlink BWPs. According to an example embodiment, the radio link failure report may further comprises a failure cause parameter indicating a cause of the connection failure. The cause may comprise an expiry of a timer. According to an example embodiment, the timer may comprise a t310. According to an example embodiment, the timer may comprise a t312.

According to an example embodiment, the plurality of BWPs may comprise one or more uplink BWPs. According to an example embodiment, the radio link failure report may further comprises a failure cause parameter indicating a cause of the connection failure. The cause may comprise a random access problem. The cause may comprise a radio link control maximum number of retransmissions.

According to an example embodiment, the first base station may determine configuration parameters of the first cell based on one or more elements of the radio link failure report. According to an example embodiment, the wireless device may transmit the radio link failure report to the first base station. According to an example embodiment, the wireless device may transmit the radio link failure report to a second base station. According to an example embodiment, the first base station may receive one or more elements of the radio link failure report from a second base station. According to an example embodiment, the radio link failure report may comprise a connection failure type indicating a radio link failure. According to an example embodiment, the radio link failure report may comprise a handover failure.

According to an example embodiment, the wireless device may select a second cell in response to detecting the connection failure. According to an example embodiment, the wireless device may transmit, via the second cell, one or more random access preambles. According to an example embodiment, the second cell may be the first cell. According to an example embodiment, the wireless device may receive, via the second cell, a random access response for the one or more random access preambles. According to an example embodiment, the wireless device may transmit a radio resource control reestablishment request message in response to receiving the random access response. According to an example embodiment, the wireless device may transmit a radio resource control setup request message in response to receiving the random access response.

According to an example embodiment, the wireless device may receive a user equipment information request message requesting information of the connection failure. The wireless device may transmit the radio link failure report in response to receiving the user equipment information request message.

According to an example embodiment, the BWP configuration parameters further comprise a BWP inactivity timer. According to an example embodiment, the configuration

parameters of the first cell may comprise a transmission power configuration parameter. According to an example embodiment, the configuration parameters of the first cell may comprise a frequency configuration parameter. According to an example embodiment, the configuration parameters of the first cell may comprise a beamforming configuration parameter. According to an example embodiment, the configuration parameters of the first cell may comprise a physical control channel scheduling parameter. According to an example embodiment, the configuration parameters of the first cell may comprise an antenna configuration parameter. According to an example embodiment, the configuration parameters of the first cell may comprise a cell selection or reselection configuration parameter for at least one second wireless device. According to an example embodiment, the configuration parameters of the first cell may comprise system information. According to an example embodiment, the configuration parameters of the first cell may comprise an interference control parameter.

According to an example embodiment, the radio link failure report may comprise a carrier frequency value of the first cell. According to an example embodiment, the radio link failure report may comprise a time value indicating a time duration that elapsed since the connection failure. According to an example embodiment, the radio link failure report may comprise a measurement result of the first cell. The measurement result may comprise a reference signal received power. The measurement result may comprise a reference signal received quality. According to an example embodiment, the radio link failure report may comprise at least one measurement result of at least one neighboring cell of the first cell. The at least one measurement result may comprise a reference signal received power. The at least one measurement result may comprise a reference signal received quality. According to an example embodiment, the radio link failure report may comprise a quality classification indication 1 bearer parameter indicating that the connection failure occurred while a bearer with a quality classification indication value being equal to 1 was configured.

According to an example embodiment, the radio link failure report may comprise a wireless device identifier of the wireless device at the first cell. The wireless device identifier may comprise a cell radio network temporary identifier. The wireless device identifier may comprise a temporary mobile subscriber identity. The wireless device identifier may comprise an international mobile subscriber identity. The wireless device identifier may comprise a globally unique temporary identifier. According to an example embodiment, the cell identifier may comprise a physical cell identifier. According to an example embodiment, the cell identifier may comprise a cell global identifier. According to an example embodiment, the cell identifier may comprise a cell index. According to an example embodiment, the cell identifier may comprise a carrier index.

FIG. 27 is an example flow diagram as per an aspect of an embodiment of the present disclosure. At 2710, a wireless device may receive a radio resource control message from a first base station. The radio resource control message may comprise bandwidth part (BWP) configuration parameters of a cell. The BWP configuration parameters may comprise at least one BWP index of at least one BWP. The BWP configuration parameters may comprise a field indicating at least one BWP bandwidth of the at least one BWP. At 2720, the wireless device may receive or transmit, transport blocks via the at least one BWP based on the BWP configuration parameters. At 2730, the wireless device may determine a

connection failure from the cell during a time that a BWP of the at least one BWP is active. At 2740, the wireless device may transmit to a second base station, a radio link failure (RLF) report may indicate the connection failure. The RLF report may comprise a cell identifier of the cell. The RLF report may comprise a BWP index of the BWP.

FIG. 28 is an example flow diagram as per an aspect of an embodiment of the present disclosure. At 2810, a wireless device may receive a first message from a first base station. The first message may comprising bandwidth part (BWP) configuration parameters of a first cell. The BWP configuration parameters may indicate a plurality of BWP indexes of a plurality of BWPs. The BWP configuration parameters may indicate a plurality of BWP bandwidths of the plurality of BWPs. The BWP configuration parameters may indicate a default BWP index of a default BWP of the plurality of BWPs. The BWP configuration parameters may indicate a BWP inactivity timer. At 2820, the wireless device may receive and/or transmit transport blocks via one of the plurality of BWPs based on the BWP configuration parameters. At 2830, the wireless device may determine a connection failure from the first cell during a time that a first BWP of the plurality of BWPs is active. At 2840, the wireless device may transmit to a second base station, a second message comprising a radio link failure (RLF) report associated with the connection failure. The RLF report may comprise a cell identifier of the first cell. The RLF report may comprise a first BWP index of the first BWP.

In this disclosure, “a” and “an” and similar phrases are to be interpreted as “at least one” and “one or more.” Similarly, any term that ends with the suffix “(s)” is to be interpreted as “at least one” and “one or more.” In this disclosure, the term “may” is to be interpreted as “may, for example.” In other words, the term “may” is indicative that the phrase following the term “may” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments.

If A and B are sets and every element of A is also an element of B, A is called a subset of B. In this specification, only non-empty sets and subsets are considered. For example, possible subsets of B={cell1, cell2} are: {cell1}, {cell2}, and {cell1, cell2}. The phrase “based on” (or equally “based at least on”) is indicative that the phrase following the term “based on” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments. The phrase “in response to” (or equally “in response at least to”) is indicative that the phrase following the phrase “in response to” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments. The phrase “depending on” (or equally “depending at least to”) is indicative that the phrase following the phrase “depending on” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments. The phrase “employing/using” (or equally “employing/using at least”) is indicative that the phrase following the phrase “employing/using” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments.

The term configured may relate to the capacity of a device whether the device is in an operational or non-operational state. Configured may also refer to specific settings in a device that effect the operational characteristics of the device whether the device is in an operational or non-operational state. In other words, the hardware, software,

firmware, registers, memory values, and/or the like may be “configured” within a device, whether the device is in an operational or nonoperational state, to provide the device with specific characteristics. Terms such as “a control message to cause in a device” may mean that a control message has parameters that may be used to configure specific characteristics or may be used to implement certain actions in the device, whether the device is in an operational or non-operational state

In this disclosure, various embodiments are disclosed. Limitations, features, and/or elements from the disclosed example embodiments may be combined to create further embodiments within the scope of the disclosure.

In this disclosure, parameters (or equally called, fields, or Information elements: IEs) may comprise one or more information objects, and an information object may comprise one or more other objects. For example, if parameter (IE) N comprises parameter (IE) M, and parameter (IE) M comprises parameter (IE) K, and parameter (IE) K comprises parameter (information element) J. Then, for example, N comprises K, and N comprises J. In an example embodiment, when one or more messages comprise a plurality of parameters, it implies that a parameter in the plurality of parameters is in at least one of the one or more messages, but does not have to be in each of the one or more messages.

Furthermore, many features presented above are described as being optional through the use of “may” or the use of parentheses. For the sake of brevity and legibility, the present disclosure does not explicitly recite each and every permutation that may be obtained by choosing from the set of optional features. However, the present disclosure is to be interpreted as explicitly disclosing all such permutations. For example, a system described as having three optional features may be embodied in seven different ways, namely with just one of the three possible features, with any two of the three possible features or with all three of the three possible features.

Many of the elements described in the disclosed embodiments may be implemented as modules. A module is defined here as an element that performs a defined function and has a defined interface to other elements. The modules described in this disclosure may be implemented in hardware, software in combination with hardware, firmware, wetware (i.e. hardware with a biological element) or a combination thereof, all of which may be behaviorally equivalent. For example, modules may be implemented as a software routine written in a computer language configured to be executed by a hardware machine (such as C, C++, Fortran, Java, Basic, MATLAB or the like) or a modeling/simulation program such as Simulink, Stateflow, GNU Octave, or LabVIEW-MathScript. Additionally, it may be possible to implement modules using physical hardware that incorporates discrete or programmable analog, digital and/or quantum hardware. Examples of programmable hardware comprise: computers, microcontrollers, microprocessors, application-specific integrated circuits (ASICs); field programmable gate arrays (FPGAs); and complex programmable logic devices (CPLDs). Computers, microcontrollers and microprocessors are programmed using languages such as assembly, C, C++ or the like. FPGAs, ASICs and CPLDs are often programmed using hardware description languages (HDL) such as VHSIC hardware description language (VHDL) or Verilog that configure connections between internal hardware modules with lesser functionality on a programmable device. The above mentioned technologies are often used in combination to achieve the result of a functional module.

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While various embodiments have been described above, it should be understood that they have been presented by way of example, and not limitation. It will be apparent to persons skilled in the relevant art(s) that various changes in form and detail can be made therein without departing from the scope. In fact, after reading the above description, it will be apparent to one skilled in the relevant art(s) how to implement alternative embodiments. Thus, the present embodiments should not be limited by any of the above described exemplary embodiments.

In addition, it should be understood that any figures which highlight the functionality and advantages, are presented for example purposes only. The disclosed architecture is sufficiently flexible and configurable, such that it may be utilized in ways other than that shown. For example, the actions listed in any flowchart may be re-ordered or only optionally used in some embodiments.

Further, the purpose of the Abstract of the Disclosure is to enable the U.S. Patent and Trademark Office and the public generally, and especially the scientists, engineers and practitioners in the art who are not familiar with patent or legal terms or phraseology, to determine quickly from a cursory inspection the nature and essence of the technical disclosure of the application. The Abstract of the Disclosure is not intended to be limiting as to the scope in any way.

Finally, it is the applicant's intent that only claims that include the express language “means for” or “step for” be interpreted under 35 U.S.C. 112. Claims that do not expressly include the phrase “means for” or “step for” are not to be interpreted under 35 U.S.C. 112.

What is claimed is:

1. A wireless device comprising:

one or more processors; and

memory storing instructions that, when executed by the one or more processors, cause the wireless device to: transmit, based on detecting a radio link failure of a cell of a base station, a radio link failure report indicating a connection failure, wherein the radio link failure report comprises:

a cell identifier of the cell; and

a subcarrier spacing parameter indicating a subcarrier spacing of a bandwidth part (BWP) associated with the connection failure.

2. The wireless device of claim 1, wherein the instructions further cause the wireless device to transmit, to the base station, a radio resource control message indicating that the wireless device has a report associated with the connection failure.

3. The wireless device of claim 2, wherein the instructions further cause the wireless device to receive, from the base station, a user equipment (UE) information request message indicating the radio link failure report.

4. The wireless device of claim 3, wherein the instructions further cause the wireless device to transmit, to the base station, a UE information response message comprising the radio link failure report.

5. The wireless device of claim 4, wherein the radio link failure report further comprises a radio link failure cause parameter indicating a cause of the connection failure is a random access problem.

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6. The wireless device of claim 5, wherein the connection failure occurs during a random access procedure of the wireless device with the base station.

7. The wireless device of claim 6, wherein the instructions further cause the wireless device to receive radio resource control (RRC) configuration parameters of the cell comprising the BWP.

8. The wireless device of claim 7, wherein:

the instructions further cause the wireless device to receive BWP configuration parameters of the BWP of the cell; and

the BWP configuration parameters indicate at least one of: the subcarrier spacing of the BWP; or an index of the BWP.

9. The wireless device of claim 8, wherein the BWP is an initial BWP of the cell for the wireless device.

10. The wireless device of claim 9, wherein:

the radio link failure report further comprises a measurement result of the cell; and

the measurement result comprises at least one of: a reference signal received power result; or a reference signal received quality.

11. The wireless device of claim 10, wherein the radio link failure report further comprises at least one of:

a time value indicating an elapsed time since the connection failure; or

a cell radio network temporary identifier (C-RNTI) of the wireless device for the cell.

12. The wireless device of claim 11, wherein the radio link failure report further comprises a second time value indicating an elapsed time since a last handover of the wireless device until the connection failure.

13. The wireless device of claim 12, wherein the radio link failure report further comprises a value indicating a carrier frequency of the cell.

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14. The wireless device of claim 13, wherein the cell is a primary cell of the wireless device when the connection failure occurred.

15. The wireless device of claim 14, wherein:

the instructions further cause the wireless device to select a second cell of the base station after the connection failure; and

the radio link failure report is transmitted via the second cell.

16. The wireless device of claim 15, wherein the connection failure comprises a handover failure.

17. The wireless device of claim 16, wherein the radio link failure report further comprises a parameter indicating a bandwidth of the BWP associated with the connection failure.

18. The wireless device of claim 4, wherein:

the instructions further cause the wireless device to receive BWP configuration parameters of the BWP of the cell; and

the BWP configuration parameters indicate at least one of: the subcarrier spacing of the BWP; or an index of the BWP.

19. The wireless device of claim 18, wherein:

the radio link failure report further comprises a measurement result of the cell; and

the measurement result comprises at least one of: a reference signal received power result; or a reference signal received quality.

20. The wireless device of claim 19, wherein the radio link failure report further comprises at least one of:

a time value indicating an elapsed time since the connection failure; or

a cell radio network temporary identifier (C-RNTI) of the wireless device for the cell.

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